

Deep Learning for Metal Artifact Reduction in Computed Tomography

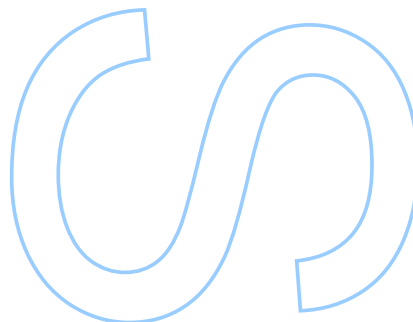
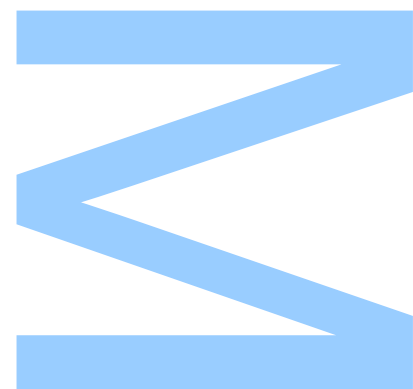
Ludmila Montorsi Queiroz

Mestrado em Física Médica

Departamento de Física e Astronomia

Faculdade de Ciências da Universidade do Porto

2026



UNIVERSIDADE DO PORTO

MASTERS THESIS

Deep Learning for Metal Artifact Reduction in Computed Tomography

Author:

Ludmila MONTORSI QUEIROZ

Supervisor:

Hélder Filipe PINTO DE OLIVEIRA

Co-supervisor:

Eduardo MATOS RODRIGUES and

Tânia PEREIRA

*A thesis submitted in fulfilment of the requirements
for the degree of MSc. Medical Physics*

at the

Faculdade de Ciências da Universidade do Porto
Departamento de Ciência de Computadores
Departamento de Física e Astronomia
INESC TEC

September 12, 2026

"The Universe is under no obligation to make sense to you."

Neil deGrasse Tyson, *Astrophysics for People in a Hurry* (2017).

"Nothing in life is to be feared; it is only to be understood."

Marie Curie, *Madame Curie: A Biography* (1937).

Acknowledgements

I'm deeply grateful to my husband for his unwavering support, patience, and encouragement throughout this journey. His love and understanding have been a constant source of strength, and I am truly thankful for his presence in my life.

Also, I thank my father-in-law and mother-in-law for their emotional presence, vibrating, fearing, and feeling each step of this journey with me while continued to encourage and support me.

I need to express my sincere gratitude to my mother, who, even though not physically present, has been a guiding light in my life. Her love, wisdom, and values continue to inspire me every day, and I carry her memory with me as I face the challenges and triumphs of this journey. I hope to honor her legacy by doing good to all who cross my path. Without her love and presence, I would not have made it this far.

I am grateful to my father, my sister, and my brother for being who they are — incredible people I am privileged to have in my life — and for their unconditional love. Their existence is fundamental to me staying strong, moving forward, and pursuing my dreams.

I would like to express my sincere love to my cats Haroldo and Luna, that were by my side every day during this work. Filling my life with joy, love, and companionship, they have been a source of comfort and emotional support, making the challenges of this journey more bearable and the achievements even sweeter.

I am also grateful to my friends and colleagues for their clarifications, companionship, encouragement, and emotional support.

Finally, I would like to express my deepest gratitude to Eduardo Matos Rodrigues, Hélder Filipe Pinto de Oliveira e Tânia Pereira for their invaluable guidance, support, and encouragement throughout this research journey. Their expertise and insights have been instrumental in shaping this thesis.

UNIVERSIDADE DO PORTO

Abstract

Faculdade de Ciências da Universidade do Porto

Departamento de Ciência de Computadores

Departamento de Física e Astronomia

INESC TEC

MSc. Medical Physics

Deep Learning for Metal Artifact Reduction in Computed Tomography

by [Ludmila MONTORSI QUEIROZ](#)

Metal artifacts remain a common and clinically significant source of image degradation in computed tomography (CT), affecting an estimated 21% of scans and potentially obscuring or mimicking pathology. This thesis develops and rigorously evaluates a reproducible deep learning pipeline for image-domain metal artifact reduction, comparing a conventional U-Net against a dual-supervision variant that adds an auxiliary artifact-mask prediction head to a shared encoder-decoder. Using the AAPM CT-MAR benchmark dataset (14,000 image pairs derived from DeepLesion and UCLH Stroke EIT), a patient-level traceability procedure was implemented to recover exam and patient identity absent from the original metadata, enabling patient-level partitioning designed to minimize cross-subset information leakage (187/41/41 patients) across three deterministic training seeds per architecture. The primary, pre-specified hypothesis, that reconstruction improves structural similarity (SSIM) specifically in the region directly affected by the artifact while leaving the Clean Zone comparatively unchanged, was confirmed on the internal test set (41 patients, 1,478 slices) for both architectures: artifact-region SSIM improved by 0.328 for the Single Model (95% CI [0.296; 0.355], Holm-adjusted $p = 2.7 \times 10^{-12}$) and by an equivalent margin for the Dual Model, against a much smaller change of about 0.06 in the Clean Zone. As secondary, complementary evidence, both architectures also significantly improved global PSNR (27.4 to ~ 35.1 dB), SSIM (0.836 to ~ 0.944) and MAE (36.6 to ~ 17.5 HU) relative to the corrupted input (patient-level sign and t -tests, $p < 0.001$). A direct paired comparison showed a small but statistically significant global advantage for the Single Model and a small but significant Clean-Zone preservation advantage for the Dual Model, indicating a correction-versus-preservation trade-off rather than a clear superiority of either design. Based on this pre-specified artifact-region criterion, the Single Model

(seed 2024) was selected, frozen, and characterized on an independent external set of 15 real exams from The Cancer Imaging Archive, where, in the absence of a clean reference, only the magnitude of input-to-output change could be quantified (mean SSIM 0.8953, PSNR 33.14 dB, MAE 21.12 HU). A completed blinded, paired evaluation by five physicians (75 case-level responses) judged the model output to better preserve the anatomical region underlying the artifact in 56.0% of presentations, versus 28.0% favoring the artifact input (95% cluster-bootstrap CI [28.0; 82.7]), with a similar pattern favoring the output for perceived artifact intensity (81.3%) and clinical-interpretation preference (56.0%). Given the small reader panel and correspondingly wide confidence intervals, these results are interpreted descriptively, as supportive but not conclusive evidence of external anatomical safety and perceived clinical benefit.

Keywords: Computed Tomography, Metal Artifact Reduction, Deep Learning, U-Net, Multi-task Learning, Reproducibility, Patient-level Data Partitioning, External Validation, Image Quality Assessment

UNIVERSIDADE DO PORTO

Resumo

Faculdade de Ciências da Universidade do Porto

Departamento de Ciência de Computadores

Departamento de Física e Astronomia

INESC TEC

MSc. Medical Physics

Aprendizado Profundo para Redução de Artefatos de Metal em Tomografia Computadorizada

por [Ludmila MONTORSI QUEIROZ](#)

Artefatos metálicos permanecem uma causa comum e clinicamente relevante de degradação de imagem em tomografia computadorizada (CT), estimados em cerca de 21% dos exames, podendo esconder ou simular patologias. Esta dissertação desenvolve e avalia rigorosamente um *pipeline* reproduzível de aprendizagem profunda para redução de artefatos metálicos no domínio da imagem, comparando uma U-Net convencional com uma variante de dupla supervisão que acrescenta uma saída auxiliar de predição da máscara de artefacto a um *encoder-decoder* partilhado. Utilizando o *dataset* de referência CT-MAR da AAPM (14.000 pares de imagens derivados do DeepLesion e do UCLH Stroke EIT), foi implementado um procedimento de rastreabilidade ao nível do paciente para recuperar a identidade do exame e do paciente, ausente nos metadados originais, permitindo um particionamento por paciente concebido para minimizar o risco de fuga de informação entre subconjuntos (187/41/41 pacientes) ao longo de três sementes de treino determinísticas por arquitetura. A hipótese primária, definida à partida, de que a reconstrução melhora a similaridade estrutural (SSIM) especificamente na região diretamente afetada pelo artefacto, mantendo o tecido limpo comparativamente inalterado, foi confirmada no conjunto de teste interno (41 pacientes, 1.478 fatias) para ambas as arquiteturas: o SSIM na região do artefacto melhorou 0,328 no modelo de supervisão única (IC 95% [0,296; 0,355], $p = 2,7 \times 10^{-12}$ ajustado por Holm) e por uma margem equivalente no modelo de dupla supervisão, contra uma alteração muito menor, de cerca de 0,06, no tecido limpo. Como evidência secundária e complementar, ambas as arquiteturas também melhoraram significativamente o PSNR global (27,4 para $\sim 35,1$ dB), o SSIM (0,836 para $\sim 0,944$) e o MAE

(36,6 para $\sim 17,5$ HU) face à entrada corrompida (testes de sinais e t ao nível do paciente, $p < 0,001$). Uma comparação pareada direta mostrou uma pequena mas estatisticamente significativa vantagem global do modelo de supervisão única e uma pequena mas significativa vantagem de preservação na região limpa do modelo de dupla supervisão, indicando um compromisso entre correção e preservação, e não uma superioridade clara de nenhuma das arquiteturas. Com base neste critério pré-definido de SSIM na região do artefacto, o modelo de supervisão única (semente 2024) foi selecionado, congelado e caracterizado num conjunto externo independente de 15 exames reais provenientes do The Cancer Imaging Archive, onde, na ausência de referência limpa, apenas foi possível quantificar a magnitude da alteração entrada-saída (SSIM médio de 0,8953, PSNR de 33,14 dB e MAE de 21,12 HU). Uma avaliação cega e pareada por cinco médicos, já concluída (75 respostas ao nível do caso), considerou que a saída do modelo preservava melhor a região anatômica subjacente ao artefacto em 56,0% das apresentações, contra 28,0% a favor da entrada com artefacto (IC 95% por *cluster bootstrap* [28,0; 82,7]), com um padrão semelhante a favor da saída quanto à intensidade percebida do artefacto (81,3%) e à preferência para interpretação clínica (56,0%). Dado o pequeno painel de leitores e os intervalos de confiança correspondentemente largos, estes resultados são interpretados de forma descritiva, como evidência de suporte, mas não conclusiva, da segurança anatômica externa e do benefício clínico percebido.

Palavras-chave: Tomografia Computadorizada, Redução de Artefatos Metálicos, Aprendizagem Profunda, U-Net, Aprendizagem Multitarefa, Reprodutibilidade, Particionamento por Paciente, Validação Externa, Avaliação da Qualidade de Imagem

Contents

Acknowledgements	ii
Abstract	iii
Resumo	v
Contents	vii
List of Figures	x
Lista de Gráficos	xi
List of Tables	xii
Lista de Abreviaturas	xiv
1 Introduction	1
1.1 Clinical and Technical Context	2
1.2 Work Motivation	3
2 Objectives and Investigation Question	4
2.1 General Objective	4
2.2 Specific Objectives	4
2.3 Hypothesis and Outcomes	5
3 Related Works	6
3.1 Metallic Artifacts in CT	6
3.2 Artifact Reduction based on Deep Learning	6
3.3 Reproducible Evaluation and Generalization	8
4 Materials and Methods	9
4.1 Data, curation and traceability	9
4.1.1 CT-MAR Dataset and Reference Sources	9
4.1.2 Partitioning at patient level	10
4.1.3 Anatomical Distribution and Stratification	13
4.2 Pre-processing	14
4.2.1 Integrity Verification	14
4.2.2 Background Removal through Body Mask	14
4.2.3 Intensity Window and Normalization	14
4.2.4 Artifact Mask for Auxiliary Supervision	14

4.3	Compared Architectures	15
4.3.1	U-Net Single Supervision	15
4.3.2	U-Net Dual Supervision	15
4.4	Reproducible Training	17
4.4.1	Settings, determinism and two-stage selection	17
4.4.2	Loss Functions	19
4.5	Internal Evaluation	20
4.5.1	ROI Metrics	21
4.5.2	Global Metrics	23
4.5.3	Statistical Analysis	23
4.6	External Evaluation	24
4.6.1	Design and Cases	24
4.6.2	Quantitative Characterization of Input–Output	26
4.6.3	Presentation and Evaluation Instrument	26
4.6.4	Outcomes and Analysis	27
4.6.5	Operational Status	27
5	Results	28
5.1	Final Group and Experimental Integrity	28
5.1.1	Final Group and Exclusion of Unmatched Pairs	28
5.1.2	Reported and Recovered Patient Counts	28
5.1.3	Automatic Approval and Manual Review Outcomes	30
5.1.4	Complementary Analysis with MS-SSIM	30
5.1.5	Anatomical Distribution and Stratification	36
5.2	Model Training	38
5.2.1	Training Convergence and Per-Run Checkpoint Selection	38
5.3	Internal Evaluation in Test Subset	41
5.3.1	Region of Interest Evaluation	42
5.3.2	Selection of the External-Evaluation Candidate	43
5.3.3	Global Outcomes	46
5.3.4	Residual Maps in Extreme Cases	50
5.4	Paired Comparison Single x Dual Model	57
5.4.1	Statistical Analysis per ROI	57
5.4.2	Global Statistical Analysis	59
5.5	External and Physician’s Evaluation	62
5.5.1	Capture Status and Participants	62
5.5.2	Quantitative Characterization of Input–Output	64
5.5.3	Primary Outcome: Anatomical Visualization	67
5.5.4	Secondary Outcomes and Safety Perception	68
5.5.5	Limited Interpretation to the Observational Design	69
6	Discussion	70
6.1	Summary of the Main Findings	70
6.2	Comparison with Prior Work	71
6.3	Architectural Comparison Interpretation	71
6.4	Anatomical Preservation and Reconstruction Safety	72
6.5	Generalization and External Evaluation	73
6.6	Reader Evaluation and Clinical Implications	74

6.7	Limitations	75
6.8	Implications and Future Work	77
7	Conclusions	79
7.1	Study Conclusions	79
7.2	Dissertation Contributions	80
	Bibliography	80
A	Supplementary Internal Evaluation	87
A.1	Training Convergence Across Runs	87
A.2	Worst Performance Cases	100
A.3	Computational Performance	103

List of Figures

4.1	Single Supervision U-Net architecture	16
4.2	Dual Supervision U-Net architecture	17
4.3	Artifact and clean ROI examples	22
5.4	Visually approved low MS-SSIM pairs	33
5.5	Visually rejected pairs	34
5.15	Greatest Artifact-ROI Δ SSIM: head_1147	51
5.16	Greatest Artifact-ROI Δ SSIM: head_1274	52
5.17	Greatest Artifact-ROI Δ SSIM: head_1280	53
5.18	Least Artifact-ROI Δ SSIM: body_11949	54
5.19	Least Artifact-ROI Δ SSIM: body_2906	55
5.20	Least Artifact-ROI Δ SSIM: body_11956	56
5.22	Least structurally changed external cases	65
5.23	Most structurally changed external cases	67

Lista de Gráficos

5.1	Final group curation	30
5.2	MS-SSIM distribution and zoom (clean CT-MAR vs DL/Stroke)	31
5.3	MS-SSIM ECDF (clean CT-MAR vs DL/Stroke)	32
5.6	Cosine similarity of MS-SSIM-audited pairs	35
5.7	MS-SSIM distribution of audited pairs	35
5.8	Score distribution of excluded pairs: DL vs Stroke EIT	36
5.9	Training convergence (results): Single Model, seed 2024	40
5.10	Training convergence (results): Dual Model, seed 2024	41
5.11	Per-image ROI metric improvements	45
5.12	Artifact-ROI correction versus clean-ROI change	46
5.13	Per-image global PSNR and SSIM improvement	49
5.14	Patient-level Bland–Altman agreement of mean HU	50
5.21	Global PSNR and SSIM improvement	62
A.1	Training convergence: Single Model, seed 42	88
A.2	Training convergence: Dual Model, seed 42	90
A.3	Training convergence: Single Model, seed 1337	91
A.4	Training convergence: Dual Model, seed 1337	93
A.5	Training convergence: Single Model, seed 2024	94
A.6	Training convergence: Dual Model, seed 2024	96
A.7	Pixel-level HU agreement	97
A.8	Per-image input-output global metrics	99
A.9	Patient-level artifact load and improvement	100
A.10	Worst-case slices by global Δ SSIM	101
A.11	Worst-case slices by global Δ PSNR	102
A.12	Worst-case slices: joint Δ PSNR vs. Δ SSIM	103
A.13	Benchmark Latency	104

List of Tables

4.1	Evaluation metrics by context	23
5.1	Initial pair-matching screening	28
5.2	Pair matching after MS-SSIM review	29
5.3	Excluded unmatched pairs before and after MS-SSIM review	29
5.4	Patient-level anatomical distribution by split	37
5.5	Patient distribution by split	37
5.6	Pair distribution by split and anatomical category	37
5.7	Pair distribution by split	38
5.8	ROI Δ SSIM	42
5.9	ROI Δ PSNR	42
5.10	ROI MAE reduction	43
5.11	Per-slice global SSIM on the test set	47
5.12	Per-slice global PSNR on the test set	47
5.13	Per-slice global MAE on the test set	47
5.14	Patient-level global Δ SSIM	47
5.15	Patient-level global Δ PSNR	47
5.16	Patient-level global Δ MAE	48
5.17	Artifact-ROI SSIM improvement	57
5.18	Artifact-ROI PSNR improvement	57
5.19	Artifact-ROI MAE reduction	57
5.20	Paired Artifact-ROI improvement comparison	58
5.21	Clean-ROI SSIM preservation magnitude	58
5.22	Clean-ROI PSNR preservation magnitude	58
5.23	Clean-ROI MAE preservation magnitude	59
5.24	Paired Clean-ROI preservation advantage	59
5.25	SSIM improvement over corrupted input	60
5.26	PSNR improvement over corrupted input	60
5.27	MAE reduction over corrupted input	61
5.28	Paired Single vs Dual Model comparison	62
5.29	Recruitment and completion status	63
5.30	Reader profile characteristics	63
5.31	Reader specialty distribution	63
5.32	External input-output global metrics	64
5.33	Illustrative external cases by input-output SSIM	64
5.34	Primary outcome: anatomical preservation preference	67
5.35	Primary outcome: output–input contrast	68
5.36	Secondary outcome: artifact intensity	68
5.37	Secondary outcome: distortion in other regions	68
5.38	Secondary outcome: clinical preference	68

5.39 Secondary outcome: complementary utility	69
5.40 Intra-reader repeat agreement	69
A.1 Best validation SSIM by model and seed	96
A.2 Validation PSNR at best validation SSIM	96
A.3 Validation MAE at best validation SSIM	96

Lista de Abreviaturas

AAPM American Association of Physicists in Medicine.

AMP Automatic Mixed Precision.

BCE Binary Cross-Entropy.

BN Batch Normalization.

CNN Convolutional Neural Network.

CT Computed Tomography.

CT-MAR CT Metal Artifact Reduction Challenge.

DICE Dice Similarity Coefficient.

DL Deep Learning.

FBP Filtered Back Projection.

FCUP Faculdade de Ciências da Universidade do Porto.

FROC Free-Response Receiver Operating Characteristic.

GAN Generative Adversarial Network.

GPU Graphics Processing Unit.

HU Unidade Hounsfield.

iMAR Iterative Metal Artifact Reduction.

IR Iterative Reconstruction.

MAE Mean Absolute Error.

MAR Metal Artifact Reduction.

MBIR Model-Based Iterative Reconstruction.

MS-SSIM Multi-Scale Structural Similarity Index Measure.

NMAR Normalized Metal Artifact Reduction.

NPY NumPy Array.

O-MAR Orthopedic Metal Artifact Reduction.

PSNR Peak Signal-to-Noise Ratio.

ReLU Rectified Linear Unit.

RMSE Root Mean Square Error.

ROC Receiver Operating Characteristic.

ROI Region of Interest.

SEMAR Single Energy Metal Artifact Reduction.

SSIM Structural Similarity Index Measure.

TCIA The Cancer Imaging Archive.

U-Net U-shaped Network.

UP Universidade do Porto.

This thesis is dedicated to my beloved late mother, who would have been filled with pride and genuine happiness with each step of this achievement.

Chapter 1

Introduction

In 1895, while repeating an experiment on cathode rays with a modified Crookes tube, the German physicist Wilhelm Conrad Röntgen observed a new, highly penetrating form of radiation and used it to produce the first radiograph of the human body. Later naming the phenomenon "X-rays" [1, 2]. The discovery spread rapidly: within months, X-rays were already being used for medical diagnosis, such as in a fracture radiograph produced in the United States in early 1896 [1]. Since then, the technique has undergone successive modernizations, diversifying into multiple modalities and devices and consolidating itself as an indispensable tool in modern healthcare [3, 4].

From these successive modernizations emerged, in 1971, the first computerized tomography scanner. This technology is based on the principle of x-rays attenuation, in which the amount of radiation absorbed or scattered by the material is related to its density [5, 6].

Computed Tomography (CT) was the first medical imaging modality to provide quantitative information about tissue attenuation, something that was not possible with conventional radiography [7]. Furthermore, allowing sectional visualization of the human body, it eliminated the overlap of anatomical structures inherent to two-dimensional images, enabling precise localization and assessment of the extent of lesions [7].

However, these advances brought a significant increase in technological complexity. The acquisition of multiple projections, combined with computational reconstruction and digital image processing, introduced new forms of degradation that were nonexistent or of little relevance in conventional radiography. Thus, computed tomography became particularly susceptible to the presence of artifacts, which can compromise the diagnostic quality and clinical reliability of images [7].

Various types of artifacts can be observed in computed tomography, including noise, beam hardening, motion, ring artifacts, and metal artifacts, which can obscure or simulate pathologies [8]. The causes of these artifacts are varied, ranging from detector failures or miscalibration,

statistical limitations associated with low photon counts, to physical phenomena such as beam hardening and scattering, which produce dark streaks between high-attenuation objects such as metal, bone, or iodinated contrast media [7].

Specifically, streak artifacts caused by metallic implants are observed in approximately 21% of computed tomography scans, being considered extremely common in clinical practice [7]. The high attenuation or even total blockage of X-rays by these implants prevents the radiation from adequately reaching the detectors, resulting in low-quality projection data and compromising image reconstruction [9].

The increased use of metallic implants, driven by the evolution of healthcare, has intensified concerns regarding the negative impact of these artifacts on image quality and the reliability of diagnostic analyses [9]. Traditionally, various [Metal Artifact Reduction \(MAR\)](#) algorithms have been proposed, including image inpainting techniques, sinogram inpainting, and [Model-Based Iterative Reconstruction \(MBIR\)](#) or combinations of these approaches [10].

Although analytical methods such as [Filtered Back Projection \(FBP\)](#) have dominated tomographic reconstruction for decades due to their computational efficiency and conceptual simplicity, their limitations become evident under non-ideal clinical conditions, requiring higher radiation doses to ensure image quality [11]. [MBIR](#) methods emerged as an alternative to overcome these limitations, at the cost of higher computational demand and the introduction of artificial textures in the reconstructed image [11].

Recently, deep learning-based approaches have been widely investigated for artifact correction in computed tomography, demonstrating superior performance and greater clinical applicability compared to conventional methods [12]. In particular, [Convolutional Neural Networks \(CNNs\)](#) have been proposed for metal artifact reduction, enabling noise suppression and preservation of anatomical and soft tissue details [12].

1.1 Clinical and Technical Context

A diagnostic mistake can have serious consequences for the patient, including treatment delays, inadequate or unnecessary treatments, and even life-threatening risks [12]. Artifacts can degrade an image to the point of making it unusable for diagnosis, in which case the scan may need to be repeated, exposing the patient to further radiation. When the artifact instead obscures nearby soft tissue rather than making the whole image unusable, it can still lead to an incorrect diagnosis or, in radiotherapy planning, an incorrect radiation dose calculation [13].

Therefore, the reduction of artifacts in CT is a research area with significant clinical importance, and the development of effective methods to reduce artifacts gains high value in medical practice.

1.2 Work Motivation

Although studies using deep learning for metal artifact reduction are not new, a lot of the work published to date does not present a rigorous and detailed evaluation of the performance of the proposed models. In particular, the evaluation of the performance of models on external datasets (which were not used during model training) or evaluation of images by qualified professionals is often lacking. Moreover, the training of supervised models depends on reference images, which are difficult to obtain in a clinical context, creating a dependency on synthetic or simulated images. This gap becomes even more relevant when considering that the performance of models trained on synthetic images may not be representative of their performance on real images. This mismatch can lead to overestimation of their effectiveness and limit their clinical applicability.

In this sense, the present study proposes a more rigorous and detailed evaluation of the performance of deep learning models for this purpose.

Chapter 2

Objectives and Investigation Question

2.1 General Objective

The general objective of this dissertation is to develop, evaluate and rigorously compare a CNN model for metal artifact reduction in CT, prioritizing anatomical preservation, robustness to domain variation and interpretability of the correction process. Additionally, the study aims to investigate the generalization of the best performing model to real and external data, with a focus on clinical applicability. Healthcare professionals will be involved in the evaluation of the model's performance on real data, providing a comprehensive assessment of its effectiveness and potential for clinical use.

2.2 Specific Objectives

The specific objectives of this study include:

- Develop a reproducible pipeline for training and evaluating metal artifact reduction models in CT in the image domain, with rigorous separation between training, validation, and test data.
- Evaluate the performance of a U-shaped Network (U-Net) for image reconstruction in metal artifact reduction and compare it with a version of itself with an additional artifact-mask supervision output.
- Evaluate the performance of the two versions using quantitative metrics like: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), Mean Absolute Error (MAE), Binary Cross-Entropy (BCE) and Dice Similarity Coefficient (DICE). Besides, performing error maps, statistical and qualitative analysis to assess the quality of the reconstructed images.

- Investigate the generalization capability of the best performing version in the internal evaluation stage on real and external data from the [The Cancer Imaging Archive \(TCIA\)](#).

2.3 Hypothesis and Outcomes

The primary hypothesis of this study is that, in the internal test partition, the correction produced by each architecture makes a real, meaningful difference to the structural quality of the image compared to the artifact input. This is measured with [SSIM](#), calculated separately in the [Region of Interest \(ROI\)](#) where the metal artifact was inserted and in the clean [ROI](#), without an artifact. The clean [ROI](#) should stay about the same or change only a little. Each patient's slices were averaged into a single value per patient ($n = 41$), and the comparison asked whether there was typically no difference at all between the input and the model.

The secondary hypotheses are threefold. First, that [PSNR](#) and [MAE](#), in HU, also improve compared to the input, using the same per-patient averaging. Second, that the same three metrics ([SSIM](#), [PSNR](#) and [MAE](#)) also improve when calculated over the whole image. Third, a direct comparison between the Single Model and the Dual Model on these same three metrics, addressing the architecture-comparison objective.

As a way to judge anatomical preservation, the change introduced in the Clean Zone was expected to be clearly smaller than the correction observed in the Artifact Zone. The overall shift in average HU values, checked with a Bland–Altman plot at the patient level, was expected to be small compared to the width of the [HU](#) window used. No fixed numeric limit was set in advance for how small the Clean-Zone change had to be, so preservation is described rather than formally tested.

Finally, since the external [TCIA](#) set has no clean reference image to compare against, the generalization hypothesis is necessarily more limited. The selected architecture is expected to produce changes of a plausible size in the external cases, without simply reproducing the input unchanged. However, only the doctors' blinded evaluation ([section 4.6](#)), not a reference-based metric, can confirm whether artifact removal is accurate and clinically safe.

Chapter 3

Related Works

3.1 Metallic Artifacts in CT

The traditional methods of artifact reduction in CT are predominantly based on physical and mathematical principles associated with image formation. Barrett and Keat[14] provide a detailed description of the origin of the main artifacts in CT, including beam hardening, photon starvation, scattering, and geometric effects, as well as classical strategies for their mitigation.

Reconstruction techniques such as FBP were widely used due to their computational efficiency, but they are highly sensitive to noise and metallic artifacts. More advanced methods, such as Iterative Reconstruction (IR) and MBIR, incorporate statistical models of the acquisition system and noise, allowing for partial artifact reduction. However, these approaches often introduce artificial textures and have limitations in complex clinical scenarios [11].

Recent revisions indicate that commercial algorithms for MAR, such as Orthopedic Metal Artifact Reduction (O-MAR), Iterative Metal Artifact Reduction (iMAR), and Single Energy Metal Artifact Reduction (SEMAR), although effective in specific situations, rely on proprietary parameters and exhibit inconsistent performance in cases of dense or multiple implants [9, 15]. These limitations have driven the investigation of deep learning-based approaches.

3.2 Artifact Reduction based on Deep Learning

The first applications of Deep Learning (DL) for MAR focused on the image domain, exploring convolutional networks to map degraded images to corrected versions. Gjestebj et al. [16] investigated metal artifact reduction in the image domain based on CNN and demonstrated significant quantitative improvements compared to conventional techniques.

Zhang and Yu [17] also proposed a supervised model based on Convolutional Neural Network (CNN) for metal artifact reduction in CT, achieving consistent gains in PSNR and Root

Mean Square Error (RMSE). These works helped consolidate the use of convolutional networks for **MAR** in the image domain. In particular, Zhang and Yu used a shallow fully convolutional **CNN** for fusing information from multiple pre-corrected images. Subsequent works increasingly adopted encoder-decoder architectures, especially those based on **U-shaped Network (U-Net)**, for correction/restoration tasks. **U-Net**-type networks and their variants are capable of capturing global contextual information through the contraction path, and recovering fine spatial details through the expansion path and *Skip Connections*[18].

Recent systematic reviews have highlighted these advances, emphasizing that most supervised methods in the image domain exhibit competitive performance, especially when trained with high-quality paired data [9, 11].

In addition to the image domain, some studies explored the sinogram domain, with linear interpolation and **Normalized Metal Artifact Reduction (NMAR)** methods. These approaches operate directly on the projection data and act directly on the physical origin of metallic artifacts[19]. However, access to raw projection data is often restricted in commercial **CT** systems, limiting the applicability of such approaches in clinical settings [16].

Beyond that, hybrid models have been proposed to leverage the advantages of each approach. The method proposed by Zhang and Yu [17] seeks to merge **CNN** in the image domain and classical **MAR** in the sinogram domain. In the adopted approach, the initial refinement occurs in the image domain, but the complete method uses the image generated by the **CNN** as a prior for replacement of affected regions in the sinogram. This characterizes a hybrid approach, but still does not employ **CNN** in both domains. More recent models, such as DuDoNet[20], integrate deep networks in both domains, demonstrating high performance and characterizing what we call Dual Domain. However, these methods present greater computational complexity and dependence on access to raw sinogram, which motivates the continuous exploration of robust solutions in the image domain.

To address the scarcity of anatomically paired data with metal artifact/metal artifact-free, self-supervised learning-based approaches have emerged. Yu et al. [21] proposed a method capable of reducing metal artifacts without the need for fully clean reference images, expanding the clinical applicability of these techniques.

Other works explored the use of **Generative Adversarial Network (GAN)** for **MAR**. Huang et al. [22] used multimodal fusion and **GANs** to improve texture preservation and reduce excessive smoothing. Despite good visual results, **GAN**-based approaches can introduce training instability and a higher risk of structural hallucination[23], raising clinical concerns.

3.3 Reproducible Evaluation and Generalization

The evaluation of DL models for CT is typically performed through quantitative metrics such as PSNR, RMSE, and structural similarity-based indices. Traditional error-based metrics, such as RMSE and PSNR, quantify pixel-wise discrepancies but do not adequately capture structural information relevant to visual perception and clinical interpretation. In this context, Wang et al. [24] introduced the SSIM as a perceptual metric that explicitly models luminance, contrast, and structure. Subsequently, the Multi-Scale Structural Similarity Index Measure (MS-SSIM) was proposed to incorporate scale-dependent structural variations, extending SSIM to multiple resolution levels [25]. Horé and Ziou [26] demonstrated that both PSNR and SSIM have limitations depending on the type of distortion considered, suggesting that the choice of metric may depend on the application context. Given that metallic artifacts manifest primarily as structural distortions affecting anatomical boundaries and long-range streak patterns, metrics sensitive to structural preservation are particularly suitable for evaluating the performance of MAR methods. This applies especially to SSIM and MS-SSIM, as they explicitly consider structural components at different spatial scales [24, 25].

Additionally, a critical challenge identified in the literature is models' generalization. Zech et al. [27] demonstrated that models with high performance on internal datasets can fail significantly when applied to external data, highlighting the impact of Domain Shift. Recent reviews reinforce the need for rigorous external and clinical validation to ensure safety, robustness and real applicability [28, 29].

The reviewed literature demonstrates significant advancements in artifact reduction in CT through DL, but also evidences persistent gaps: lack of experimental standardization, limited external validation, and risk of anatomical hallucination.

Chapter 4

Materials and Methods

4.1 Data, curation and traceability

4.1.1 CT-MAR Dataset and Reference Sources

The main dataset used in this work comes from the [CT Metal Artifact Reduction Challenge \(CT-MAR\)](#) [30, 31] promoted by the [American Association of Physicists in Medicine \(AAPM\)](#) and consists of 14,000 training pairs containing [CT](#) images with simulated metal artifacts and their original artifact-free versions. The data was generated through a hybrid simulation framework [32], which incorporates publicly available clinical images and simulated metallic objects with the CatSim simulator, integrated into the open-source XCIST toolkit [33]. The dataset is distributed into two anatomical categories:

- **Head:** 1,626 (11.61%) samples derived from the *UCLH Stroke EIT Data* [34, 35];
- **Body:** 12,374 (88.38%) samples derived from the *NIH DeepLesion Data* [36].

The folders in the *body* category contained trunk [CTs](#), covering axial, thoracic, and abdominal slices. The data is organized into subfolders, each containing up to 1,000 samples. Only the following files were used for training and validation:

1. **Metal-Free images (reference):** Files contained in the *Target* folder of size 512 $\tilde{\text{A}}$ —512 $\tilde{\text{A}}$ —1 pixels;
2. **Images with metal artifacts (corrupted):** Files contained in the *Baseline* folder of size 512 $\tilde{\text{A}}$ —512 $\tilde{\text{A}}$ —1 pixels.

All image files (.raw) in the dataset are presented in binary format without a header, with float32 precision in little-endian order. The images were already converted from linear attenuation coefficients (in mm^{-1} units) to [Unidade Hounsfield \(HU\)](#), using the normalization value of

0.02 mm^{-1} for water. There was not additional information about which exam each slice belonged to, if they were from the same patient, or which specific anatomical regions were represented. The only available information was the organization by folder, indicating the anatomical category (Head or Body). The lack of detailed metadata imposed the need for a careful pre-processing approach to ensure that the data was properly organized and used during model training and validation. Therefore, a significant effort was dedicated to tracking the images to the original datasets (UCLH Stroke EIT Data and NIH DeepLesion Data) to obtain additional information about the exams and patients, allowing for an appropriate partitioning of the data.

4.1.2 Partitioning at patient level

The data partitioning was stratified by patients, ensuring that slices from the same patient were not distributed across training, validation, and test sets. This approach prevents information leakage and ensures a more realistic assessment of the models' generalization capability.

Due to the lack of detailed clinical metadata in the [CT-MAR](#) dataset, an algorithmic screening of the slices to the original datasets (UCLH Stroke EIT and NIH DeepLesion Data) was necessary to enable the association of the provided metadata with the [CT-MAR](#) slices. This process allowed for the identification of which exam and patient from DeepLesion/UCLH Stroke EIT each [CT-MAR](#) image belonged to. This information was subsequently used to perform patient-level partitioning of the data into training, validation, and test sets, reducing the risk of data leakage between experimental sets.

The correct separation of the data is particularly important in medical imaging studies based on [DL](#), since slices belonging to the same exam often present high anatomical similarity. Thus, the simultaneous presence of images from the same patient in training and testing can artificially inflate the model's performance and compromise the assessment of its generalization capability [37].

The Deep Lesion images [36] are stored in 16-bit PNG format, while the [CT-MAR](#) images are provided in .raw format directly in [HU](#). The UCLH Stroke EIT [34] provides the exams in DICOM format. Therefore, regardless of the original format, all images were converted to [HU](#) before comparison, ensuring uniformity across the different datasets.

After the conversion, all images were represented on the same intensity scale to allow direct comparison.

The screening algorithm was based on the fact that [CT](#) images preserve quantitative information in [HU](#), which directly reflects the tissue attenuation properties. Since anatomical structures exhibit high spatial continuity between consecutive slices of the same acquisition,

images corresponding to the same anatomical slice tend to present similar HU distributions and highly consistent global structural patterns [24, 38].

That way, the method was not designed to verify absolute pixel-to-pixel equality between images. Rather, it was designed to determine whether the overall structural organization of the anatomy remains extremely similar across different representations of the same slice. This behavior is particularly important because small local variations in intensity, numerical discretization, and linear transformations associated with storage format conversion tend to preserve the global anatomical structure of the image. This holds even when small local intensity variations occur [24, 38].

Thus, the adopted methodology seeks to identify correspondences between images based on global structural similarity, allowing for a margin of local variation that does not compromise the correct identification of the exam and patient of origin. Initially, each image was reduced to a compact representation through *downsampling*, preserving mainly the relevant global anatomical information. This step significantly reduces the computational cost without compromising the overall structure of the image.

Subsequently, a representative vector (*fingerprint*) was calculated for each image. The fingerprints of the CT-MAR images were then compared with the fingerprints of the DeepLesion and UCLH Stroke EIT images using a normalized dot product-based vector similarity metric (cosine similarity). It was chosen for being invariant to global scale differences between vectors, a relevant property given that the images come from different formats with different intensity normalizations. For each CT-MAR image, the image with the highest similarity score was selected.

Since the majority of the CT-MAR images originated from DeepLesion, the screening process was initially prioritized on that dataset. Subsequently, the same methodology was applied to the DICOM exams from UCLH Stroke EIT to identify the remaining images. In this way, it was possible to implement a unified traceability strategy regardless of the storage format of the original datasets.

The matching was considered valid only when the similarity score exceeded an empirically defined threshold. Cases with very high scores were automatically marked as safe matches, as it was observed that all pairs visually analyzed with a score above 0.993 effectively corresponded to the same image. To maintain an additional safety margin, the match was considered automatically valid only when the score exceeded 0.994.

Scores below this threshold indicated potentially ambiguous matches and were subjected to manual visual inspection. Scores below approximately 0.98 indicated clearly incorrect matches

and were automatically discarded without the need for manual review. Thus, all pairs with scores between 0.98 and 0.994 were visually analyzed to confirm the validity of the match. In order to avoid unnecessary exclusion of pairs, all pairs with a score below 0.98 for both datasets (DeepLesion and UCLH Stroke EIT) were subjected to visual inspection to ensure they were not erroneously discarded.

For the 530 pairs that remained without approved matches after direct search in both source datasets, a complementary indirect matching step was applied. In this step, each unmatched CT-MAR image was compared with already traced CT-MAR images. The most similar neighbors were used to select candidate exams from DeepLesion and then all *slices* of those exams were searched with 64×64 dimension *fingerprints*. The 530 proposals were visually inspected: 95 matches were approved and incorporated into the consolidated manifest, while 435 remained without a safe match and were excluded.

Moreover, the algorithm was designed to automatically ignore images that had already been verified as correct matches, reducing redundancies and accelerating future reprocessing. Given the high number of images present in the datasets, the processing was parallelized and optimized through the use of vectorized matrix operations. When available, the algorithm utilizes GPU acceleration to reduce execution time. The calculated fingerprints were also cached to avoid unnecessary recomputation in subsequent runs.

Although the compact representation based exclusively on the mean of block intensities HU is efficient, it can theoretically generate false positives when different patients' exams are acquired with identical acquisition protocols or present very similar anatomical structures. This occurs because these conditions tend to produce similar intensity distributions and, consequently, fingerprints with high scores. To mitigate this risk, a conservative automatic acceptance threshold (0.994) was adopted and a systematic visual validation of all pairs with scores between 0.98 and 0.994 was performed. Additionally, as a complementary safeguard, all automatic matches were analyzed in terms of MS-SSIM, where the pairs with the most critical scores were subjected to an additional visual review to ensure the reliability of the matches.

Despite that, this limitation does not compromise the main objective of the methodology, even though consecutive slices from the same exam often present high spatial redundancy and can therefore obtain very similar similarity scores. The purpose of the algorithm was not to perform rigorous voxel-to-voxel alignment, but rather to correctly identify the exam and patient of origin of each CT-MAR image. In this way, even potential ambiguities between adjacent slices remain restricted to the same anatomical volume and are unlikely to introduce a relevant risk of data leakage between training, validation, and test sets.

The literature demonstrates that the quality of the pairs used in supervised studies directly influences the performance and generalization capability of [MAR](#) models [9, 20]. In addition, structural metrics of image quality, such as [SSIM](#), have been widely used in medical imaging due to their higher sensitivity to structural preservation [24]. Although the implemented method does not directly use [SSIM](#), the adopted strategy seeks to precisely preserve the global structural information of the image during the matching process.

Moreover, the data utilization without any screening treatment would condition the entire project to an inherent data leakage from the dataset, compromising the evaluation of the models' performance and the validity of the results. Thus, even if the screening method is not perfect, it represents a significant improvement over the total absence of screening, ensuring a more reliable basis for the retained cohort. This reliability claim is limited to leakage control among recovered identities: the source publication reports approximately 1,230 DeepLesion patients and 10 Stroke EIT patients [30], whereas image-based tracing recovered 258 and 11 identifiers, respectively ([subsection 5.1.2](#)). This discrepancy does not show that the remaining source patients are absent, but it does show that the screening is not an exhaustive patient-identification procedure. It strengthens provenance for retained cases without establishing complete recovery of the source cohort.

Taking this into account, this step constituted an important quality control procedure for the experimental dataset and was fundamental to ensure an appropriate partitioning of the data used in this work.

4.1.3 Anatomical Distribution and Stratification

Stratification was implemented within each subset (train/val/test) to maintain the anatomical proportion at the patient level, ensuring that the relative proportion between categories is preserved across all subsets. In this way, the anatomical balance is preserved, ensuring that the models are trained and evaluated on an appropriate representation of the anatomical diversity present in the original dataset.

The patient-level partition was defined only once, with a fixed random seed, maintaining approximately 70% of the patients for training, 15% for validation, and 15% for testing. This fixed partition ensured independent subsets for validation and testing and was kept unchanged across all experimental repetitions.

The resulting anatomical and split distributions are presented in [subsection 5.1.5](#).

4.2 Pre-processing

4.2.1 Integrity Verification

All pairs were validated through an integrity report, which checks for the existence of both sides of the pair (artifact image and clean image). There were no pairs with missing data, ensuring that the dataset was complete and ready for subsequent pre-processing and model training steps.

4.2.2 Background Removal through Body Mask

The CT images often contain large background areas that do not contribute to the task of metal artifact removal. That creates a class imbalance in the training data, where irrelevant background voxels dominate over the anatomical regions of interest. This imbalance, in turn, hinders the learning and deviates the model focus from clinically relevant structures [39, 40]. To address this challenge, a pre-processing step also based on DL was implemented. A U-Net was trained to segment the body tissue from the background, generating binary masks that were applied to the original images to isolate the regions of interest. This model was trained using $\sim 1,500$ samples from the CT-MAR, where the segmentation masks were automatically generated through a thresholding process. Additionally, a fraction of the generated masks was subjected to iterative manual refinement to improve data quality and was used to retrain the model. This process was repeated until masks of sufficient quality for the intended purpose were obtained. This step was essential to reduce the influence of background artifacts and improve the quality of features extraction during training.

4.2.3 Intensity Window and Normalization

The CT-MAR source image intensities were expressed in HU. A common intensity window of $[-1000, 400]$ HU was applied to all images, followed by linear normalization to $[0, 1]$. Each clean/artifact image pair was then independently resized using bicubic interpolation and stored as a single-channel two-dimensional tensor of 256×256 pixels.

4.2.4 Artifact Mask for Auxiliary Supervision

Binary difference masks of the regions affected by artifacts were produced and used as an additional supervision signal in the dual architecture. These masks are exclusively used during

the training and are not required during inference. The binary difference mask is formally defined as:

$$M_{diff}(i, j) = \begin{cases} 1; & \text{if } |I_{artifact}(i, j) - I_{clean}(i, j)| > \tau \\ 0; & \text{otherwise.} \end{cases} \quad (4.1)$$

where $I_{artifact}$ represents the image with artifacts, I_{clean} the clean reference image, and τ is the normalized difference threshold. The value of this threshold was empirically defined to capture regions with perceptible differences while maintaining robustness to low-amplitude noise.

4.3 Compared Architectures

4.3.1 U-Net Single Supervision

The first model consist of an architecture based on [U-Net \[18\]](#), composed of encoder-decoder with skip connections. Li et al. [\[41\]](#) apply blocks of [Batch Normalization \(BN\)](#) and [Rectified Linear Unit \(ReLU\)](#) activation functions after each dynamic convolution. This work does something similar, but chooses to use standard convolutions with 3×3 kernel, followed by [BN](#) blocks and [ReLU](#) activation functions. The choice of standard convolutions is justified by simplicity and computational efficiency, in addition to maintaining the approach used by Ronneberger et al. [\[18\]](#). Like Li et al. [\[41\]](#), the decoder also uses bilinear upsampling. This architecture has only one output.

4.3.2 U-Net Dual Supervision

A variation of the model described above was conceived, maintaining the shared *encoder* and *decoder* and adding an independent output in the last layer. Thus, there are two distinct outputs in the decoder, each with its own loss function. The first output is responsible for reconstructing the corrected image, while the second output is dedicated to predicting the artifact mask.

The rationale for adding the auxiliary supervision of the artifact mask is based on a simple premise. The presence of an additional loss focused on segmenting regions affected by artifacts can provide complementary information to the model, aiding in learning more robust and discriminative representations. This approach is inspired by previous works that have shown that multi-task supervision can improve performance in related tasks [\[42, 43\]](#).

The two architectures were evaluated against each other to determine the impact of the additional supervision with artifact mask on reconstruction quality and artifact reduction capability. All model bases are identical, with the presence of the artifact mask prediction output being the

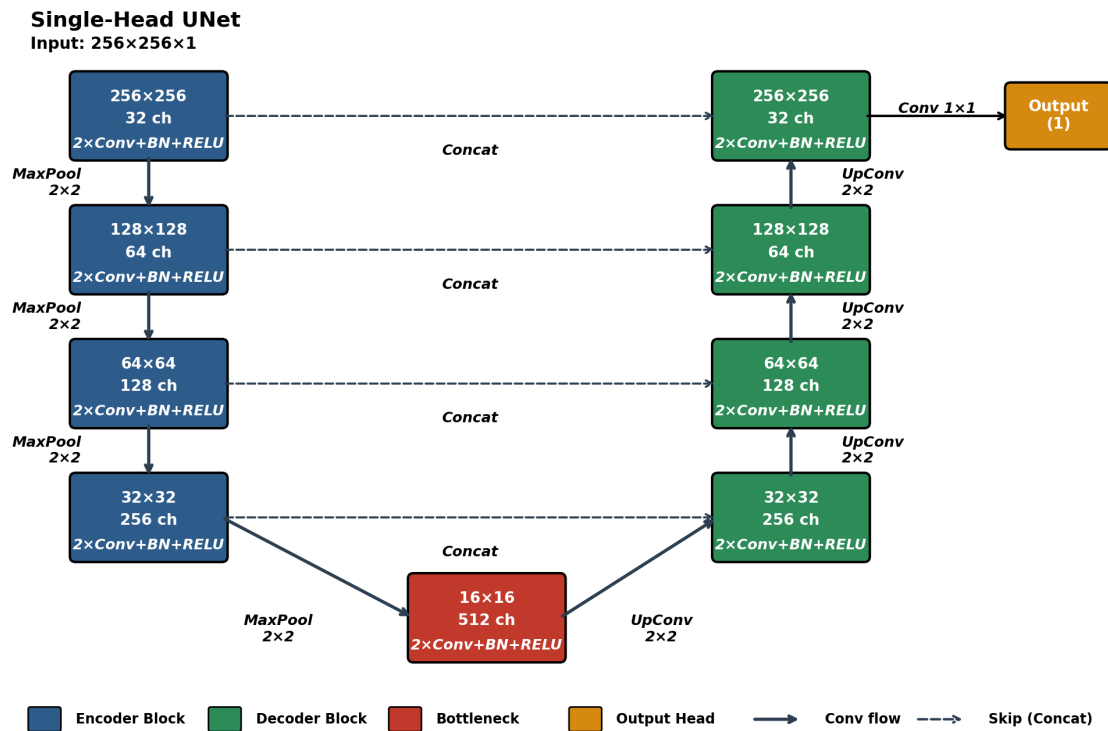


Fig.4.1 Single supervision architecture diagram.

only difference in the second version. Thus, the comparison between the two architectures is fair and isolated, allowing for an exclusive evaluation of the impact of the additional supervision of the artifact mask.

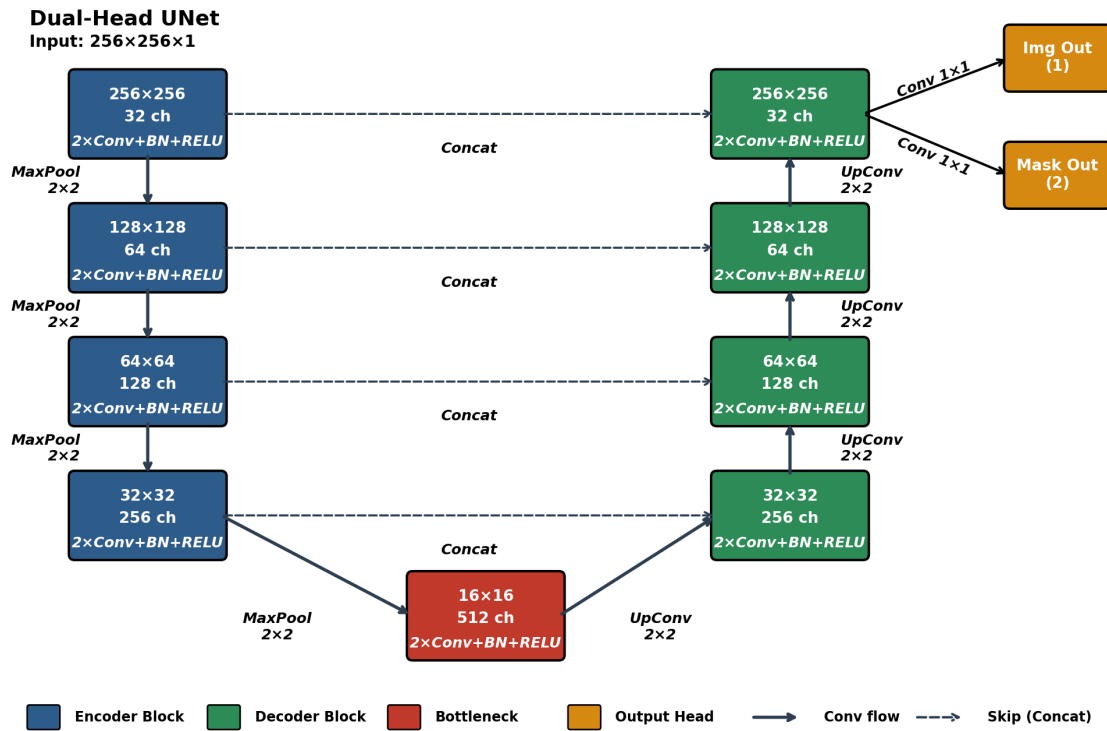


Fig.4.2 Dual supervision architecture diagram.

4.4 Reproducible Training

4.4.1 Settings, determinism and two-stage selection

Both models were trained under the same settings, with *float32* precision. To assess stability against different initializations and batch presentation orders, each architecture was trained with deterministic operations and with seeds 1337, 42, and 2024, always on the same patient partition. **Automatic Mixed Precision (AMP)** was disabled and batches were kept complete, including the last batch of each epoch, to avoid samples being excluded from training, validation, or evaluation.

The AdamW optimizer was used, with an initial learning rate of 2.26×10^{-4} , weight decay of 1.00×10^{-4} and gradient clipping with norm 1.00, identical for both architectures. The learning rate was adjusted by the ReduceLROnPlateau scheduler, with a reduction factor of 0.50, patience of two epochs, cooldown of one epoch and minimum value of 1.00×10^{-6} , monitoring the reconstructed image **SSIM** on the validation set in both models. The training had a maximum limit of 300 epochs and early stopping could only be triggered after at least five epochs and occurred when the monitored metric did not improve by at least 1.00×10^{-4} for 15 consecutive

epochs. In each run, the checkpoint corresponding to the best value of the monitored validation metric was retained.

Only the best checkpoint per run was preserved, saved in `.keras` format with a filename encoding the architecture, epoch, and monitored metric. The training CSV explicitly flags the corresponding row with `best_checkpoint=1`. Checkpoint selection relied exclusively on the validation metric chosen in advance: `val_ssim_hu` for the Single Model and `val_img_out_ssim_hu` for the reconstruction output of the Dual Model.

Each of the six validation-selected checkpoints was evaluated on the fixed internal test partition. The candidate advanced to external evaluation was selected from this frozen set by the highest mean patient-level $\Delta SSIM$ in the Artifact Zone. As a preservation safeguard, the corresponding changes in the Clean Zone were required to be substantially smaller than the correction observed in the Artifact Zone. No specific numeric limit was set in advance for how small this Clean-Zone change had to be, so this comparison is only descriptive and is not, by itself, a formal safety check. Global PSNR, SSIM, and MAE, together with the remaining region-of-interest metrics, were complementary and did not override this selection rule. Thus, the test partition was used only to compare the six already frozen candidates. It was not used to choose a new checkpoint, tune hyperparameters, alter weights, or trigger further training. Once selected, the architecture, seed, weights, and hyperparameters of the external-evaluation candidate were frozen.

Run traceability was ensured through reproducibility snapshots in JSON format, preserved in `outputs/reproducibility/`. Six final internal evaluations were frozen, corresponding to the three random initializations (seeds 42, 1337, and 2024) of the Single Model and Dual Model. All snapshots were generated from commit `a950be7276a5585de45e8ae874c59c065a1985be`, on branch `projeto_final`, with a clean working tree (`is_dirty: false`).

Each snapshot records the operating system, Python version, and relevant dependency versions. The execution environment used Python 3.11.9 on Windows 10, including TensorFlow 2.15.0, NumPy 1.26.4, Pandas 2.2.2, scikit-image 0.21.0, and Matplotlib 3.7.5. In addition, SHA-256 hashes were computed for the configuration files, the canonical pair manifest, the patient-level partition definition, the tensor generation metadata, and the environment requirements file. Thus, every value, table, or figure reported can be unambiguously traced to the code, environment, input data, checkpoint, and results that produced it.

During the training, the following quantities were monitored:

- **SSIM, PSNR, and MAE in HU scale:** calculated in the body region to track the structural and radiometric fidelity of the reconstructions;

- **Reconstruction loss:** to track the optimization of the corrected image;
- **Segmentation loss:** only in the dual model, to track the auxiliary supervision of the artifact mask.

4.4.2 Loss Functions

- **Reconstruction:** Despite L2 being widely used for image restoration, Zhao et al. [44] highlight the best results studying different loss functions for this task and show that it consistently underperforms perceptually-motivated alternatives. The authors studied loss functions based on L1, SSIM and MS-SSIM, with the second achieving equal or slightly inferior results to L1, while the third achieves better performance than SSIM but still does not clearly outperform L1. The paper also tests a mix of L1 and MS-SSIM, and obtains superior results in all metrics evaluated by the authors. This combination is justified by the complementarity between the two functions, where L1 favors local fidelity and MS-SSIM promotes structural preservation, resulting in a more balanced correction of artifacts. Although the best result reported by the authors involved L1 and MS-SSIM, in this work the weighted sum of L1 and SSIM was adopted. This choice maintains the general principle of combining a pixel fidelity component with a structural component, without reproducing the specific formulation or weights used by Zhao et al.

The weights used for the experimental protocol were 0.7 for L1 and 0.3 for SSIM. The relation adopted for the combination of the loss functions is as follows:

$$\mathcal{L}_{Recon} = 0.7 \cdot \mathcal{L}_{L1} + 0.3 \cdot (1 - \text{SSIM})$$

Zhao et al. [44] applied a Gaussian spatial weighting to the L1 component in the hybrid loss, in order to make it compatible with the spatial distribution of the gradients of the MS-SSIM component. In this work, the L1 component was calculated directly as the mean absolute error in the body region, without additional Gaussian weighting. Since the normalization used is linear with respect to HU values, this component maintains a penalty directly proportional to the radiometric error, while SSIM contributes to local structural preservation.

This combination of loss functions is adopted for the Single Model and also for the reconstruction output of the Dual Model. In this way, it is ensured that the comparison between the two versions is fair, isolating the effect of the additional supervision of the mask.

- **Mask auxiliary supervision:** The artifact-mask prediction output, present in the Dual Model, was trained with the sum of [Dice Similarity Coefficient \(DICE\)](#) loss and [Binary Cross-Entropy \(BCE\)](#). This combination provides simultaneous supervision for global overlap between masks and pixel-wise classification, being suitable for segmentation problems with class imbalance [39]. The relation adopted for the combination of the loss functions is as follows:

$$\mathcal{L}_{Seg} = \mathcal{L}_{Dice} + \mathcal{L}_{BCE}$$

The total loss function for the Dual Model is composed of the weighted sum of the reconstruction and segmentation losses, and is defined as:

$$\mathcal{L} = 1.0 \cdot \mathcal{L}_{Recon} + 0.045 \cdot \mathcal{L}_{Seg}$$

4.5 Internal Evaluation

In the inference phase, the models were evaluated using quantitative metrics widely adopted in the literature, in comparison with the Ground Truth. The metrics used to evaluate reconstructed image quality were:

- **PSNR** ;
- **SSIM** ;
- **MAE**.

Recent studies have shown that [MS-SSIM](#) and [SSIM](#) capture image fidelity better than other image quality metrics [45]. Especially for medical images, where the fidelity of predicted images is crucial, the choice of metrics that reflect human perception of quality and structural preservation is fundamental. Therefore, [SSIM](#) was selected as the primary metric for quantitative evaluation of the quality of reconstructed images, ensuring a robust and relevant analysis for the clinical context. And [PSNR](#) was used as a complementary metric to assess the radiometric fidelity of reconstructions, especially important in computed tomography where the accuracy of [HU](#) values is essential for diagnosis and clinical planning.

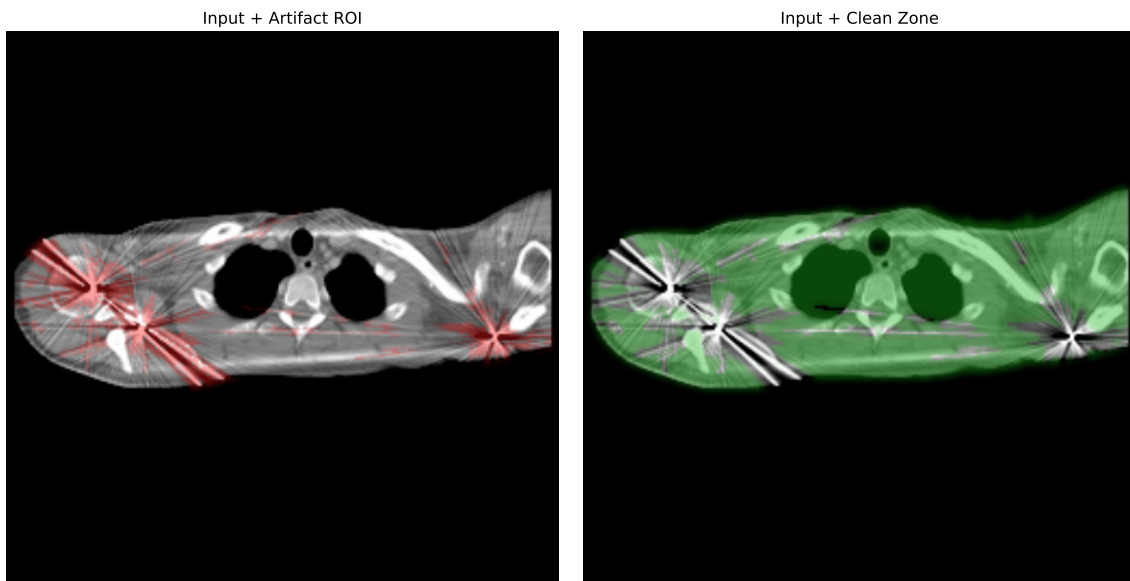
The internal validation was conducted on the validation and test subsets, evaluating consistency, stability, and intra-dataset generalization capability.

4.5.1 ROI Metrics

Because the global evaluation does not distinguish the model's behavior in regions directly affected by artifacts from that in body regions distant from the implant, each slice was divided into two complementary zones. These zones are based on how much the artifact-corrupted input differs from the clean reference. Metal artifacts do not only cause strong local streaks near the implant. They are typically accompanied by a faint, almost uniform intensity shift across the whole slice, a side effect of beam hardening. To avoid mistaking this global shift for a local artifact, the pixel-wise difference between the clean reference and the input was first computed. Its typical (median) value over the body region was then estimated and subtracted out, so that only deviations beyond this baseline shift were considered.

The Artifact Zone was then defined as the set of body pixels whose bias-corrected difference exceeds a fixed threshold, equivalent to about 42 HU on the normalized intensity scale used ($\tau = 0.03$). Connected-component filtering (using SciPy's image-labeling utilities) was applied to discard components smaller than 20 pixels, which correspond to isolated CT quantum-noise pixels rather than genuine artifact streaks. No additional morphological dilation was applied, preserving the observed geometry of the *streaks*. The Clean Zone was defined simply as the remaining body region outside the Artifact Zone. This zone quantifies changes outside the directly affected region.

The [Figure 4.5.1](#) illustrates the application of the two zones in representative slices of the dataset, covering different artifact patterns.



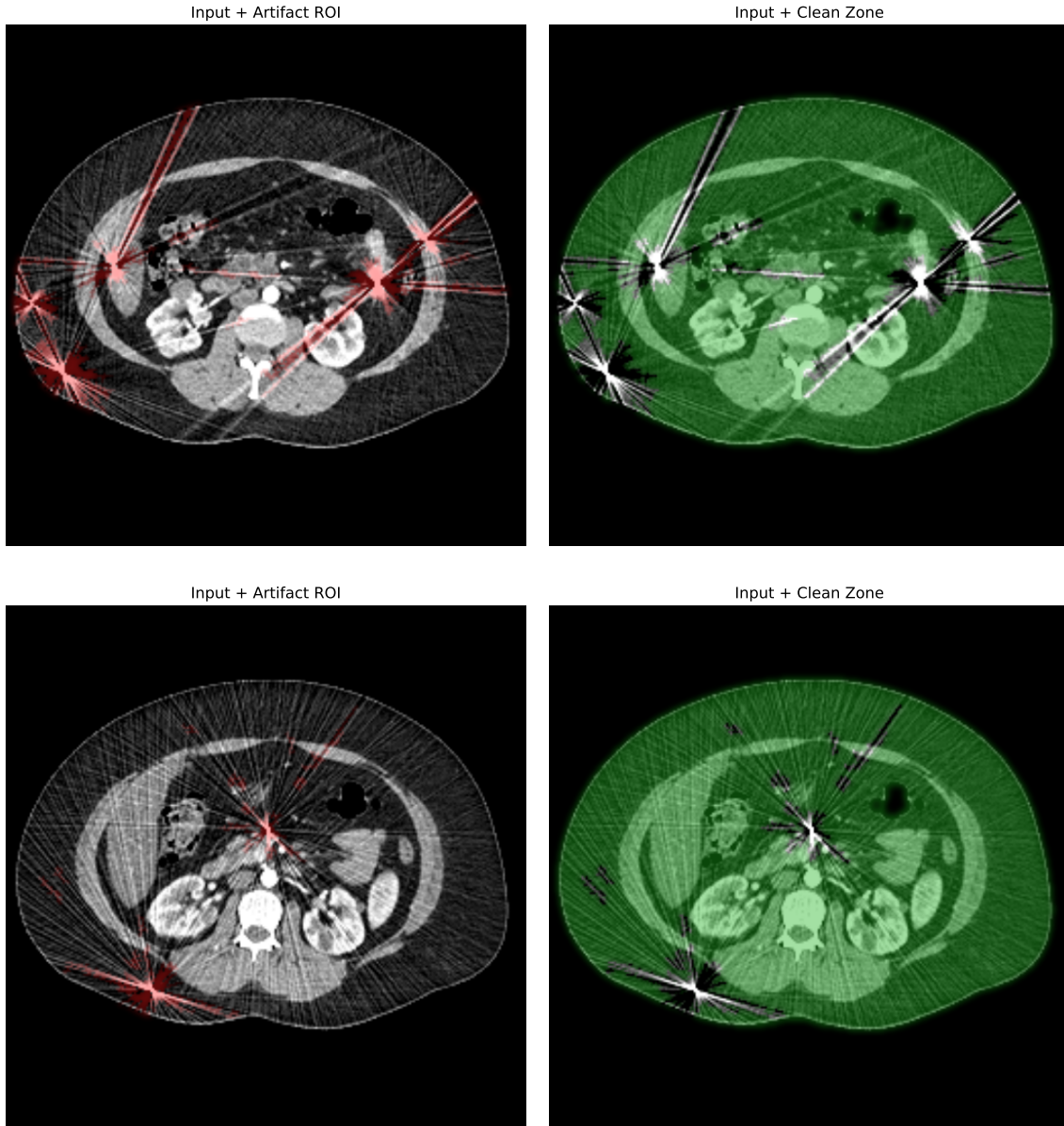


Fig.4.3 Artifact Zone ROI (left) and Clean Zone ROI (right) examples.

In every evaluation context, the metrics were expressed as variations relative to the performance of the corrupted input. Global metrics were also calculated, as a complementary resource. Then, let Ω be the evaluation domain, which can correspond to the global region defined in the next subsection (Ω_{Global}), the Artifact Zone ($\mathcal{R}_{\text{Art}}$), or the Clean Zone ($\mathcal{R}_{\text{Clean}}$), and let $\hat{I}$ be the reconstruction produced by the model. The variations were defined by:

$$\Delta\text{PSNR}_{\Omega} = \text{PSNR}(\hat{I} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega}) - \text{PSNR}(I_{\text{in}} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega})$$

$$\Delta\text{SSIM}_{\Omega} = \text{SSIM}(\hat{I} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega}) - \text{SSIM}(I_{\text{in}} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega})$$

$$\Delta\text{MAE}_\Omega = \text{MAE}(I_{\text{in}} |_\Omega, I_{\text{ref}} |_\Omega) - \text{MAE}(\hat{I} |_\Omega, I_{\text{ref}} |_\Omega).$$

Thus, $\Delta\text{PSNR} > 0$ and $\Delta\text{SSIM} > 0$ indicate improvement over the corrupted input. Since lower MAE represents better fidelity, the order of subtraction was reversed for this metric: $\Delta\text{MAE} > 0$ indicates error reduction and, therefore, improvement. $\Delta\text{MAE} < 0$ indicates error increase. Regular SSIM was used, calculated with a local window of 11×11 pixels and cropping with context around the evaluated region.

Metric	Train	Global	Artifact Zone	Clean Zone
PSNR	✓	✓	✓	✓
SSIM	✓	✓	✓	✓
MAE	✓	✓	✓	✓

Table 4.1 Evaluation metrics by context.

4.5.2 Global Metrics

After denormalization, the global evaluation region is cropped to the smallest bounding box of the body mask before SSIM evaluation, eliminating scan windows composed exclusively of background “” which would artificially score $\text{SSIM} = 1.0$ and inflate the global metric.

As an additional, purely descriptive check for any overall shift in average HU values, a Bland–Altman plot was made comparing the reconstruction with the clean reference. For each patient, the mean HU of each image was calculated over all valid pixels of the body mask of the respective slices, weighted by the number of pixels in each slice. The graph plots the mean difference per patient (Model – GT) against the average of the two measurements, together with the average difference (bias) and the range within which most differences fall ($\bar{d} \pm 1.96 SD_d$, the limits of agreement). This plot is descriptive only, one point per patient, and does not replace the accuracy metrics above or the region-based analysis above.

4.5.3 Statistical Analysis

Every statistical comparison in this work used the patient, not the slice, as the unit of analysis. A patient with more slices should not count more than a patient with fewer slices, so before any comparison, the per-slice metrics were averaged within each patient, giving one paired value per patient.

To evaluate the improvement over the corrupted input, for each patient the output’s value was compared with the corrupted input’s value, for both PSNR and SSIM, so that a positive

difference always represents improvement. For [MAE](#), whose reduction represents better performance, the subtraction was reversed so that a positive difference also represents improvement. In every test, the question asked was whether there was typically no difference at all between the paired values. The direct comparison between the two architectures followed the same logic, subtracting the Single Model's value from the Dual Model's value for each patient, so that negative differences favor the Single Model.

The p -values presented in the comparison tables were adjusted using the Holm sequential procedure, which controls the probability of at least one false positive across a family of related tests without assuming that the tests are independent: the smallest raw p -values are penalized the most, while the relative ordering between tests is preserved. This correction was computed in Python with the `multipletests` function from the `statsmodels` library, using the 'holm' method.

In the reconstruction improvement table, the correction was applied jointly to the four corresponding tests for [PSNR](#) and [SSIM](#) for each of the two architectures. In the paired comparison Single Model vs Dual Model, the correction was applied jointly to the two tests, one for [PSNR](#) and another for [SSIM](#).

The confidence interval of 95% reported for the mean differences was obtained by patient-level percentile *bootstrap*. For each metric, the vector of $n = 41$ paired differences per patient was resampled with replacement, maintaining n observations in each resample. The mean difference was then calculated in each of the 10,000 resamples. The lower and upper limits of the interval correspond, respectively, to the 2.5 and 97.5 percentiles of the empirical distribution of these means. This procedure does not assume normality of the differences and preserves the patient, rather than the slice, as the unit of inference. Seed 1337 was used to make the resampling deterministic.

4.6 External Evaluation

The external evaluation was realized in independent data provided by the [The Cancer Imaging Archive \(TCIA\)](#) [46], allowing to assess the impact of domain shift and the robustness of the model in scenarios not seen during training.

4.6.1 Design and Cases

The study involves doctors comparing the images produced by the model with the original images containing artifacts. Every doctor looks at the same set of cases, one at a time, and for

each one says which of the two images they prefer on a few simple questions. The evaluation was designed to be blind, with the doctors not knowing which image is the original and which is the model output. A set of 15 samples from diverse TCIA datasets was used, with 2 additional hidden samples repeated for consistency check and 2 training samples excluded from the analysis. The 2 hidden repeats were shown again later in the same session, at least five cases after they first appeared, without telling the reader that it was a repeat.

A web application was developed to present the images in a randomized and counterbalanced manner, allowing the doctors to jointly zoom in on the pair for better visualization. It also enabled two anatomical windowing options (soft tissue and lung). Whether the original or the model output appeared on the left or the right was shuffled between doctors, but kept the same when a case was repeated, so that this choice could not by itself explain any preference.

The link to the application was sent to the physicians invited by the researcher, rather than through an open call, with a target of seven participants and at least five required for the results to be useful. To take part, a doctor only had to confirm that they use CT in their clinical practice at least once a month. No name, contact details, professional license number, or other proof of qualification was collected. The only information kept about each doctor is their country, medical specialty, how often they use CT, years of experience, previous experience with artifact reduction, and monitor type, none of which can identify a specific person.

The cases used in this evaluation were chosen from a larger pool of 48 TCIA exams, mostly from the chest (31 cases), with a smaller number from the abdomen (6), the pelvis (1), and other or mixed regions (10). Each case corresponds to a single axial CT slice. The 17 cases used (15 main and 2 training) were drawn from five TCIA collections: TCGA-LUAD [47], LIDC-IDRI [48], VAREPOP-APOLLO [49], CPTAC-SAR [50], and CPTAC-PDA [51].

In accordance with the data-usage policy of each collection, this work acknowledges the following. The results shown here are in whole or in part based upon data generated by the TCGA Research Network (<http://cancergenome.nih.gov/>), used in the TCGA-LUAD cases. It also acknowledges the National Cancer Institute and the Foundation for the National Institutes of Health, for their critical role in the creation of the free publicly available LIDC-IDRI Database, used in the LIDC-IDRI cases. It further acknowledges the Veterans Health Administration's Research for Precision Oncology Program and the Applied Proteogenomics Organizational Learning and Outcomes (APOLLO) Research Network, whose data were used in the VAREPOP-APOLLO cases. Finally, it acknowledges the National Cancer Institute Clinical Proteomic Tumor Analysis Consortium (CPTAC), whose data were used in the CPTAC-SAR and CPTAC-PDA cases.

4.6.2 Quantitative Characterization of Input–Output

As a complementary characterization of the alteration introduced by the model in external cases, **SSIM**, **PSNR**, and **MAE** were calculated directly between the input image and the respective output of the model selected for external evaluation. The metrics were evaluated only in the 15 main cases of the frozen external set. The two training cases were excluded. The calculation domain was the effective body mask, manually corrected during case preparation. For **PSNR**, the intensity window $[-1000, 400]$ HU with amplitude 1400 HU was used, identical to the intensity window applied to the internal training and evaluation data (subsection 4.2.3), so that no additional normalization discrepancy is introduced when comparing internal and external results.

There is no clean reference for these real cases. Thus, higher **SSIM** and **PSNR**, as well as lower **MAE**, describe less global alteration between input and output, and not greater accuracy of artifact removal or improvement compared to a reference. The statistics were summarized per case, such as mean and standard deviation, and the 95% confidence intervals for the mean were obtained by percentile *bootstrap* with 10,000 resamplings and seed 1337. The interpretation of the clinical acceptability of these alterations remained reserved for blind evaluation by readers.

4.6.3 Presentation and Evaluation Instrument

For each case, the two images, the artifact input and the model output, are shown side by side at the same time, labelled only “Image A” and “Image B”, with nothing on screen indicating which is which. After looking at a case, the doctor answers four short questions, always with the same three possible answers: “Image A”, “Image B”, or “No difference”. In turn, the four questions ask the following. First, which image has less visible metal artifact. Second, which image better shows the anatomy hidden behind the artifact, the main question of the study. Third, which image has fewer artifacts or distortions in the rest of the picture, away from the artifact. Fourth, which image the doctor would prefer to use for a clinical read. The third question exists specifically to catch a possible unwanted effect of the model: if doctors tend to prefer the artifact input in regions away from the implant, that would suggest the model is introducing changes where none were needed. At the very end of the session, each doctor answers one extra question, on a five-point scale from “Definitely not” to “Definitely yes”, about whether a processed version like this could be a useful addition to a normal **CT** exam. The doctor may also optionally leave a short written comment.

4.6.4 Outcomes and Analysis

The main question of the study, the primary outcome, is which image better shows the anatomy under the metal artifact, taken from the corresponding question described above. Each displayed A/B answer was recoded to input, output, or no difference using the saved left/right allocation. To account for the fact that every reader judged every case, response proportions and the primary output–input contrast were accompanied by 95% percentile confidence intervals from a two-way non-parametric cluster bootstrap, resampling readers and cases independently (10,000 resamples; seed 2024). The other three per-case questions, together with the final usefulness question, are treated as secondary outcomes and are summarized using the same descriptive approach. No multiplicity correction was applied.

Because two cases are shown a second time to each doctor later in the same session, without saying so, it is possible to check how consistent each doctor is with themselves. As these are nominal three-category responses and only two repeat pairs are available per reader, consistency is reported descriptively as exact agreement between the original and repeated answer for each question. Given the small number of doctors expected to take part, the confidence intervals and repeat agreement are interpreted descriptively rather than as definitive clinical evidence.

4.6.5 Operational Status

Before any real doctor took part, the application was first tested with a technical pilot using placeholder data, to check that the questions, the images, and the randomization all worked as intended. Moving from this rehearsal stage to inviting real doctors was only allowed once a documented checklist was fully satisfied, covering an approved and tested hosting setup, a defined data retention and deletion plan, an approved participant information text, and a working, secure deployment of the application.

Chapter 5

Results

5.1 Final Group and Experimental Integrity

5.1.1 Final Group and Exclusion of Unmatched Pairs

It is important to clarify the final results of the tracing and matching process. First of all, not all slices could be traced for the matching algorithm, but most of the data was successfully associated with the original exams. At the end of the direct and indirect matching steps, and before the complementary analysis with [MS-SSIM](#), 11,942 of the 14,000 pairs provided by the [CT-MAR](#) Dataset were traced to DeepLesion (including the 95 recovered by indirect matching), and 1,623 were traced to UCLH Stroke EIT. This totaled 13,565 pairs with associated metadata [5.1](#).

Dataset	Traced	Matched	Unmatched
DeepLesion	12,374	11,942	432
UCLH Stroke	1,626	1,623	3
Total	14,000	13,565	435

Table 5.1 Screening results before the complementary [MS-SSIM](#) analysis.

5.1.2 Reported and Recovered Patient Counts

The challenge paper reports that 12,227 body images were derived from approximately 1,230 DeepLesion patients and that 1,773 head images were derived from 10 UCLH Stroke EIT patients [\[30\]](#). The official downloaded package used in this study instead labels its 14,000 cases as 12,374 body and 1,626 head. Its accompanying `README_training_data.txt` identifies these labels as the archive-folder organisation [\[31\]](#). The difference is exactly 147 cases in

each stratum while preserving the total of 14,000. It is therefore compatible with some head-anatomy images having been packaged under `body`. This possibility is recorded as a dataset-level labelling discrepancy and is not treated as a reclassification made by this study.

The image-based traceability procedure recovered 258 distinct DeepLesion identifiers among the 11,939 retained `body` pairs and 11 Stroke EIT identifiers among the 1,623 retained `head` pairs, for 269 recovered source-patient identities. These are recovered identities, rather than an estimate of the complete source cohort. In particular, the discrepancy from the approximately 1,230 DeepLesion patients cited by the authors does not demonstrate that the remaining patients are absent: unmatched slices, conservative rejection of ambiguous candidates, source-image transformations, and the documented discrepancy between archive labels and the publication can all limit recovery. The published patient count was never used to accept a candidate or to change the patient-level partition.

In the evaluation step through [MS-SSIM](#), among all pairs with a score below 0.90, only 3, previously associated with Deep Lesion, were rejected in the visual inspection [5.2](#).

Dataset	Traced	Matched	Unmatched
DeepLesion	12,374	11,939	435
UCLH Stroke	1,626	1,623	3
Total	14,000	13,562	438

Table 5.2 Pair matching results after the complementary MS-SSIM analysis.

Due to the fact that a significant amount of the data could be traced with high reliability, the remaining cases that could not be safely grouped were excluded from the training and validation process. In total, 438 pairs were removed for not having sufficient information to ensure safe grouping at the patient level. Among them were the 3 pairs rejected for being false positives in the evaluation step via [MS-SSIM](#). All pairs removed from this study were visually inspected to confirm the absence of correspondence, ensuring that the exclusion was justified.

	Excluded	
	Before	After
Total	435	438

Table 5.3 Unmatched pairs data that was excluded before and after the complementary MS-SSIM analysis.

Thus, out of the 14,000 pairs originally provided by the [CT-MAR](#) Dataset, 13,562 were included in this work, while 438 were excluded to preserve the integrity of the experimental partitioning and minimize the risk of data leakage.

5.1.3 Automatic Approval and Manual Review Outcomes

The final curation outcome of every pair in the [CT-MAR](#) dataset was recovered from the canonical manifest. Retained pairs were classified as manually reviewed when either source-level decision was `manual_approved`. All remaining retained pairs were classified as automatically retained. Pairs not included in the final group were classified as excluded after verification, matching the 438 pairs already identified above.

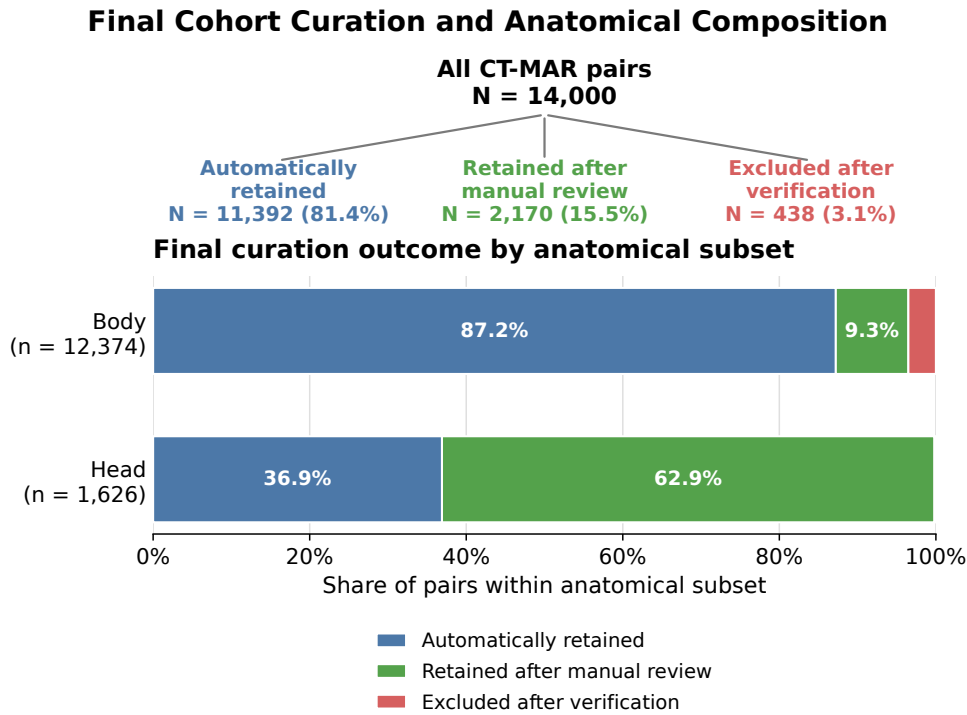


Fig.5.1 Final curation outcomes for all 14,000 [CT-MAR](#) pairs. The upper panel reports the global distribution of outcomes. The lower panel reports the percentage within each anatomical subset. The manifest labels distinguish *Body* and *Head* subsets, rather than the precise anatomical location of a metallic implant.

5.1.4 Complementary Analysis with [MS-SSIM](#)

Before advancing to the analysis of the results of the artifact reduction models, another evaluation of the [CT-MAR](#) x DeepLesion/Stroke EIT results was conducted to verify the quality of the automatic matches. This step is crucial to ensure an extra layer of confidence in the results obtained by the structural similarity comparison algorithm. Every pair that had its match automatically approved due to its score being above 0.994 was reviewed by a new algorithm that calculates the [MS-SSIM](#) between the corresponding images. A histogram of the [MS-SSIM](#) distribution was generated, and pairs with scores below 0.9 were manually reviewed and either approved or rejected based on human evaluation.

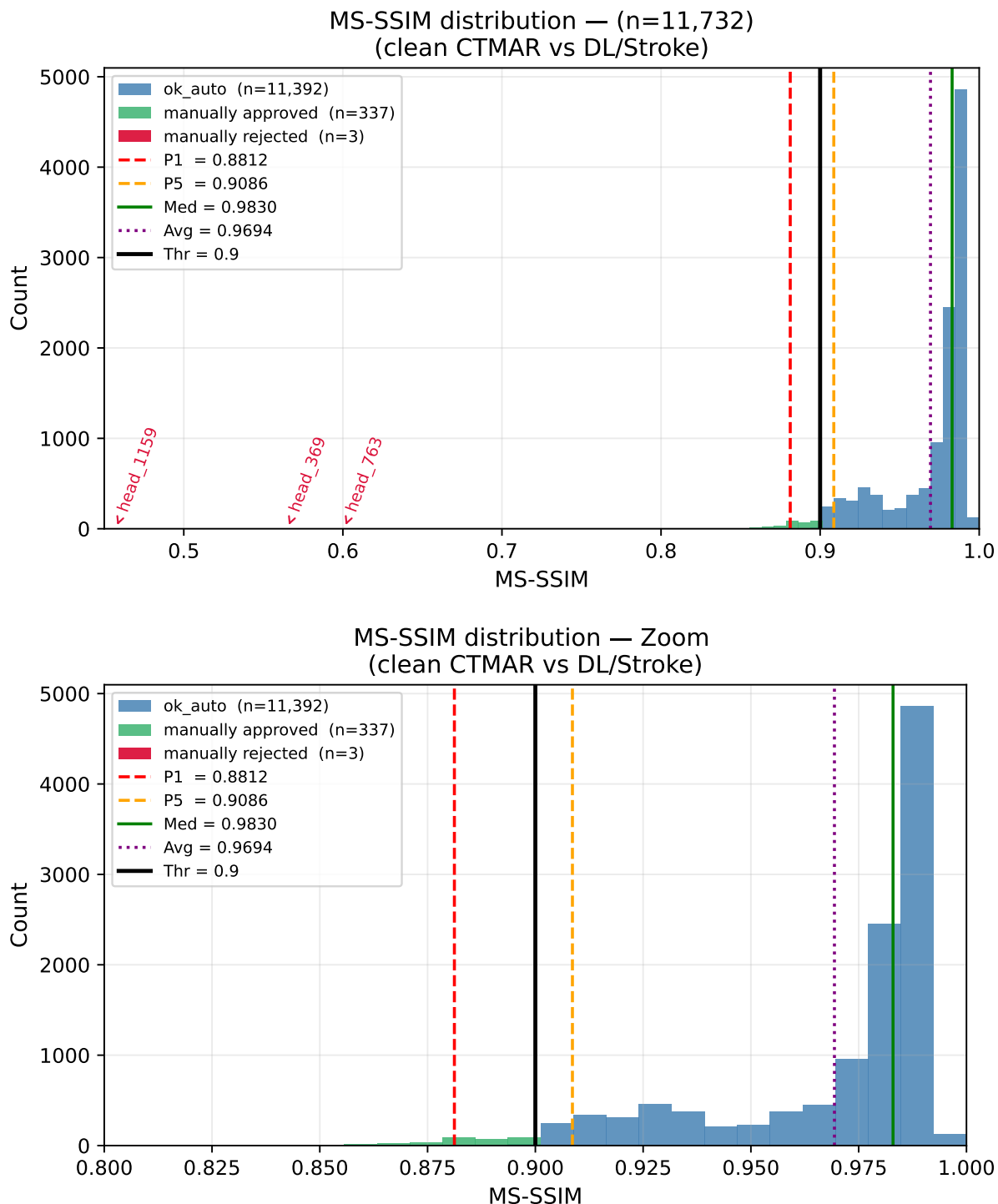


Fig.5.2 MS-SSIM distribution between corresponding pairs (CT-MAR vs DL/Stroke, automatically approved). Above: histogram with percentiles P1 and P5, median, mean, and threshold of 0.9. Below: zoom of the histogram for the region of greatest interest, highlighting pairs with MS-SSIM below 0.9, which were subject to manual review.

The distribution analysis between the corresponding pairs reveals that most pairs exhibit high structural similarity, with a mean value of 0.9694 and a median of 0.9830. This indicates that the automatic matching process is generally reliable and effective in identifying corresponding pairs between the CT-MAR and DeepLesion/Stroke EIT datasets.

The pairs with **MS-SSIM** scores below 0.9 totaled 340, of which only 3 were rejected after human visual inspection 5.3. In this inspection, it was observed that even cases with quite low **MS-SSIM** scores were not false positives. Instead, they were due to differences in post-processing or general image treatment that generate local alterations to which **MS-SSIM** is more sensitive. This can be visualized in figure 5.4. This fact highlights the effectiveness of the global structural similarity matching strategy adopted, which managed to identify pairs accurately, with only 3 false positives identified even after rigorous analysis 5.5. This shows that the methodology is robust and performed better in detecting slices in this work than using **MS-SSIM** itself. While local differences affected **MS-SSIM** more significantly, the cosine similarity-based methodology was able to identify correct matches with high global scores, even in cases of pronounced local differences.

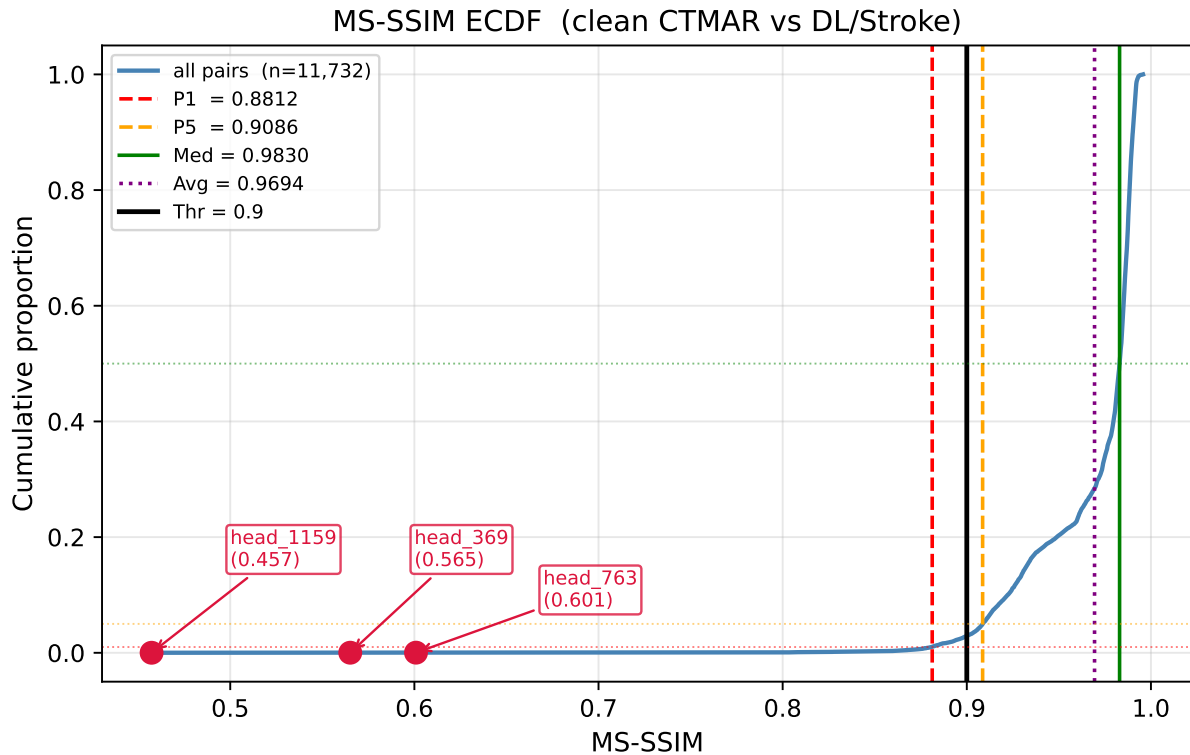


Fig.5.3 **MS-SSIM** distribution between corresponding pairs (**CT-MAR** vs DL/Stroke, automatically approved). Left: histogram with percentiles P1 and P5, median, mean, and threshold of 0.9. Right: corresponding ECDF. Pairs with **MS-SSIM** below 0.9 were subject to manual review.



Fig.5.4 Examples of corresponding pairs with low **MS-SSIM** values (0.7597, 0.7694, and 0.7717, respectively) approved after human visual inspection despite low scores.

Clearly, the reprovved pairs presented **MS-SSIM** values significantly below the standard (< 0.61), with extremely low scores, as shown in figure 5.5. Additionally, they represented pairs with scores closer to the automatic acceptance threshold of 0.994 5.6. This demonstrates that the cases where the method fails are clear outliers in **MS-SSIM** and remain precisely in the gray area prone to errors in the empirical determination of the automatic acceptance threshold. Furthermore, the reprovved pairs have very particular visual characteristics, with

backgrounds dominating the image and few visible anatomical structures, making structural similarity assessment more challenging due to the high homogeneity of visual features.

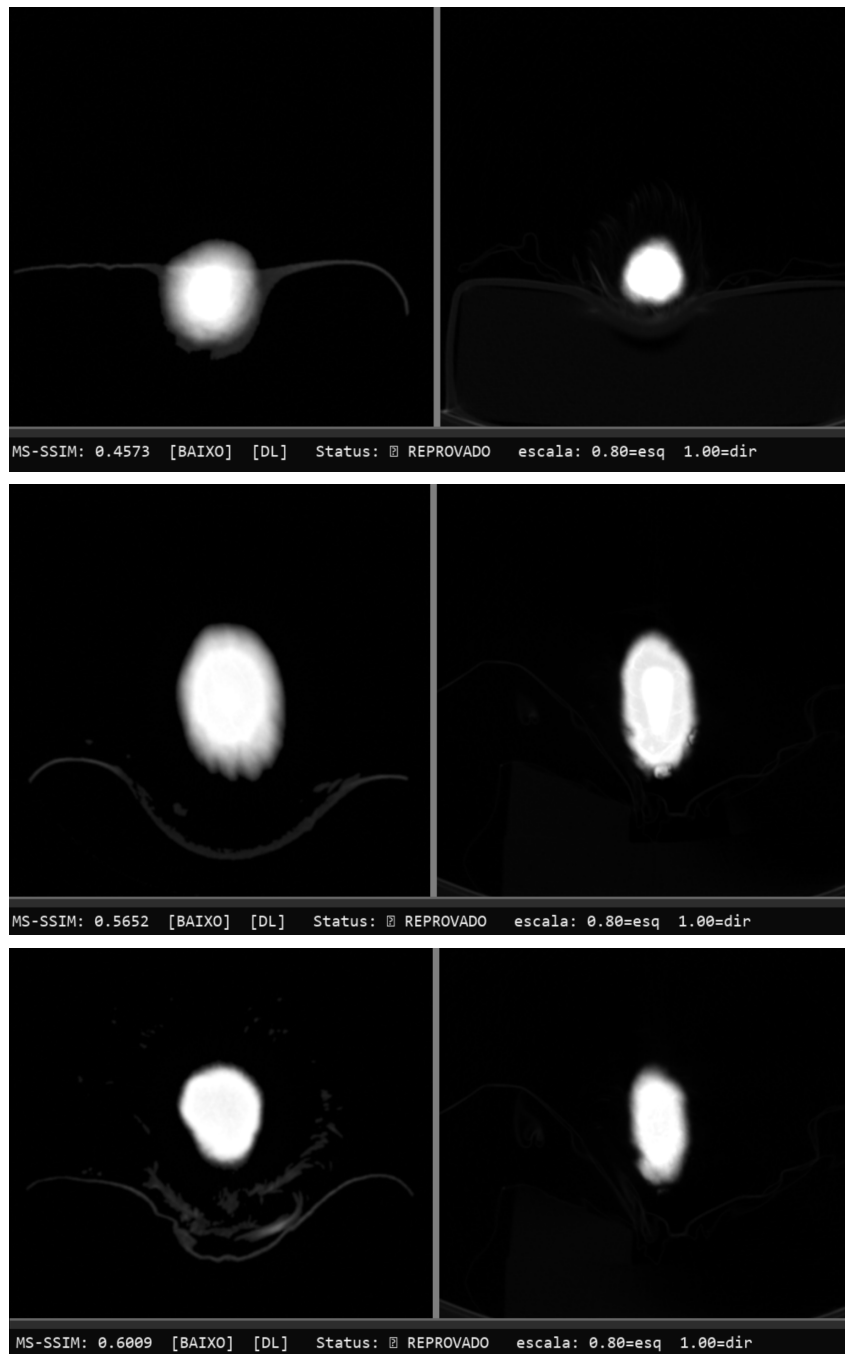


Fig.5.5 Three reprovod pairs after human visual inspection with [MS-SSIM](#) values (head_1159: 0.4573, head_369: 0.5652, and head_763: 0.6009, respectively). The cosine similarity scores for these pairs were head_1159: 0.9960, head_369: 0.9952, and head_763: 0.9951.

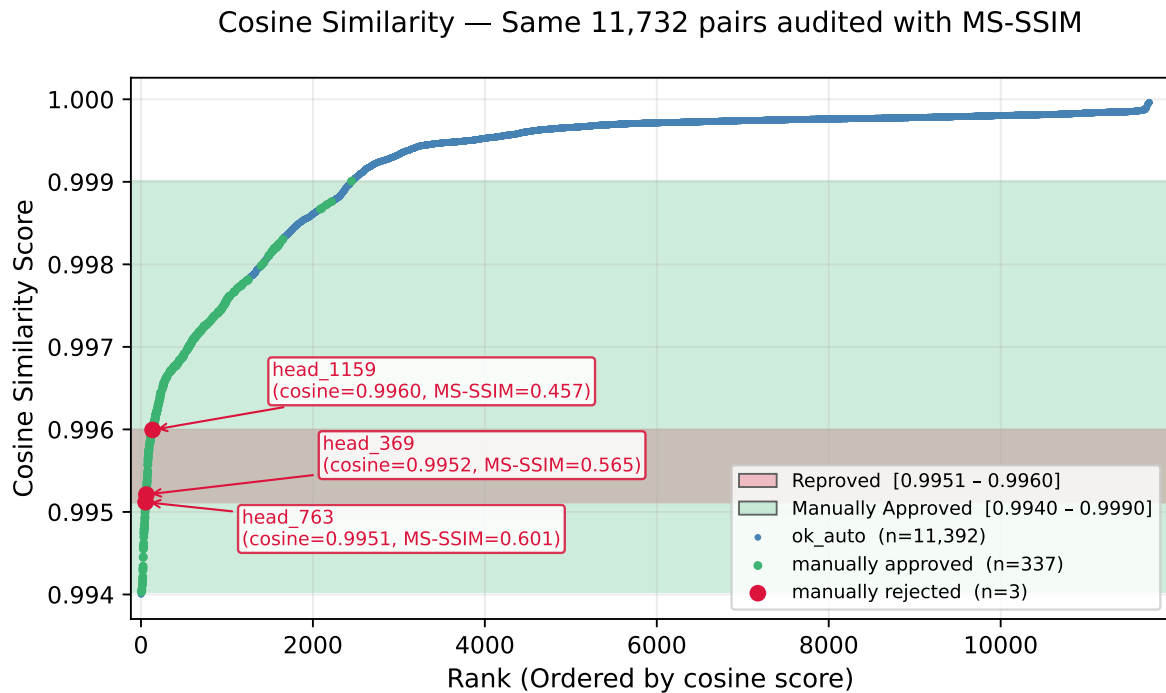


Fig.5.6 Cosine similarity distribution of visually inspected pairs.

The pairs with the most critical **MS-SSIM** scores just above the reproved ones had values greater than 0.75 [5.4](#), demonstrating a clear distance between false positives and visually approved pairs. All inspected pairs with scores above this **MS-SSIM** value passed the visual inspection test [5.7](#). Since the pairs that did not undergo this scrutiny had scores above 0.90 in **MS-SSIM**, it is safe to assume that automatically approved pairs with scores above this threshold are reliable.

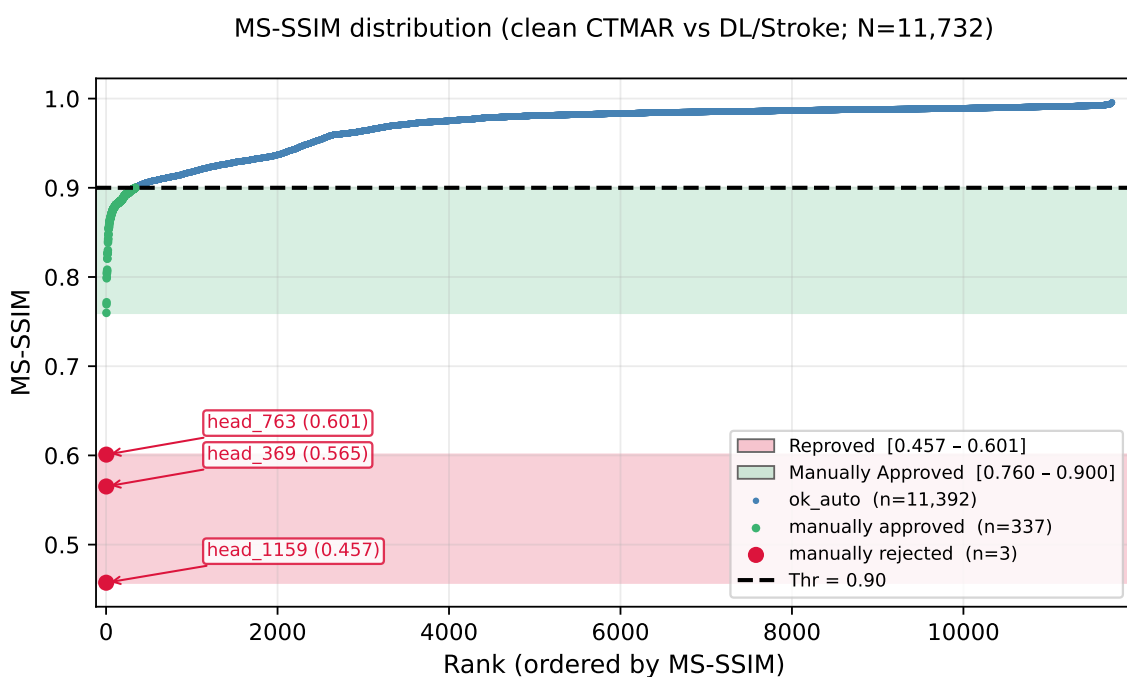


Fig.5.7 **MS-SSIM** values distribution for visually inspected pairs.

In the direct search, 530 pairs did not obtain approved matches in either of the two source datasets. The complementary indirect matching step recovered 95 of these cases after visual inspection, leaving 435 pairs without secure matches.

With relation to possible false negatives that could have resulted in unfair exclusion of pairs in this study, the adopted methodology was even more robust. Since there were few pairs that did not receive approval in either of the two source datasets, it was easy to visually verify each of these cases to confirm whether there was really no match or if the method failed to identify it. Of the 435 excluded pairs, 245 were excluded due to the automated criterion based on the threshold of 0.98. None of these cases were really false negatives, confirming once again the effectiveness of the adopted automatic matching method. The other 190 pairs fell into the low-reliability score range (0.98 – 0.994) and were marked as false matches at this stage. This highlights the importance of having determined a gray area of scores for which the automatic matching method is not efficient, and where human visual review becomes indispensable 5.8.

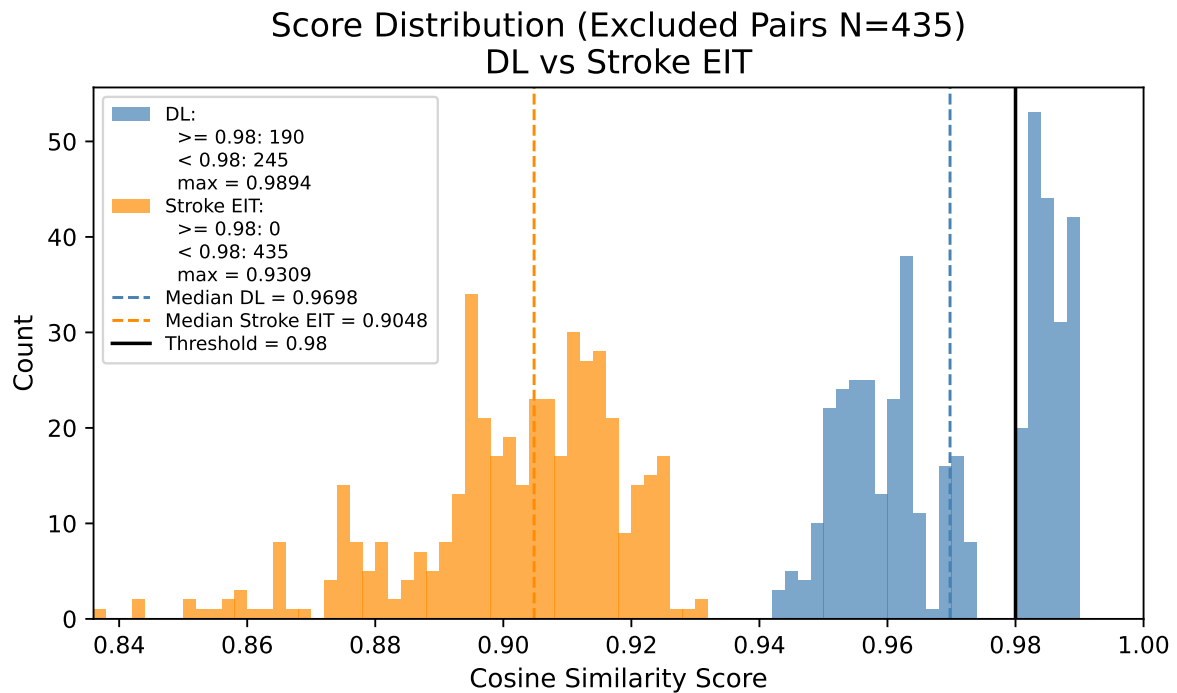


Fig.5.8 Score distribution of excluded pairs (DL x Stroke EIT) for pairs without approved matches in either of the two source datasets (N=435). The orange histogram shows the distribution of scores for the 435 pairs rejected in the Stroke EIT Dataset. The blue histogram shows the distribution of the same 435 pairs rejected in the Deep Lesion Dataset.

5.1.5 Anatomical Distribution and Stratification

The anatomical proportion of 14,000 pairs in the CT-MAR Dataset changed little after the screening and exclusion process of pairs with low reliability. Of the 438 removed cases, 435

were of the *body* type and 3 of the *head* type. The anatomical distribution of the pairs at the slice level included in the study is as follows:

- **Body:** approximately 88.03% ;
- **Head:** approximately 11.97% .

The anatomical distribution of the pairs at the patient level included is found below:

- **Body:** approximately 95.9% (258/269 patients);
- **Head:** approximately 4.1% (11/269 patients).

After partitioning, the train, validation, and test subsets had the following distribution:

Split	Body	Head	% Body	% Head
Train	180	7	96.3%	3.7%
Val	39	2	95.1%	4.9%
Test	39	2	95.1%	4.9%
Total	258	11	–	–

Table 5.4 Patient-level anatomical distribution by split.

Split	Patients	% Total
Train	187	69.52%
Val	41	15.24%
Test	41	15.24%
Total	269	100%

Table 5.5 Patient distribution by split.

Due to the large variation in the number of slices per patient, the anatomical distribution at the pair level is quite different from the distribution at the patient level (5.6 and 5.7). The same occurs for the total pairs distribution. Thus, the proportion of $\sim 70/15/15$ was maintained at the patient level, but at the pair level it does not follow exactly the same proportion.

Split	Body	Head	% Body	% Head
Train	8 491	953	89.91%	10.09%
Val	2 255	385	85.42%	14.58%
Test	1 193	285	80.72%	19.28%
Total	11 939	1 623	–	–

Table 5.6 Pair distribution by split and anatomical category.

Split	Pairs	% Total
Train	9 444	69.64%
Val	2 640	19.47%
Test	1 478	10.90%
Total	13 562	100%

Table 5.7 Pair distribution by split.

5.2 Model Training

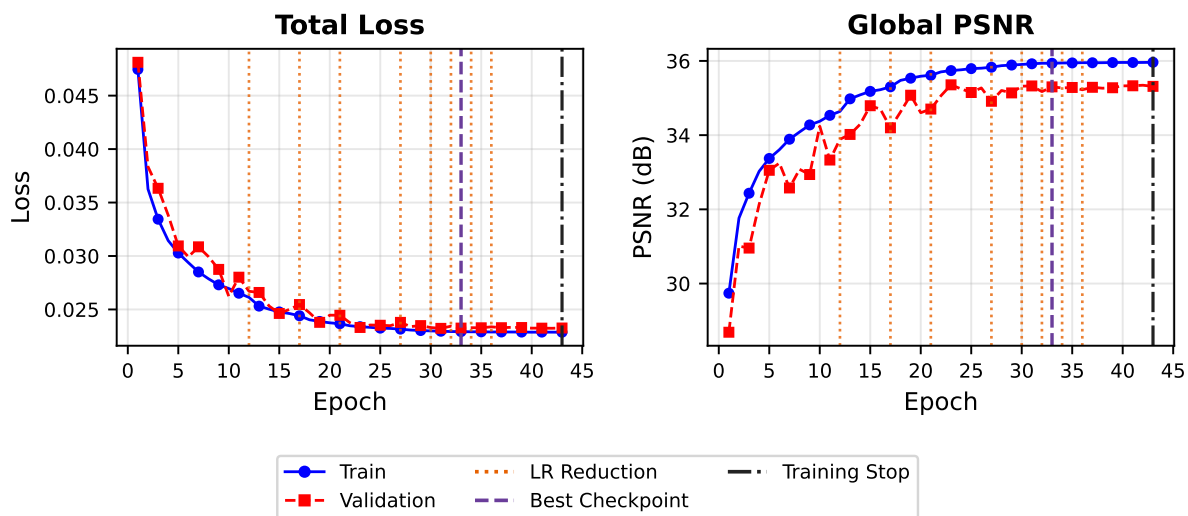
5.2.1 Training Convergence and Per-Run Checkpoint Selection

All six deterministic runs converged and terminated between epochs 38 and 47, after the learning rate reached its lower bound. The best validation [SSIM](#) values occurred between epochs 23 and 44, before training termination, supporting restoration of the best checkpoint weights rather than use of the final epoch.

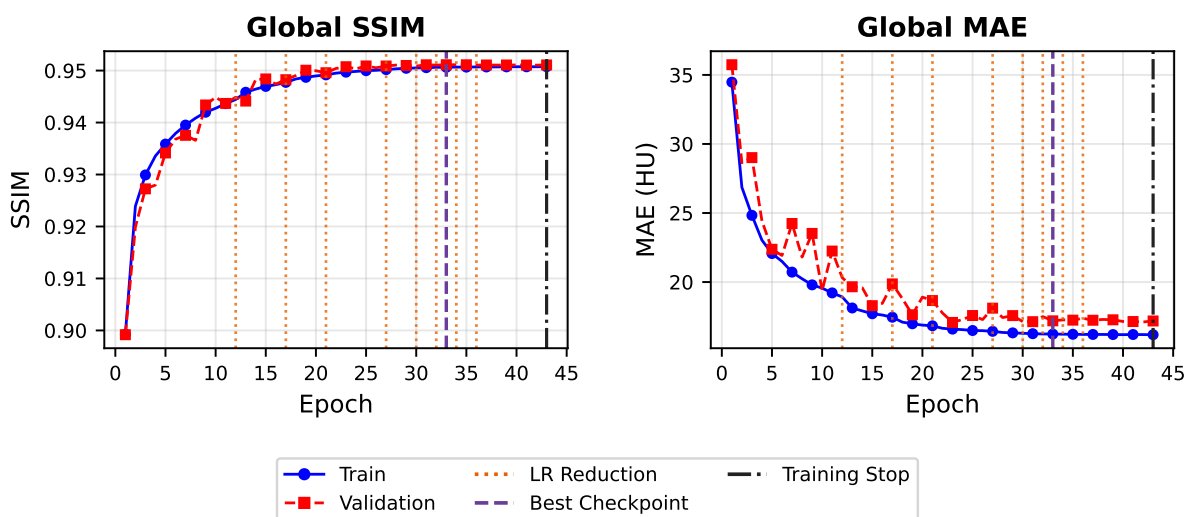
The highest validation reconstruction [SSIM](#) was 0.95113, obtained by the Single Model with seed 2024 at epoch 33. At that same checkpoint, validation [PSNR](#) was 35.30099 dB and validation [MAE](#) was 17.19180 HU. These values show which checkpoint was chosen for this individual run. On their own, they did not decide which run would go forward to external evaluation: that choice was made afterwards, by comparing all six checkpoints on the internal test partition using the Artifact-ROI rule agreed on in advance.

The convergence curves for the two seed-2024 architectures are shown below. The complete set of six runs and the validation tables for all model–seed combinations are provided in [section A.1](#).

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

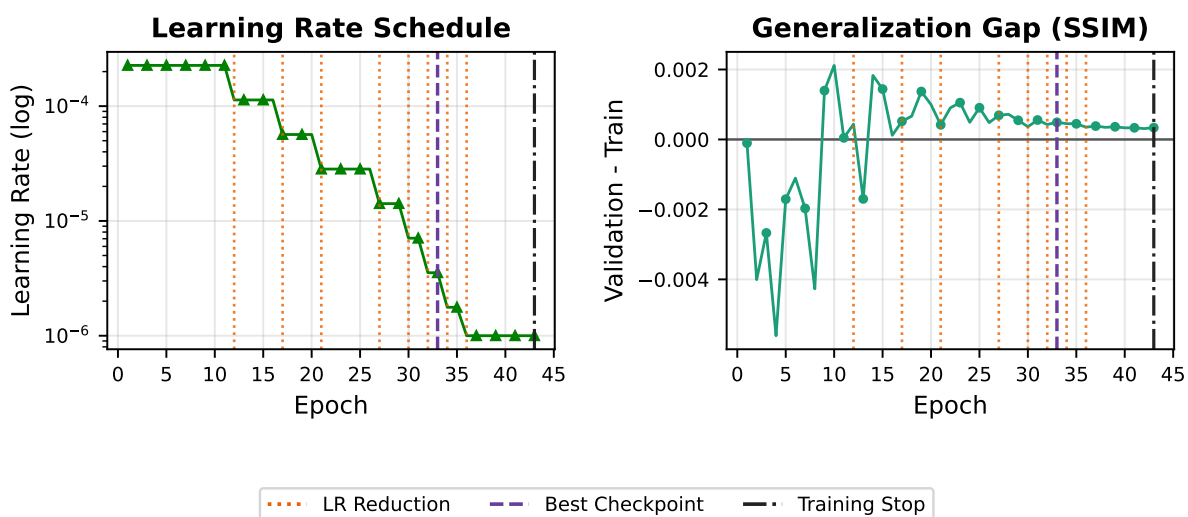
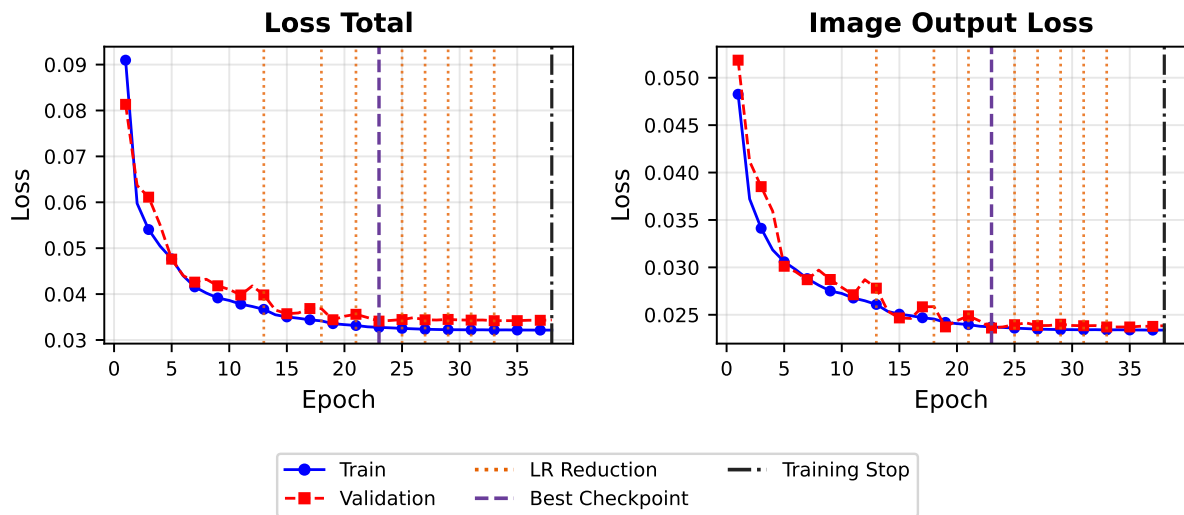
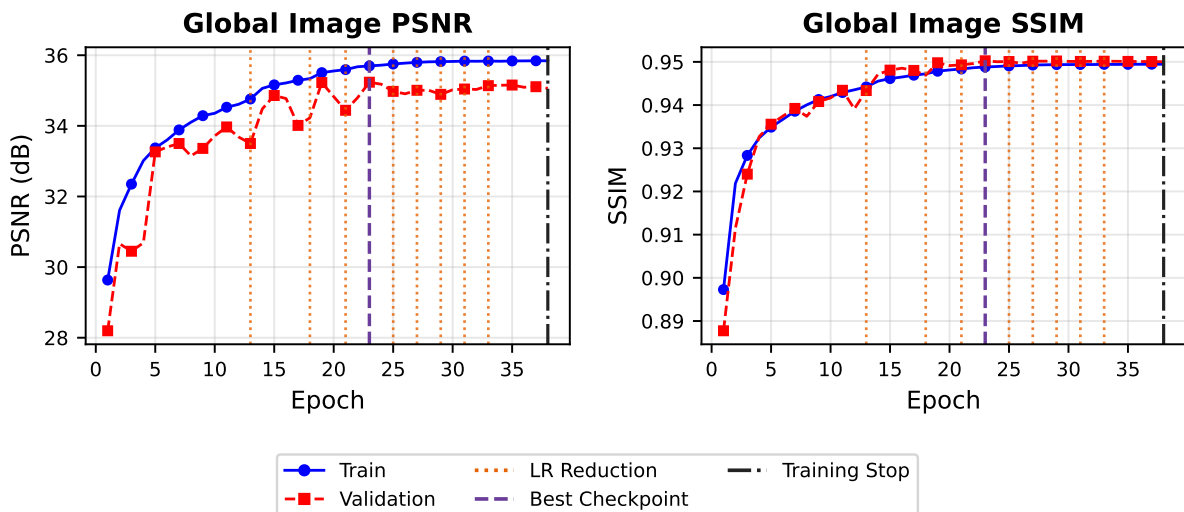


Fig.5.9 Training convergence for the Single Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

Dual Model — Training Convergence



Dual Model — Training Convergence



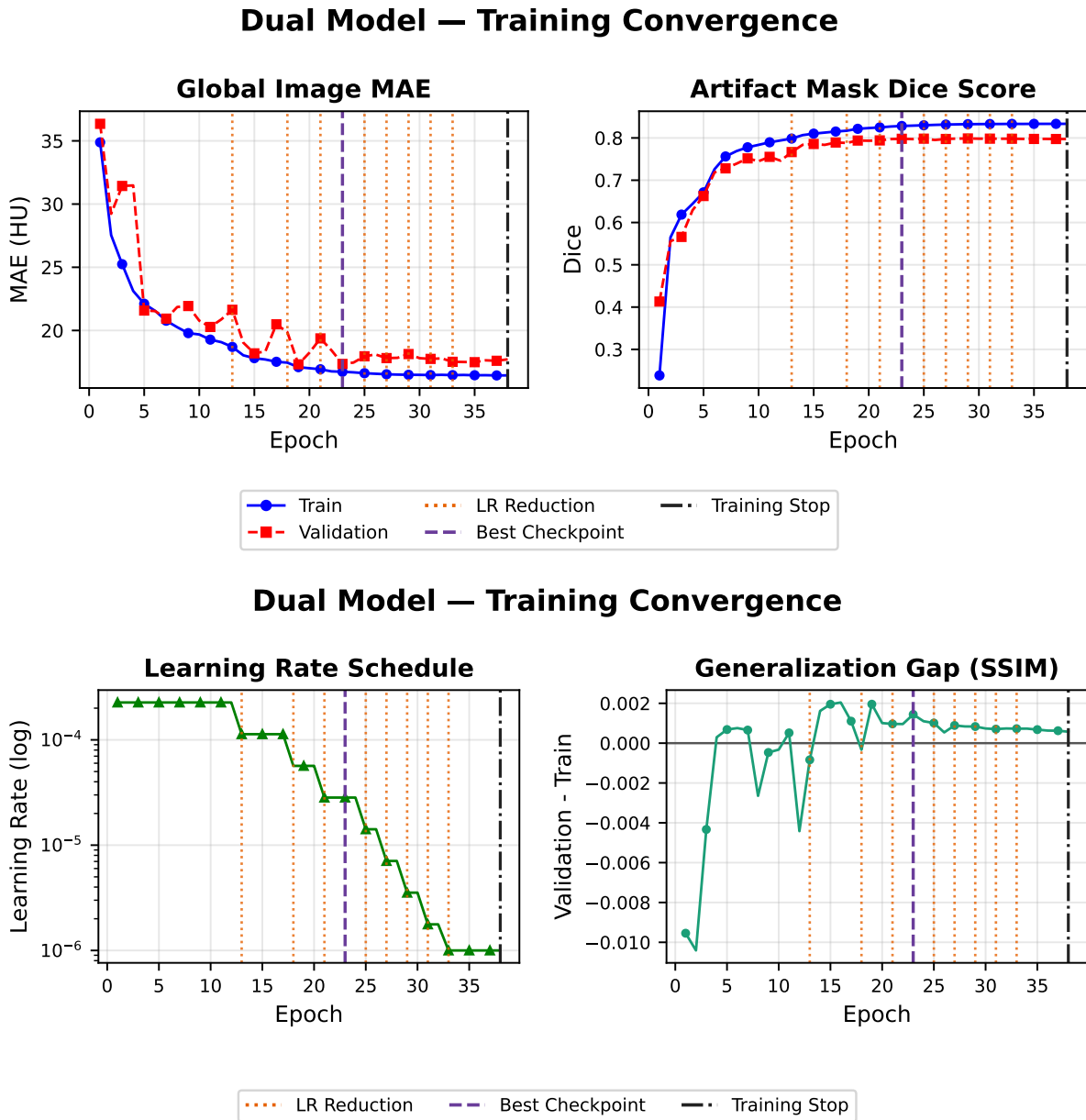


Fig.5.10 Training convergence for the Dual Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

5.3 Internal Evaluation in Test Subset

The internal evaluation was conducted on the fixed test partition, composed of 1478 slices from 41 independent patients. For each checkpoint previously selected in validation, PSNR, SSIM, and MAE were calculated between the reconstruction and the clean reference. All statistical comparisons used the patient, not the slice, as the unit of analysis: metrics were first averaged

within each patient, and 95% confidence intervals were obtained by resampling patients 10,000 times (a bootstrap procedure). The test set was not used to change any run’s checkpoint. It was used once to compare the six validation-frozen candidates for the external-evaluation decision described in [subsection 5.3.2](#).

5.3.1 Region of Interest Evaluation

The regions of interest and the per-zone metric calculation followed the procedure described in [subsection 4.5.1](#). This stratification makes it possible to separately quantify the correction in the region directly affected by artifacts and the changes introduced in body regions distant from the implant.

The final results, first aggregated per patient and then summarized over 41 independent patients, are presented in [Table 5.8](#), [Table 5.9](#), and [Table 5.10](#). All six runs showed substantial improvement in the Artifact Zone: Δ PSNR between 11.47 and 11.86 dB, Δ SSIM between 0.322 and 0.328, and MAE reduction between 96.9 and 98.5 HU. In the Clean Zone, the mean changes were smaller (Δ PSNR between 1.36 and 1.61 dB, Δ SSIM between 0.057 and 0.058, MAE reduction between 3.2 and 3.6 HU). These changes should be interpreted as a quantitative description rather than as clinical validation of anatomical preservation.

Δ SSIM			
Seed	Model	Artifact ROI	Clean ROI
42	Single Model	0.3276 ± 0.0954	0.0579 ± 0.0426
	Dual Model	0.3233 ± 0.0937	0.0568 ± 0.0421
1337	Single Model	0.3278 ± 0.0933	0.0582 ± 0.0421
	Dual Model	0.3224 ± 0.0953	0.0566 ± 0.0422
2024	Single Model	0.3279 ± 0.0955	0.0581 ± 0.0422
	Dual Model	0.3266 ± 0.0954	0.0576 ± 0.0422

Note: Δ SSIM = SSIM(Model, GT) – SSIM(Input, GT) within the indicated ROI.

Table 5.8 Δ SSIM by ROI on the test set. Values are mean $\pm$ standard deviation across patients ($n = 41$).

Δ PSNR (dB)			
Seed	Model	Artifact ROI	Clean ROI
42	Single Model	11.75 ± 2.59	1.46 ± 1.43
	Dual Model	11.65 ± 2.53	1.48 ± 1.43
1337	Single Model	11.84 ± 2.54	1.53 ± 1.35
	Dual Model	11.47 ± 2.57	1.36 ± 1.52
2024	Single Model	11.86 ± 2.69	1.61 ± 1.41
	Dual Model	11.84 ± 2.62	1.53 ± 1.48

Note: Δ PSNR = PSNR(Model, GT) – PSNR(Input, GT) within the indicated ROI.

Table 5.9 Δ PSNR by ROI on the test set. Values are mean $\pm$ standard deviation across patients ($n = 41$).

$\Delta\text{MAE (HU)}$

Seed	Model	Artifact ROI	Clean ROI
42	Single Model	97.8 ± 19.6	3.3 ± 3.2
	Dual Model	97.5 ± 19.1	3.4 ± 3.1
1337	Single Model	98.5 ± 19.6	3.5 ± 3.1
	Dual Model	96.9 ± 19.9	3.2 ± 3.3
2024	Single Model	98.4 ± 20.2	3.6 ± 3.1
	Dual Model	98.3 ± 19.8	3.5 ± 3.3

Note: $\Delta\text{MAE} = \text{MAE}(\text{Input}, \text{GT}) - \text{MAE}(\text{Model}, \text{GT})$ within the indicated ROI. Positive values indicate error reduction.

Table 5.10 MAE reduction by ROI on the test set. Values are mean $\pm$ standard deviation across patients ($n = 41$).

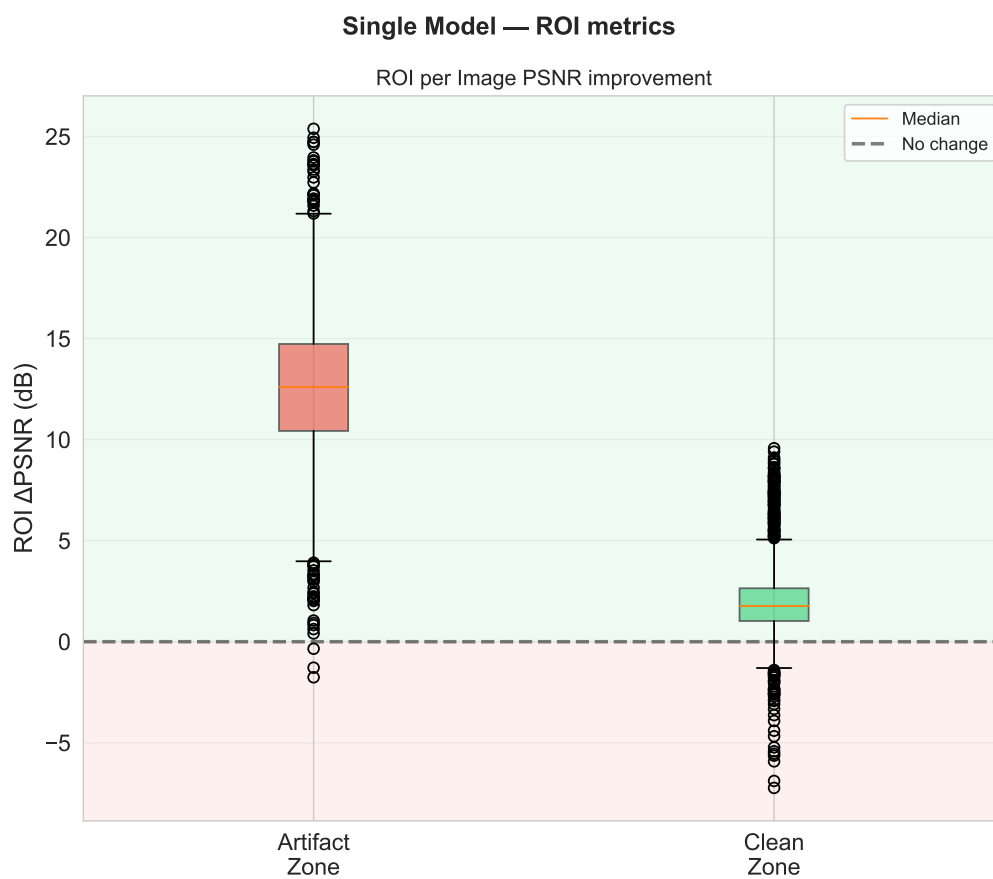
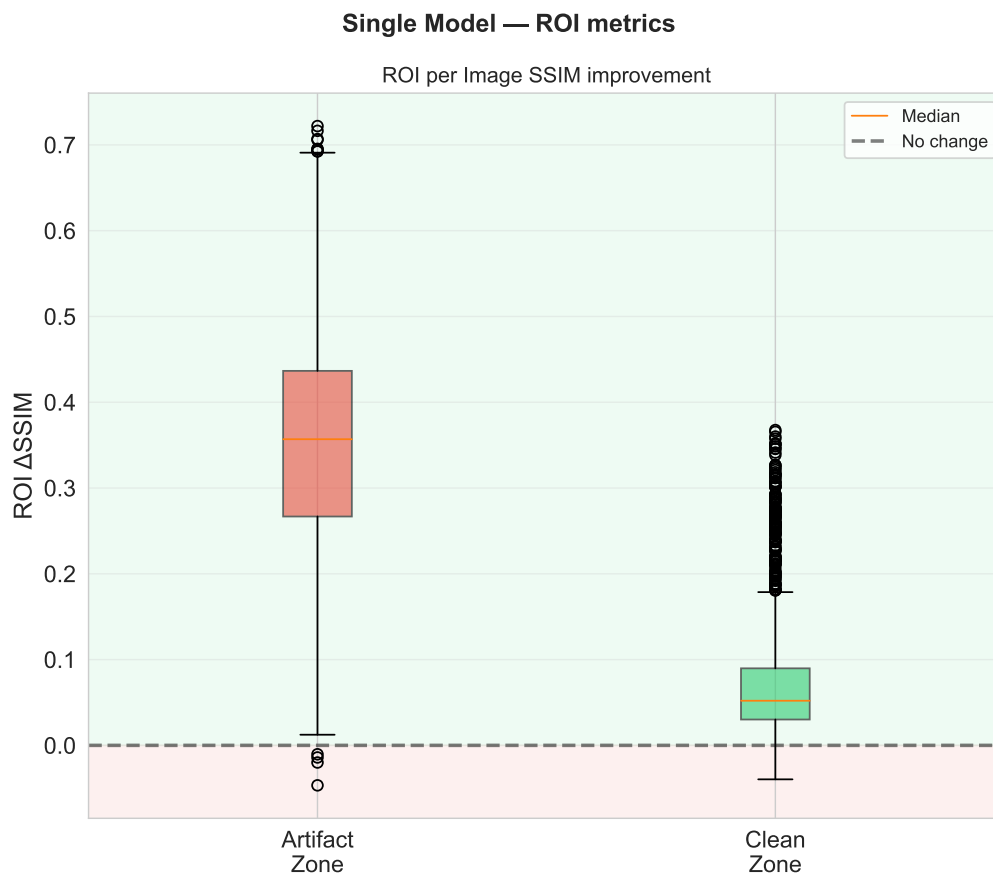
Positive values correspond to error reduction.

5.3.2 Selection of the External-Evaluation Candidate

Among the six checkpoints previously frozen by validation SSIM, the Single Model with seed 2024 had the highest mean patient-level Artifact-ROI ΔSSIM (0.3279 ± 0.0955). Its mean Clean-ROI ΔSSIM was 0.0581 ± 0.0422 , substantially smaller than the improvement observed in the Artifact Zone. This meets the preservation check that was planned in advance. But since no specific numeric limit was set for how small the Clean-Zone change should be, it should not be read as formal proof of anatomical safety. The Single Model, seed 2024, was therefore selected for external evaluation, after which its architecture, seed, weights, and hyperparameters remained unchanged. The other ROI metrics reported above are complementary and did not change this decision. The global metrics, reported next for completeness, are likewise complementary.

The Figure 5.11 and Figure 5.12 show the image-level distributions for the selected run. For clarity, the remaining plots present only this model. Data for all six runs are available in Appendix A.

The box plots highlight the separation between the correction in the Artifact Zone and the smaller change in the Clean Zone. The scatter plot allows identification of heterogeneity between slices, including individual cases with unfavorable changes in the Clean Zone.



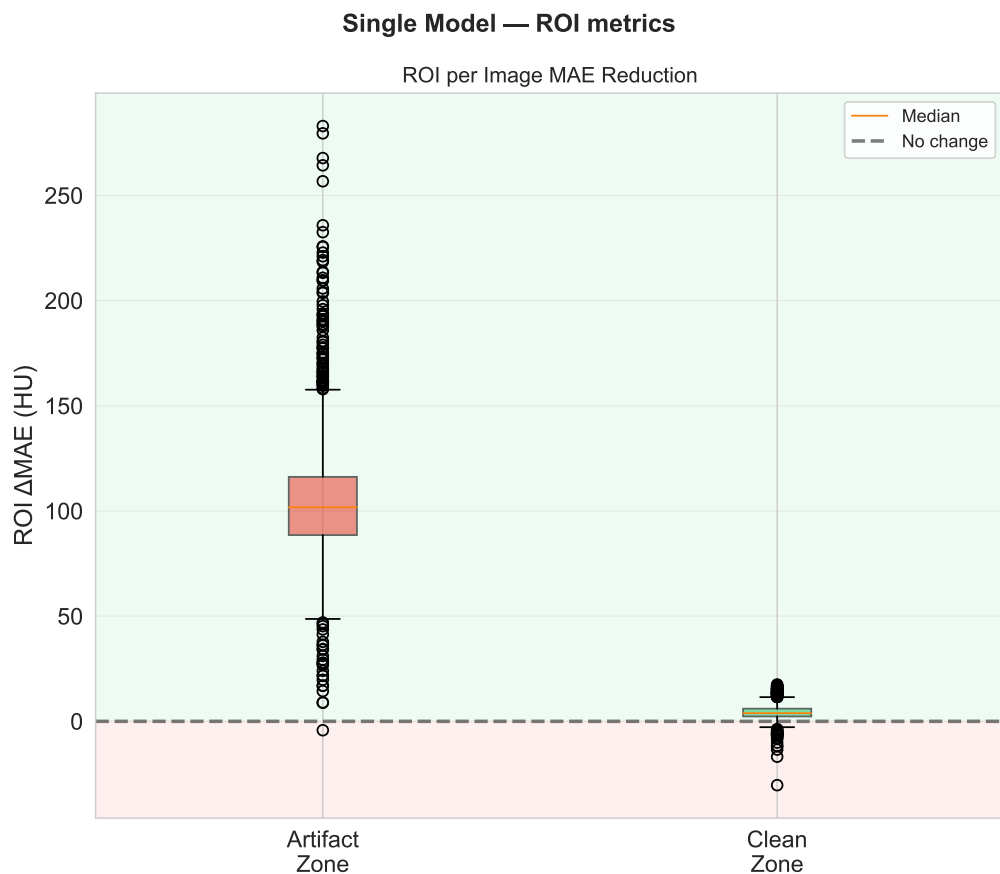
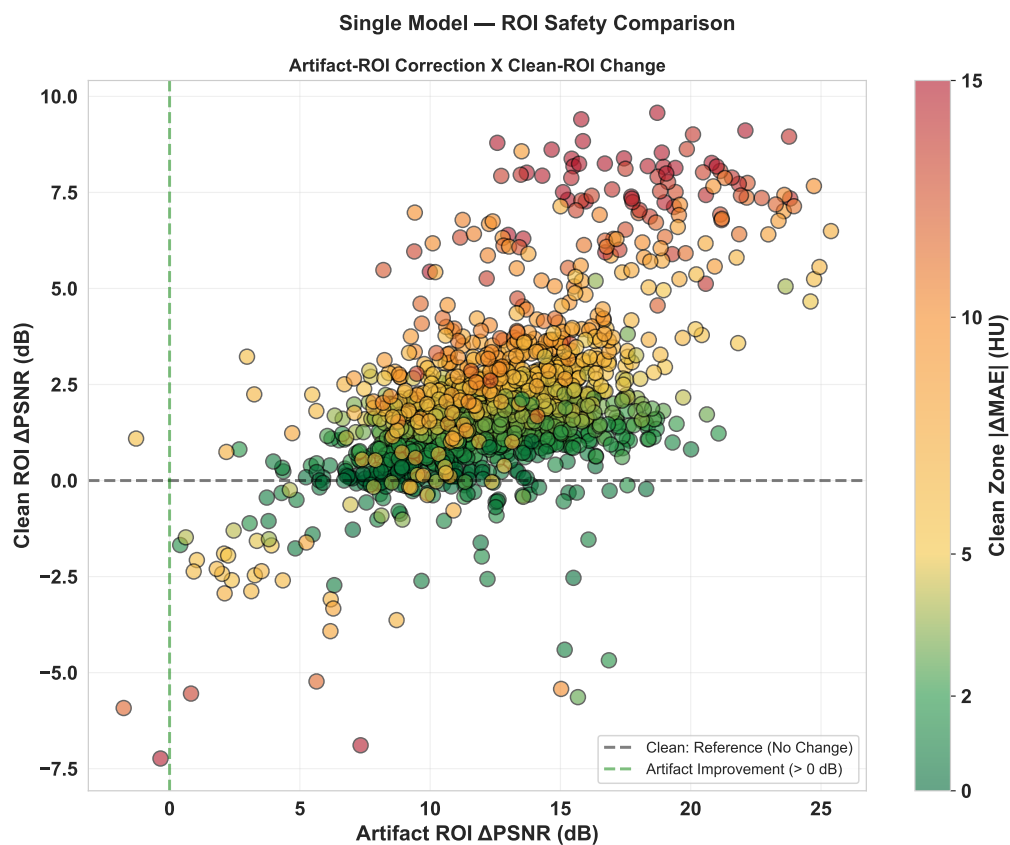


Fig.5.11 Per-image distributions of SSIM, PSNR, and MAE changes in the artifact and clean zones for the Single Model, seed 2024.



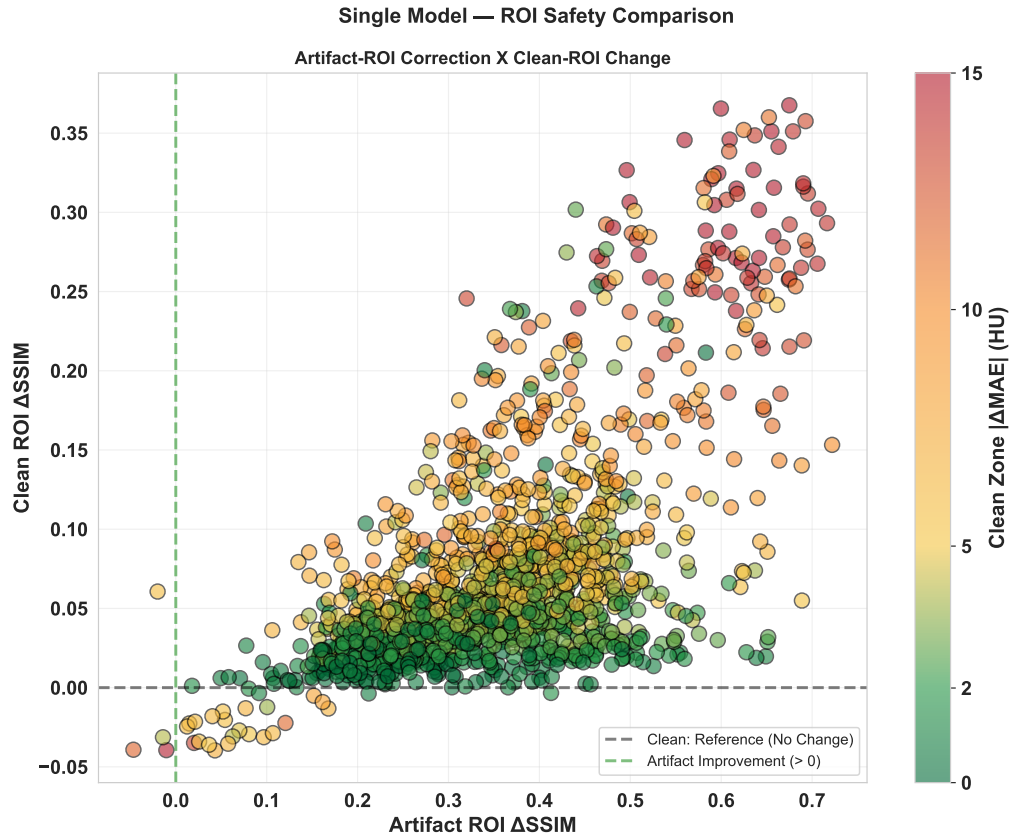


Fig.5.12 Per-image relationship between the $\text{SSIM} \setminus \text{PSNR}$ improvement in the Artifact Zone and the $\text{SSIM} \setminus \text{PSNR}$ change in the Clean Zone for the Single Model, seed 2024. Color encodes Clean Zone $|\Delta\text{MAE}|$ in HU: green at 0–2 HU, yellow around 5 HU, orange around 10 HU, and red at or above 15 HU. Both plots use the same 0–15 HU scale, with values above 15 HU saturated in red. These color anchors are descriptive, not clinical thresholds.

5.3.3 Global Outcomes

Having established the Single Model, seed 2024, as the run selected for external evaluation, this subsection reports the global metrics, computed over all body-mask pixels without ROI stratification, as complementary evidence of overall reconstruction quality. They were not part of the basis for the selection decision above.

The Table 5.11, Table 5.12, and Table 5.13 present the slice-wise means. Since all runs were evaluated on the same test partition, the input value is common to both models for each seed. In all runs, reconstruction increased the global PSNR from 27.39 dB to 34.95–35.20 dB, increased SSIM from 0.8361 to 0.9421–0.9442, and reduced MAE from 36.6 HU to 17.3–17.8 HU.

SSIM global

Seed	Input	Single Model	Dual Model
42	0.8361 ± 0.0880	0.9439 ± 0.0289	0.9424 ± 0.0292
1337	0.8361 ± 0.0880	0.9441 ± 0.0286	0.9421 ± 0.0289
2024	0.8361 ± 0.0880	0.9442 ± 0.0288	0.9432 ± 0.0292

All input and model values are computed against the clean reference (*ground truth*).

Table 5.11 Global **SSIM** per slice in test subset ($n = 1478$), expressed as mean $\pm$ standard deviation.**PSNR global (dB)**

Seed	Input	Single Model	Dual Model
42	27.39 ± 4.14	35.09 ± 2.82	35.06 ± 2.84
1337	27.39 ± 4.14	35.13 ± 2.81	34.95 ± 2.82
2024	27.39 ± 4.14	35.20 ± 2.84	35.16 ± 2.82

All input and model values are computed against the clean reference (*ground truth*).

Table 5.12 Global **PSNR** per slice in test subset ($n = 1478$), expressed as mean $\pm$ standard deviation.**MAE global (HU)**

Seed	Input	Single Model	Dual Model
42	36.569 ± 21.906	17.649 ± 5.689	17.654 ± 5.905
1337	36.569 ± 21.906	17.485 ± 5.624	17.842 ± 5.789
2024	36.569 ± 21.906	17.341 ± 5.573	17.475 ± 5.563

All input and model values are computed against the clean reference (*ground truth*).

Table 5.13 Global **MAE** per slice in test subset ($n = 1478$), expressed as mean $\pm$ standard deviation.

The seed-wise global tables show that the Single Model, seed 2024, was not an isolated run: the three Single Model runs had very similar global performance. Seed 2024 was at the favorable extreme for global **SSIM**, **PSNR**, and **MAE**.

The patient-level deltas and their confidence intervals are shown in Table 5.14, Table 5.15, and Table 5.16. All six runs showed positive mean improvement for the three global metrics, and the corresponding confidence intervals excluded the null value.

 Δ SSIM Global (95% CI)

Seed	Single Model	Dual Model
42	0.0887 [0.0728; 0.1063]	0.0870 [0.0713; 0.1043]
1337	0.0891 [0.0734; 0.1065]	0.0867 [0.0710; 0.1041]
2024	0.0891 [0.0732; 0.1066]	0.0879 [0.0719; 0.1053]

Note: Δ SSIM = SSIM(Model, GT) – SSIM(Input, GT).

Table 5.14 Δ SSIM global aggregated per patient in the test subset ($n = 41$). CI: 95% confidence interval. **Δ PSNR Global (dB, 95% CI)**

Seed	Single Model	Dual Model
42	6.96 [6.22; 7.68]	6.96 [6.24; 7.65]
1337	7.05 [6.33; 7.75]	6.85 [6.12; 7.55]
2024	7.11 [6.36; 7.83]	7.06 [6.31; 7.78]

Note: Δ PSNR = PSNR(Model, GT) – PSNR(Input, GT).

Table 5.15 Δ PSNR global aggregated per patient in the test subset ($n = 41$). CI: 95% confidence interval.

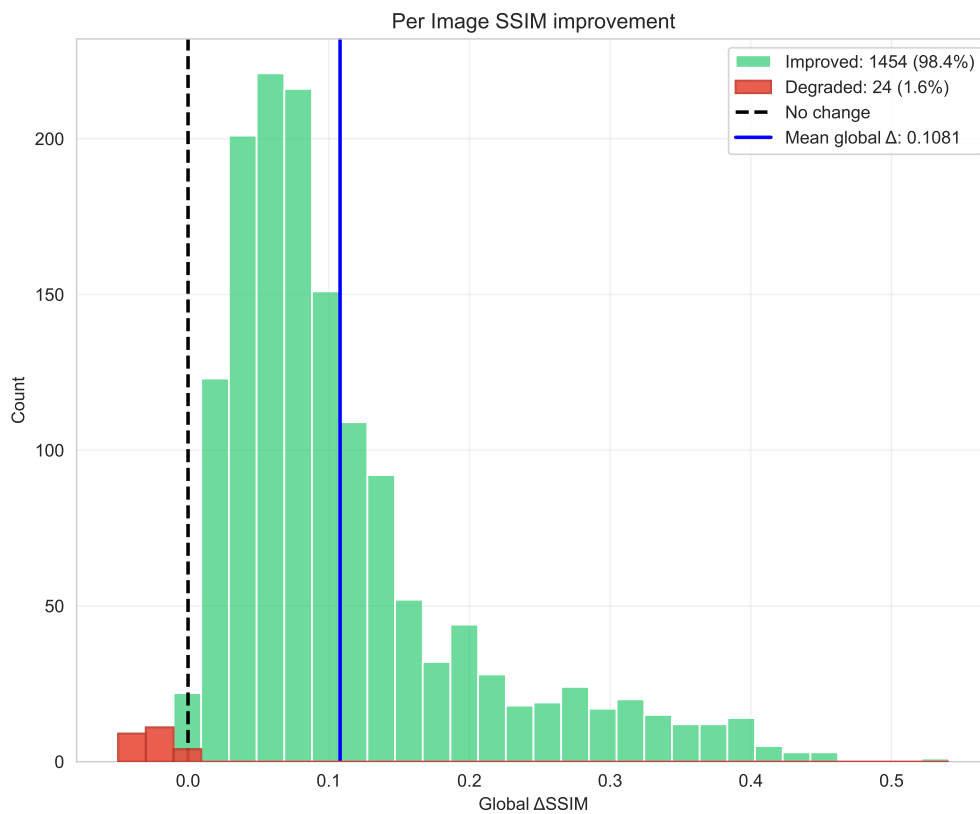
Δ MAE Global (HU, 95% CI)

Seed	Single Model	Dual Model
42	15.3 [12.2; 19.0]	15.3 [12.3; 18.9]
1337	15.6 [12.4; 19.3]	15.2 [12.1; 18.8]
2024	15.7 [12.5; 19.5]	15.5 [12.4; 19.3]

Note: Δ MAE = MAE(Input, GT) – MAE(Model, GT). Positive values indicate error reduction.

Table 5.16 Δ MAE global aggregated per patient in the test subset ($n = 41$). Values > 0 indicate improvement. CI: 95% confidence interval.

For the selected run, the histograms in Figure 5.3.3 show the per-slice distribution of the global deltas. Although these plots are descriptive and slices from the same patient are not independent, they make the few unfavorable cases visible: 12 of 1478 slices (0.8%) showed a negative Δ PSNR and 24 (1.6%) showed a negative Δ SSIM. Statistical inference remains based on the patient-level means presented in the preceding tables.

Single Model — Global per Image Improvement

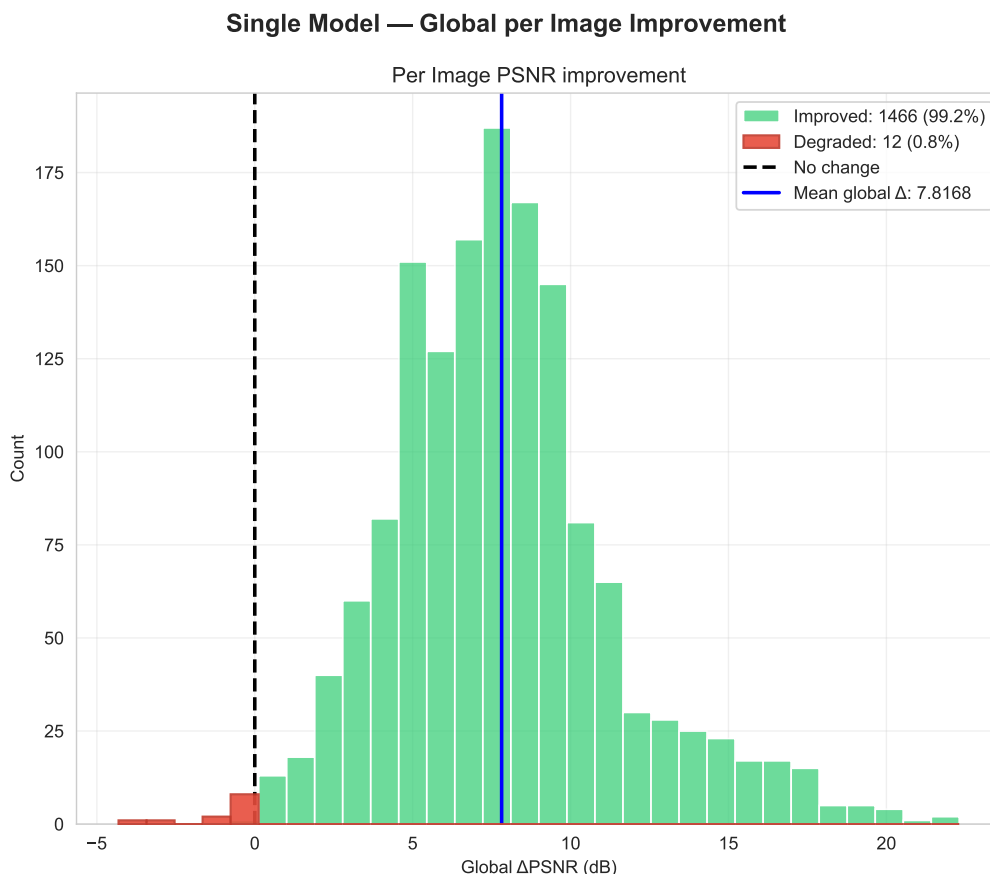


Fig.5.13 Per-image global improvement over the corrupted input for the Single Model, seed 2024: Δ SSIM (top) and Δ PSNR (bottom).

As an additional, purely descriptive check of the overall HU calibration, Figure 5.14 presents the agreement between the mean HU of the reconstruction and that of the clean reference, aggregated per patient. Each point corresponds to one of the 41 patients, each patient's mean was calculated over all valid body-mask pixels, weighted by the number of pixels in each slice. The mean bias was +4.19 HU, with 95% limits of agreement between +1.45 and +6.93 HU. This analysis describes a small positive shift in global mean intensity, but does not quantify local artifact removal, nor does it replace the global and ROI metrics.

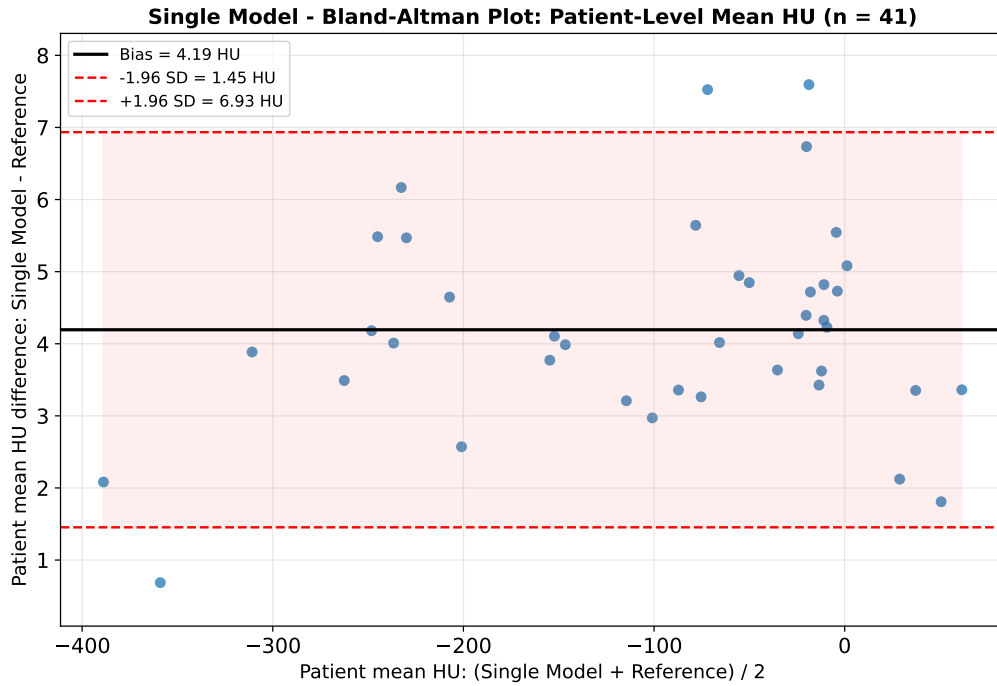


Fig.5.14 Bland-Altman agreement between the Single Model reconstruction and the clean reference for patient-level mean HU ($n = 41$). Each patient mean was calculated over all valid body-mask pixels. Positive differences indicate higher mean HU in the reconstruction.

5.3.4 Residual Maps in Extreme Cases

To complement the results, six individual cases from the Single Model, seed 2024, were inspected in the test partition. The selection was made exclusively by the per-slice $\Delta SSIM$ in the Artifact Zone. The three highest values correspond to the greatest structural improvement in that region relative to the input, while the three lowest values correspond to the smallest structural improvement. Thus, the terms “greatest” and “smallest” improvement in this subsection refer only to this quantitative criterion.

In each map, the first row shows the change required relative to the clean reference ($r_{\text{target}} = \text{GT} - \text{Input}$), the change produced by the model ($r_{\text{model}} = \text{Output} - \text{Input}$), and the remaining residual error ($r_{\text{diff}} = r_{\text{model}} - r_{\text{target}} = \text{Output} - \text{GT}$). The second row shows the absolute error of the input, the absolute error of the output, and their difference, where green indicates error reduction and red indicates worsening. The yellow outline delimits the Artifact Zone. Color scales are adjusted individually for each slice to preserve detail in cases with small residuals. However, comparison between cases should prioritize the spatial patterns and the statistics presented in each panel, rather than the apparent color intensity alone. Below each map, the corresponding visualization of the artifact input, the model output, and the clean reference is shown.

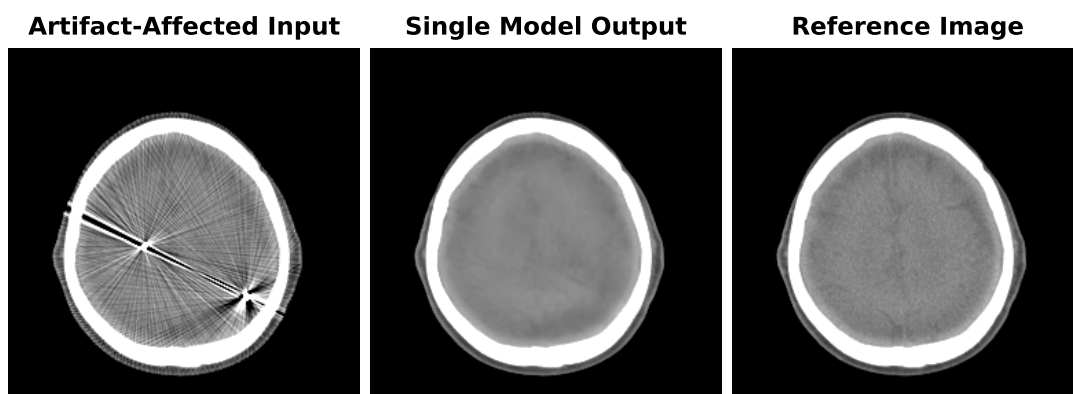
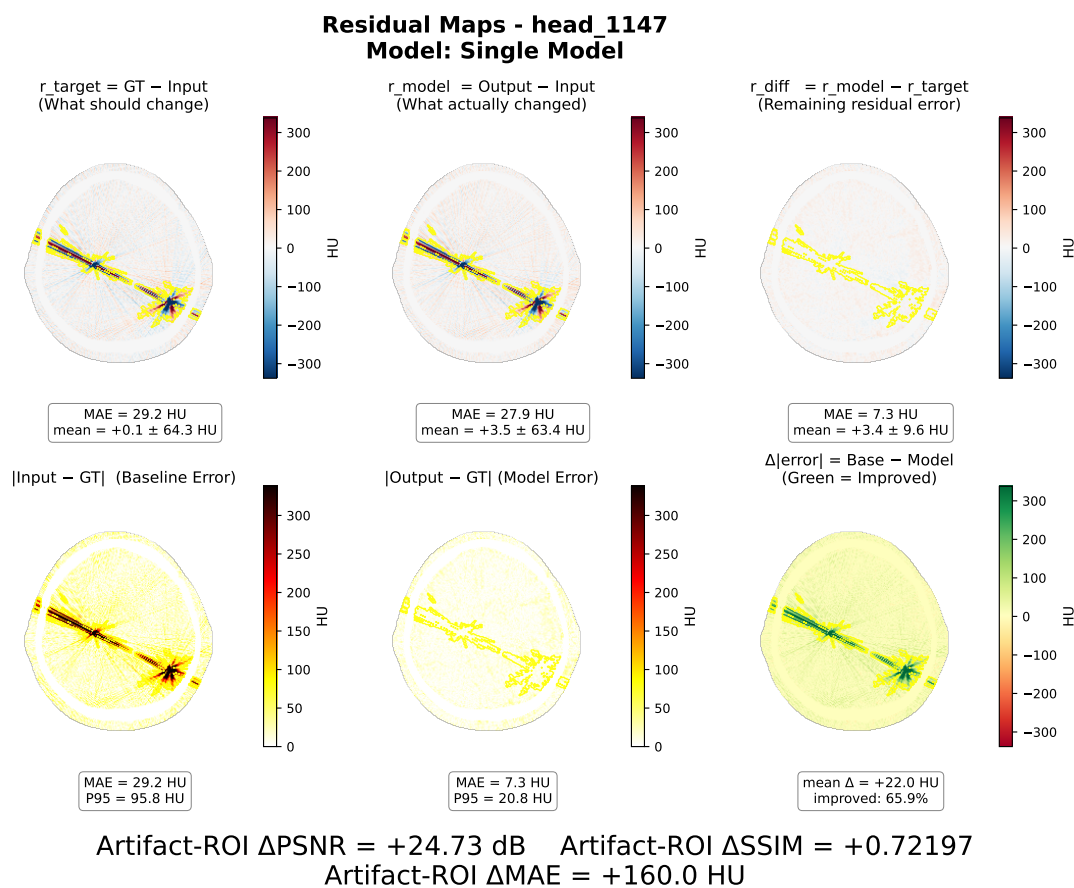


Fig.5.15 Residual map (top) and reconstruction visualization (bottom) for case head_1147 (test-set index 1250), selected among the three highest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = +0.7220$; global $\Delta\text{SSIM} = +0.1649$; $\Delta\text{PSNR} = +15.99$ dB). All error maps are restricted to the body mask.

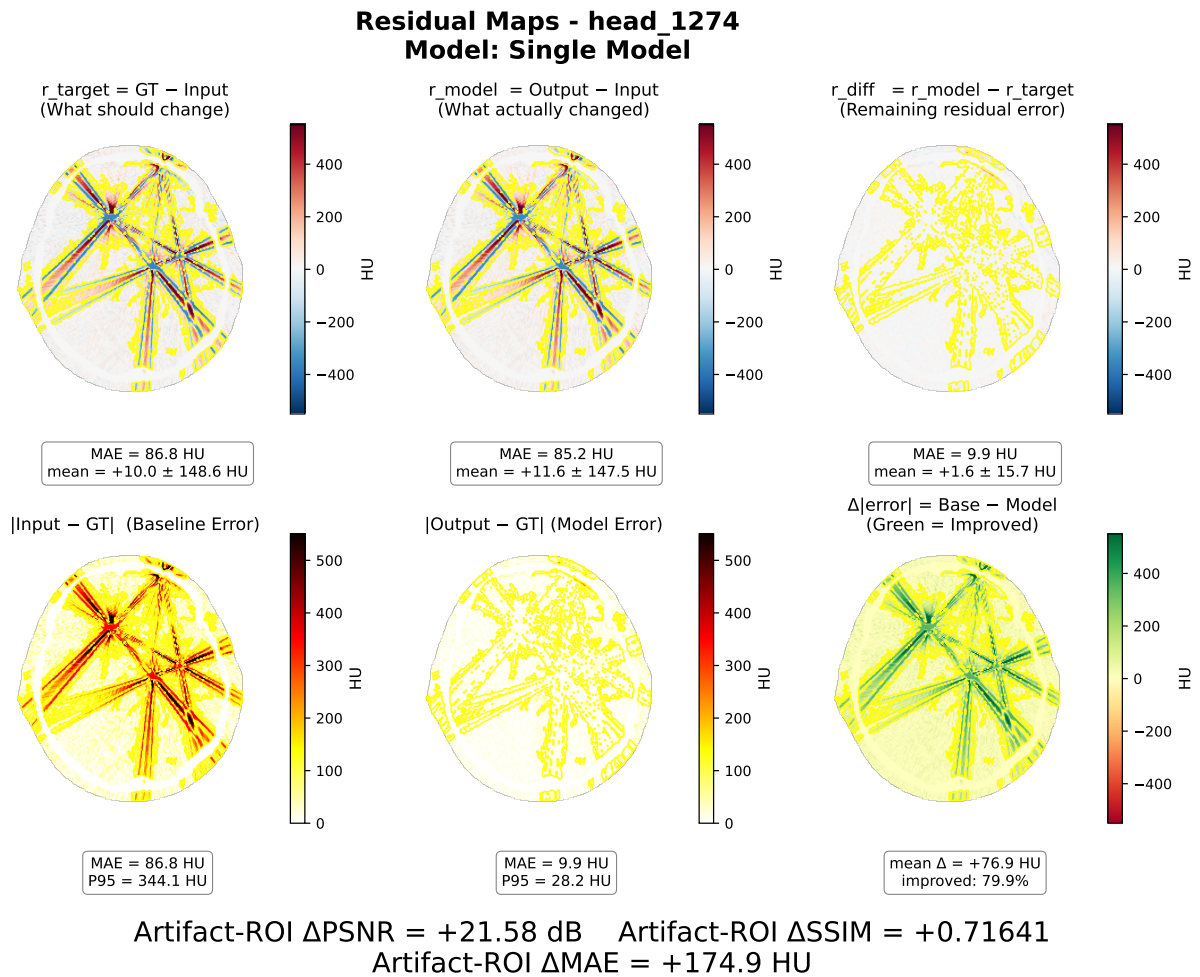


Fig.5.16 Residual map (top) and reconstruction visualization (bottom) for case head_1274 (test-set index 1364), selected among the three highest artifact-ROI $\Delta SSIM$ values ($\Delta SSIM_{ROI} = +0.7164$; global $\Delta SSIM = +0.3752$; $\Delta PSNR = +19.48$ dB). All error maps are restricted to the body mask.

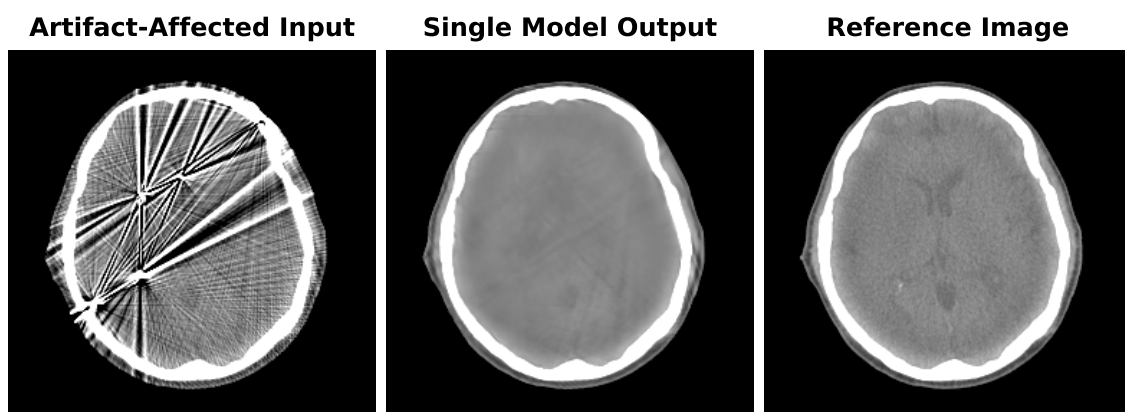
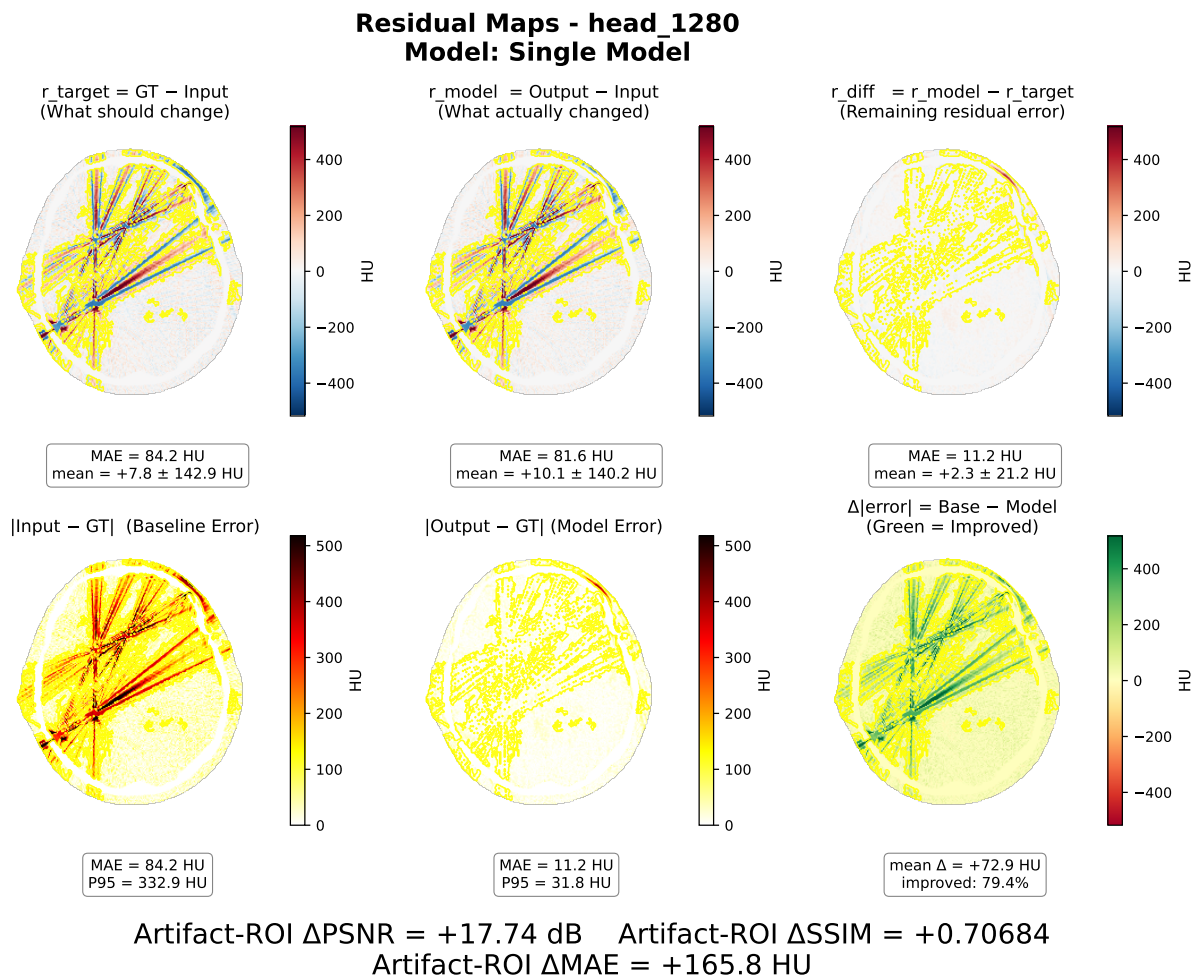


Fig.5.17 Residual map (top) and reconstruction visualization (bottom) for case head_1280 (test-set index 1369), selected among the three highest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = +0.7068$; global $\Delta\text{SSIM} = +0.3837$; $\Delta\text{PSNR} = +16.53$ dB). All error maps are restricted to the body mask.

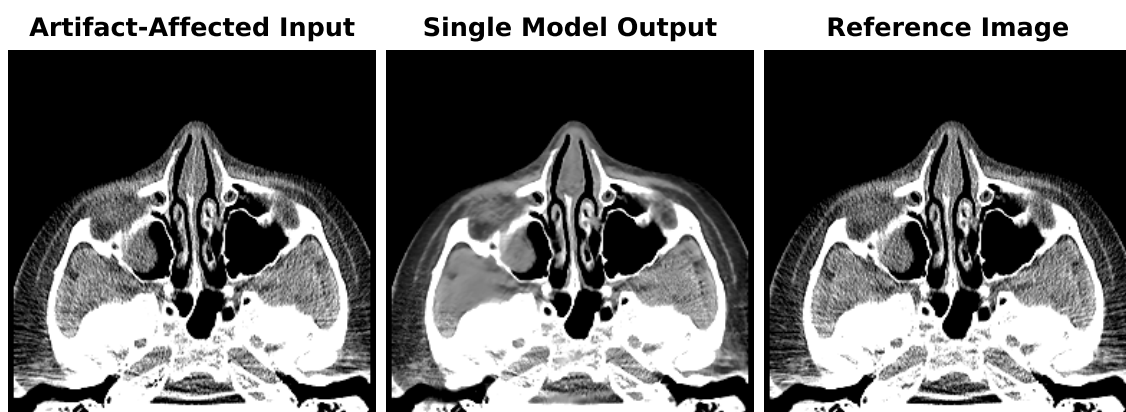
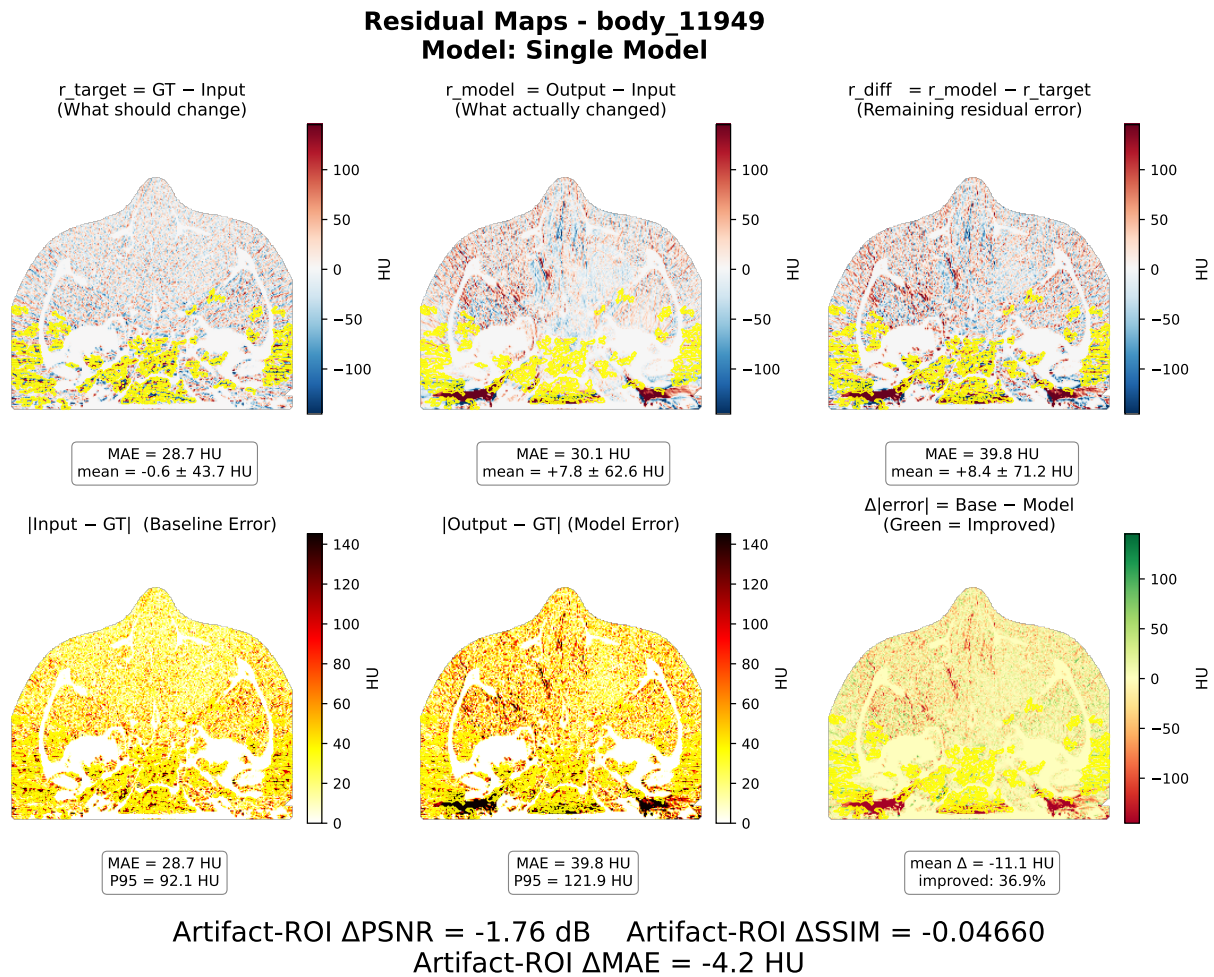


Fig.5.18 Residual map (top) and reconstruction visualization (bottom) for case body_11949 (test-set index 0716), selected among the three lowest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = -0.0466$; global $\Delta\text{SSIM} = -0.0497$; $\Delta\text{PSNR} = -4.31$ dB). All error maps are restricted to the body mask.

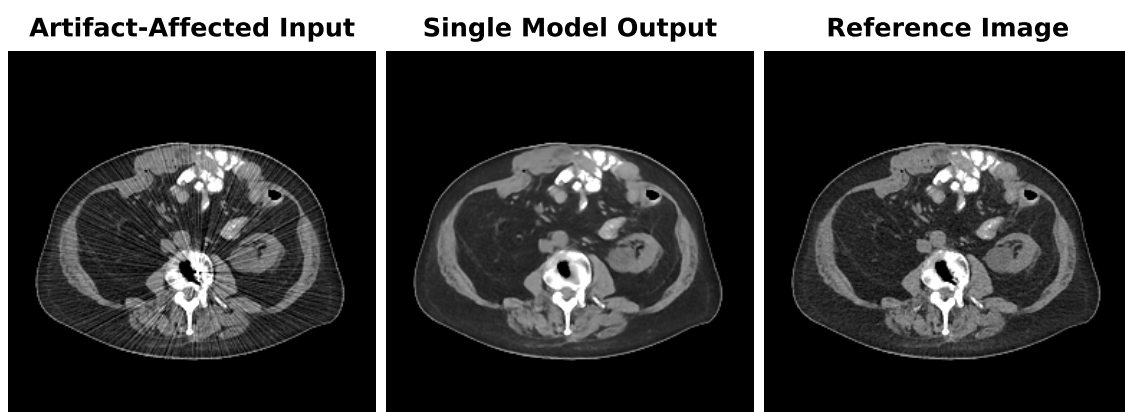
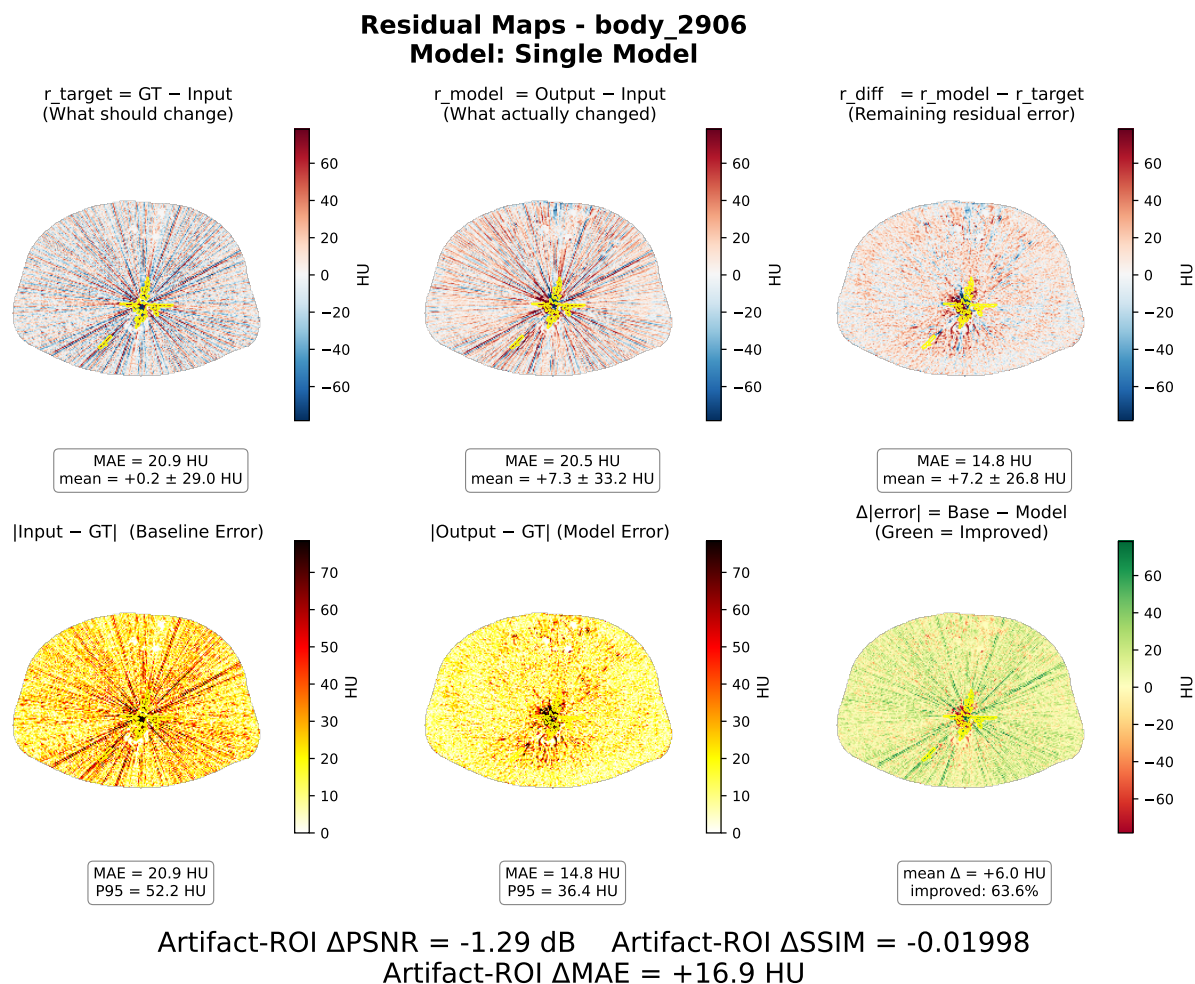


Fig.5.19 Residual map (top) and reconstruction visualization (bottom) for case body_2906 (test-set index 1122), selected among the three lowest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = -0.0200$; global $\Delta\text{SSIM} = +0.0740$; $\Delta\text{PSNR} = +0.38$ dB). All error maps are restricted to the body mask.

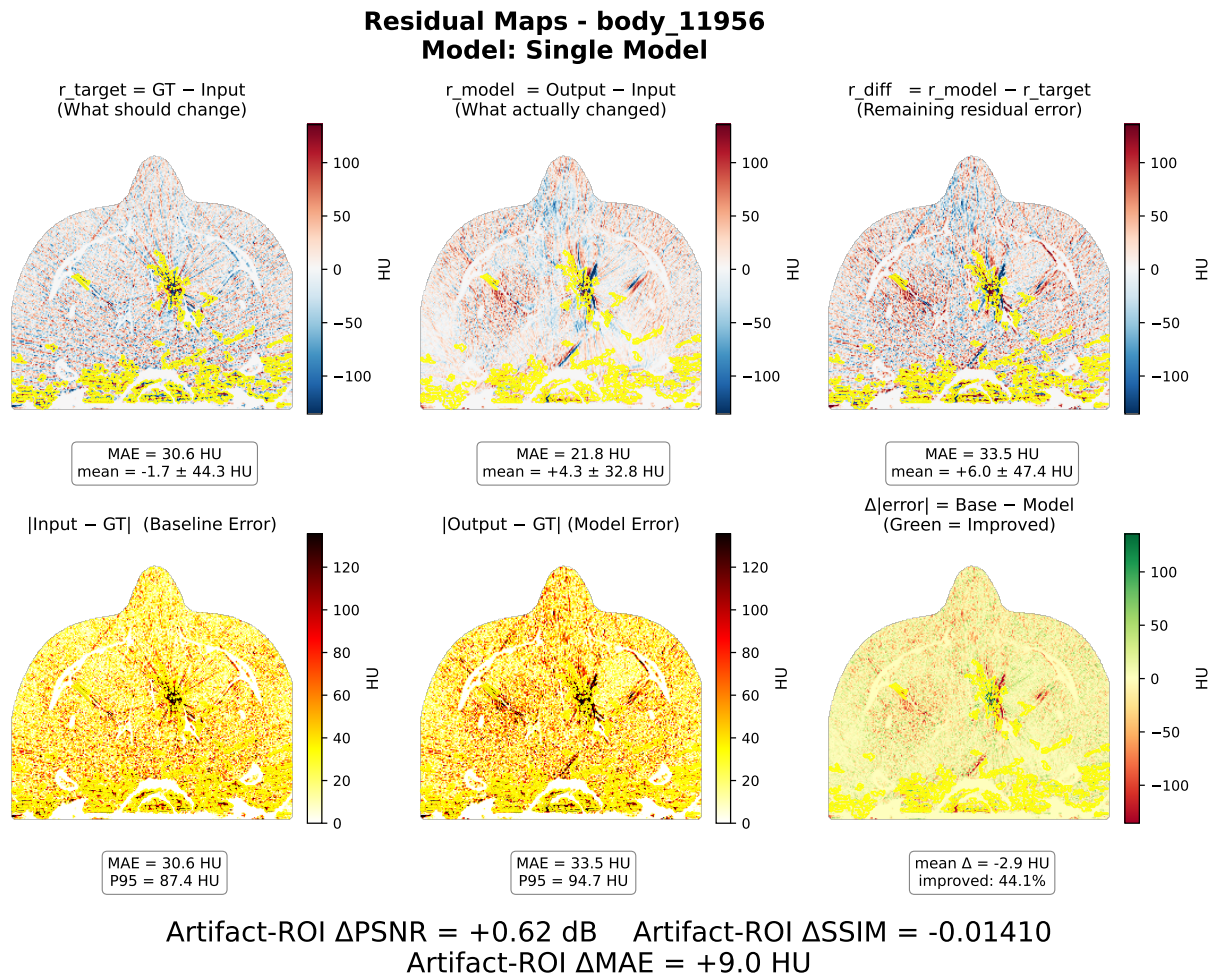


Fig.5.20 Residual map (top) and reconstruction visualization (bottom) for case body_11956 (test-set index 0723), selected among the three lowest artifact-ROI $\Delta SSIM$ values ($\Delta SSIM_{ROI} = -0.0141$; global $\Delta SSIM = -0.0345$; $\Delta PSNR = -0.64$ dB). All error maps are restricted to the body mask.

5.4 Paired Comparison Single x Dual Model

5.4.1 Statistical Analysis per ROI

This section formally tests this study’s main hypothesis, that structure improves in the Artifact Zone, and describes preservation in the Clean Zone as a complement. As before, per-slice metrics were always averaged per patient before any summary or test ($n = 41$), and 95% confidence intervals were obtained by resampling patients 10,000 times. In the Artifact Zone, the question is simply whether the model improved on the input, so the signed changes in PSNR, SSIM, and MAE were analyzed. For each model, the sign test was applied to each of the three metrics, with a joint Holm correction across them.

Artifact ROI: SSIM

Model	Mean improvement	95% BCa CI	p (Holm)
Single Model	0.3279	[0.2963; 0.3548]	2.73×10^{-12}
Dual Model	0.3266	[0.2947; 0.3534]	2.73×10^{-12}

Patient-level Artifact-ROI improvement ($n = 41$). Two-sided exact binomial sign-test p -values, Holm-adjusted across PSNR, SSIM, and MAE separately for each model.

Table 5.17 SSIM improvement over the corrupted input in the Artifact ROI.

Artifact ROI: PSNR

Model	Mean improvement (dB)	95% BCa CI	p (Holm)
Single Model	11.86	[10.90; 12.59]	2.73×10^{-12}
Dual Model	11.84	[10.91; 12.55]	2.73×10^{-12}

Patient-level Artifact-ROI improvement ($n = 41$). Two-sided exact binomial sign-test p -values, Holm-adjusted across PSNR, SSIM, and MAE separately for each model.

Table 5.18 PSNR improvement over the corrupted input in the Artifact ROI.

Artifact ROI: MAE

Model	Mean reduction (HU)	95% BCa CI	p (Holm)
Single Model	98.38	[91.75; 104.24]	2.73×10^{-12}
Dual Model	98.33	[91.84; 104.06]	2.73×10^{-12}

Patient-level Artifact-ROI improvement ($n = 41$). Positive Δ MAE denotes error reduction. Two-sided exact binomial sign-test p -values, Holm-adjusted across PSNR, SSIM, and MAE separately for each model.

Table 5.19 MAE reduction over the corrupted input in the Artifact ROI.

Table 5.17, Table 5.18, and Table 5.19 show improvement in the three metrics in the Artifact Zone for both models. The mean gains were approximately 0.32 for SSIM, 11.8 dB for PSNR, and 98 HU for MAE. These results confirm, for both architectures, this study’s primary hypothesis of statistically significant structural improvement in the Artifact Zone.

Metric	$\Delta_{\text{Dual}} - \Delta_{\text{Single}}$	95% BCa CI	p (Holm)
ΔSSIM	-0.0013	[-0.0025; 0.0002]	0.0345
$\Delta\text{PSNR (dB)}$	-0.019	[-0.085; 0.060]	0.533
$\Delta\text{MAE (HU)}$	-0.049	[-0.318; 0.345]	0.422

Patient-level paired comparison of input-to-output improvement deltas ($n = 41$). Differences are $\Delta_{\text{Dual}} - \Delta_{\text{Single}}$. Negative differences favor the Single Model for all metrics. The 95% BCa bootstrap CIs use 10 000 resamples. Two-sided exact binomial sign-test p -values are Holm-adjusted across the three Artifact-ROI metrics.

Table 5.20 Patient-level paired comparison of improvement deltas in the Artifact ROI.

In the direct comparison of improvement deltas in the Artifact Zone (Table 5.20), the difference in SSIM was small but remained significant after the Holm correction ($p_{\text{Holm}} = 0.0345$), favoring the Single Model. The differences in the PSNR and MAE deltas were not significant after adjustment.

In the Clean Zone, the question is different: the goal is to see how well the tissue away from the artifact was left unchanged, since ideally there should be no change at all relative to the input. So the average per-slice $|\Delta|$ was calculated for each patient, where lower values mean the reconstruction stayed closer to the input. Because no specific “close enough to zero” limit was set in advance, it is not possible to formally test or conclude that the change is equivalent to zero. Thus, Table 5.21, Table 5.22, Table 5.23 report the size of the change and its confidence intervals, without a p -value against zero.

Clean ROI: SSIM

Model	Mean $ \Delta $	95% BCa CI
Single Model	0.0599	[0.0503; 0.0754]
Dual Model	0.0595	[0.0499; 0.0748]

Patient-level mean absolute per-slice input-to-output change ($n = 41$). Lower values are closer to zero. No equivalence margin or hypothesis test against zero was prespecified.

Table 5.21 Magnitude of SSIM input-to-output change in the Clean ROI.

Clean ROI: PSNR

Model	Mean $ \Delta $ (dB)	95% BCa CI
Single Model	1.99	[1.66; 2.34]
Dual Model	1.97	[1.64; 2.33]

Patient-level mean absolute per-slice input-to-output change ($n = 41$). Lower values are closer to zero. No equivalence margin or hypothesis test against zero was prespecified.

Table 5.22 Magnitude of PSNR input-to-output change in the Clean ROI.

Clean ROI: MAE

Model	Mean $ \Delta $ (HU)	95% BCa CI
Single Model	4.30	[3.60; 5.16]
Dual Model	4.24	[3.54; 5.08]

Patient-level mean absolute per-slice input-to-output change ($n = 41$). Lower values are closer to zero. No equivalence margin or hypothesis test against zero was prespecified.

Table 5.23 Magnitude of MAE input-to-output change in the Clean ROI.

To compare the architectures on preservation in the Clean Zone, the Dual Model's advantage was defined for each patient as $|\Delta|_{\text{Single}} - |\Delta|_{\text{Dual}}$, and the sign test was applied, with a joint Holm correction across the three metrics. This is a separate comparison from the Artifact-Zone improvement above: here, what matters is the size of the change, and the better value is the one closest to zero. Positive values indicate a smaller change introduced by the Dual Model. This test answers only which model introduced a smaller change, not whether that change is clinically negligible.

Metric	Dual preservation advantage	95% BCa CI	p (Holm)
SSIM	0.000390	[0.000056; 0.000642]	3.37×10^{-4}
PSNR (dB)	0.020	[-0.104; 0.069]	0.0275
MAE (HU)	0.066	[-0.050; 0.155]	0.00865

Patient-level paired preservation advantage ($n = 41$), defined as $|\Delta|_{\text{Single}} - |\Delta|_{\text{Dual}}$. Positive values favor the Dual Model because its mean absolute input-to-output change is smaller. The 95% BCa bootstrap CIs use 10 000 resamples, two-sided exact binomial sign-test p -values are Holm-adjusted across the three Clean-ROI metrics.

Table 5.24 Relative preservation advantage of the Dual Model in the Clean ROI.

In the relative comparison (Table 5.24), the Dual Model showed a smaller absolute change in PSNR, SSIM, and MAE after the Holm correction. This suggests the Dual Model preserved the clean tissue somewhat better than the Single Model, but it does not prove the change was equivalent to none at all.

The computational performance (inference latency and throughput) of the model selected for external evaluation is characterized in section A.3.

5.4.2 Global Statistical Analysis

As in the ROI analysis, global per-slice metrics were averaged per patient before any summary or statistical test, giving $n = 41$ independent patients. So the means, 95% confidence intervals, and p -values in this section's tables all refer to per-patient averages, with each patient counted equally.

The tables in this section analyze the global metrics, calculated over all pixels of the body mask of each slice (background excluded), as a complementary confirmation of the main hypothesis of structural improvement.

Checking the distribution of the per-patient values for Δ SSIM and Δ MAE showed that they were skewed, not evenly spread around the middle. Because of this, the main evidence for these two metrics came from the sign test, which simply asks how often the model did better than the input, without assuming the improvements are symmetrically distributed. For Δ PSNR, whose per-patient values showed no such skew, the two-sided one-sample t -test was used instead, to test the average improvement directly.

This distribution check was done separately for each model before applying the same rule to both. For Δ SSIM, the Shapiro–Wilk test showed the Single Model’s values were not evenly distributed ($W = 0.9158$, $p = 0.0050$, skewness = 1.087). The same was true for the Dual Model ($W = 0.9162$, $p = 0.0052$, skewness = 1.056), supporting the use of the sign test for both architectures. For Δ MAE, the same pattern was found in the Single Model ($W = 0.7607$, $p = 8.95 \times 10^{-7}$, skewness = 2.281) and in the Dual Model ($W = 0.7686$, $p = 1.27 \times 10^{-6}$, skewness = 2.200). For Δ PSNR, neither model’s values showed this kind of unevenness (Single Model: $W = 0.9742$, $p = 0.4675$; Dual Model: $W = 0.9734$, $p = 0.4427$), which supported using the t -test for the average improvement in both models.

Reconstruction Quality: SSIM

Model	Mean improvement	95% BCa CI	p (Holm)
Single Model	0.0891	[0.0744; 0.1075]	7.64×10^{-11}
Dual Model	0.0879	[0.0731; 0.1062]	7.64×10^{-11}

Global body-mask-aware metric, aggregated at the patient level ($n = 41$). No ROI restriction. Two-sided exact binomial sign-test p -values, Holm-adjusted separately within each model across SSIM, PSNR, and MAE.

Table 5.25 Global SSIM improvement over the corrupted input on the test set.

Reconstruction Quality: PSNR

Model	Mean improvement (dB)	95% t CI	p (Holm)
Single Model	7.11	[6.35; 7.87]	4.53×10^{-21}
Dual Model	7.06	[6.30; 7.81]	4.42×10^{-21}

Global body-mask-aware metric, aggregated at the patient level ($n = 41$). No ROI restriction. Two-sided one-sample t -test p -values, Holm-adjusted separately within each model across SSIM, PSNR, and MAE.

Table 5.26 Global PSNR improvement over the corrupted input on the test set.

Reconstruction Quality: MAE			
Model	Mean reduction (HU)	95% BCa CI	p (Holm)
Single Model	15.72	[12.94;20.14]	7.64×10^{-11}
Dual Model	15.54	[12.79;19.88]	7.64×10^{-11}

Global body-mask-aware metric, aggregated at the patient level ($n = 41$). No ROI restriction. Positive Δ MAE denotes error reduction. Two-sided exact binomial sign-test p -values, Holm-adjusted separately within each model across SSIM, PSNR, and MAE.

Table 5.27 Global MAE reduction over the corrupted input on the test set.

Table 5.25, Table 5.26, and Table 5.27 show that both architectures substantially improved the global reconstruction quality relative to the corrupted input. The Single Model showed a mean Δ SSIM of 0.0891 (95% BCa CI: [0.0744;0.1075]), a mean Δ PSNR of 7.11 dB (95% t CI: [6.35;7.87]), and a mean MAE reduction of 15.72 HU (95% BCa CI: [12.94;20.14]). The Dual Model showed a Δ SSIM of 0.0879 (95% BCa CI: [0.0731;0.1062]), a Δ PSNR of 7.06 dB (95% t CI: [6.30;7.81]), and a mean MAE reduction of 15.54 HU (95% BCa CI: [12.79;19.88]). All six tests of global improvement over the input remained significant after the Holm adjustment applied separately within each architecture. The p -values for Δ SSIM and Δ MAE measure how consistently the model did better than the input from patient to patient, while those for Δ PSNR measure the average improvement. So their sizes should not be compared directly.

These tests confirm that each architecture improves upon the corrupted input, but they do not constitute a direct comparison between architectures. This distinction is essential: small p -values in Table 5.25, Table 5.26, Table 5.27 do not, by themselves, allow one to claim that one architecture is superior to the other.

The direct comparison between architectures is presented in Table 5.28. For this analysis, the improvement difference (Dual Model minus Single Model) was calculated for each patient, for Δ SSIM, Δ PSNR, and Δ MAE, and the sign test was applied to each, with a joint Holm correction across the three tests. This asks whether one architecture wins in more patients than would be expected by chance, without assuming the differences are symmetrically distributed. The 95% confidence intervals, obtained by resampling patients 10,000 times, describe the size of the average difference, while the p -values describe how consistently one architecture won. For all metrics, negative values favor the Single Model.

Dual Model – Single Model: Improvement Deltas

Metric	Mean Difference	95% BCa CI	p (sign test; Holm)
Δ SSIM	−0.0012	[−0.0015; −0.0010]	2.35×10^{-9}
Δ PSNR (dB)	−0.054	[−0.107; −0.017]	2.75×10^{-2}
Δ MAE (HU)	−0.186	[−0.338; −0.091]	8.62×10^{-4}

Differences of patient-level input-to-output improvement deltas (Dual Model – Single Model), with 95% BCa bootstrap CIs (10 000 resamples). Two-sided exact binomial sign-test p -values are Holm-adjusted across PSNR, SSIM, and MAE. Negative differences favor the Single Model for all three metrics.

Table 5.28 Patient-level paired comparison of improvement deltas between architectures ($n = 41$).

The Single Model showed a mean advantage of 0.054 dB in Δ PSNR, 0.0012 in Δ SSIM, and 0.186 HU in Δ MAE. The confidence intervals did not include zero, and the adjusted p -values were 2.75×10^{-2} for PSNR, 2.35×10^{-9} for SSIM, and 8.62×10^{-4} for MAE. The differences favored the Single Model in 28 of the 41 patients for PSNR, in 39 for SSIM, and in 32 for MAE. Although these differences are statistically significant, they are small in absolute terms. So the results point to a modest overall numerical advantage for the Single Model, but do not, on their own, show that it is clinically better. The bar chart summarizes the similar global improvement obtained by both architectures.

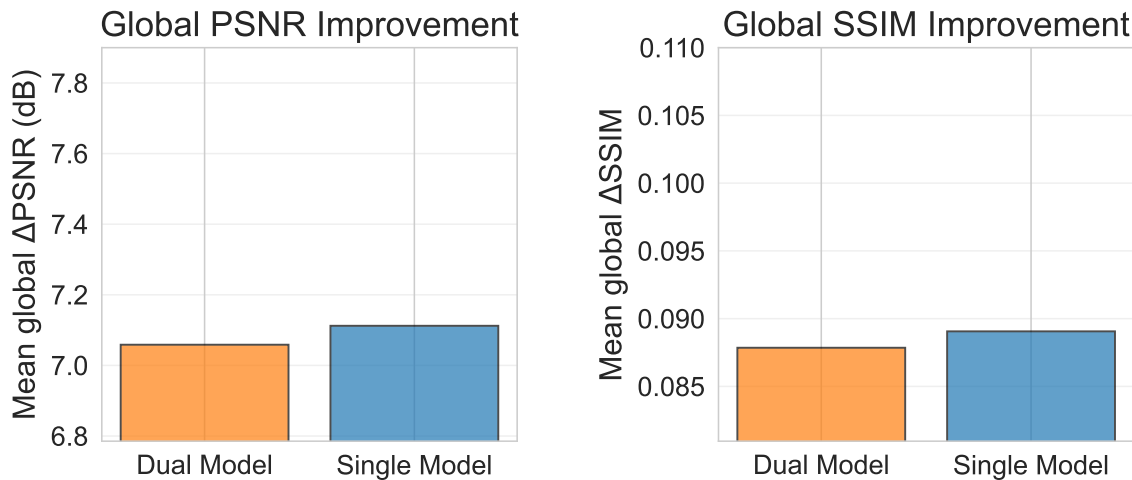


Fig.5.21 Global PSNR and SSIM improvement over the corrupted input for the Single and Dual Models on the test set.

5.5 External and Physician's Evaluation

5.5.1 Capture Status and Participants

The external evaluation consists of a paired comparison blinded to the processing label: in each case, readers simultaneously view the artifact input and the model output, identified only as “Image A” and “Image B”. The left/right position is counterbalanced across readers and kept constant between a case and its hidden repeat. Each session presents 19 comparisons: 2

training cases (excluded from the analysis), 15 main cases, and 2 hidden repeats of previously presented cases, used exclusively to assess intra-reader consistency. Physicians were recruited directly by the researcher, with self-declaration of clinical CT use with a minimum monthly frequency. No name, contact information, professional registration number, or proof of qualification was collected.

Table 5.29 summarizes the status of recruitment and response completion.

Indicator	Value
Total Readers	5
Total valid responses (readers $\times$ 15 main cases)	75
Data collection start date	27-08-2026
Data collection closing date	10-09-2026

Table 5.29 Status of recruitment and response completion for the external evaluation.

The self-reported characteristics of the readers who completed the evaluation are presented in Table 5.30. None of these variables allow individual identification of a participant.

Variable	Category	N	%
Jurisdiction	Brazil	5	100
CT usage frequency	Monthly	4	80
	Daily	1	20
CT experience (years)	< 2	2	40
	2–5	1	20
	6–10	1	20
	> 10	1	20
Experience with artifact reduction	None	4	80
	Occasional	1	20
Monitor type	Conventional	3	60
	Unknown	2	40

Table 5.30 Self-reported characteristics of the readers who completed the external evaluation.

The distribution by medical specialty is presented separately in Table 5.31, given the larger number of categories.

Specialty	N	%
Radiology and Diagnostic Imaging	1	20
Internal Medicine or General Medicine	1	20
Orthopedics or Traumatology	1	20
Infectious Diseases	1	20
Other medical specialty	1	20

Table 5.31 Distribution by medical specialty of the readers who completed the external evaluation.

5.5.2 Quantitative Characterization of Input–Output

Before the professional evaluation, the global change produced by the *Single Model* was characterized in the 15 main cases of the external set. The two training cases were excluded from this summary. Since these real exams do not have a clean reference, the metrics in Table 5.32 compare exclusively the input image with the model output, within the body mask.

Input–Output Similarity and Change		
Metric	Mean $\pm$ SD	95% CI
SSIM (Input, Output)	0.8953 $\pm$ 0.0729	[0.8572; 0.9290]
PSNR (Input, Output) (dB)	33.14 $\pm$ 6.49	[29.95; 36.21]
MAE (Input, Output) (HU)	21.12 $\pm$ 15.26	[14.54; 29.26]

No artifact-free reference is available. Metrics were calculated within the manually corrected effective body mask and quantify input–output similarity or change; they do not measure artifact-removal accuracy or improvement.

Table 5.32 Global input–output metrics in the frozen external main set ($n = 15$) for `singlehead-033-0.9511-ssim_hu.keras`. Values are mean $\pm$ standard deviation; 95% CIs are percentile bootstrap intervals for the mean.

In the external set, the mean input–output similarity was 0.8953 for SSIM and 33.14 dB for PSNR, while the mean magnitude of change was 21.12 HU in MAE. The dispersion between cases, expressed by the standard deviations and confidence intervals in the table, evidences heterogeneity in the degree of change produced by the reconstruction. These results describe the preservation and overall modification of the external images, but do not allow inference, without a clean reference, of artifact-removal accuracy or clinical benefit.

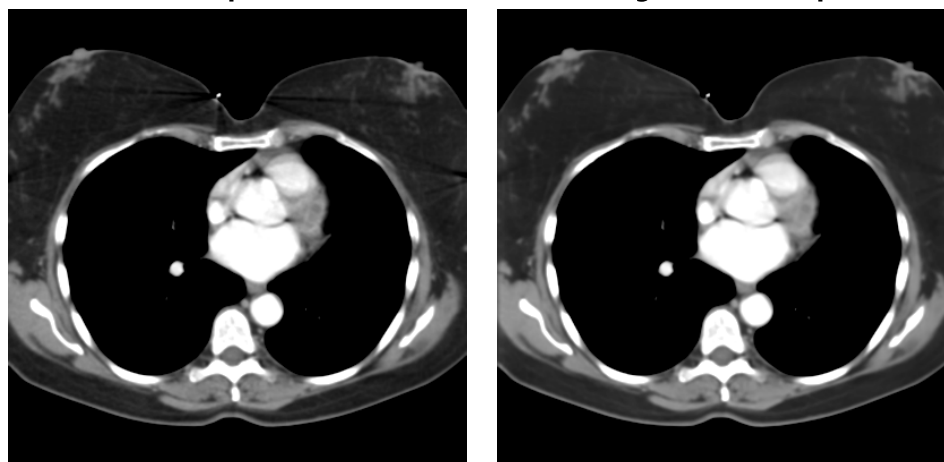
To make this heterogeneity visible, the three cases with the highest and the three with the lowest input–output SSIM were selected descriptively. The former represent the pairs with the smallest overall structural change, while the latter represent those with the greatest change.

Illustrative External Cases Selected by Input–Output SSIM				
Group	Case	SSIM	PSNR (dB)	MAE (HU)
Least changed	Case 003	0.9809	41.19	8.76
	Case 002	0.9756	42.35	8.10
	Case 012	0.9586	37.91	9.64
Most changed	Case 015	0.7278	23.32	43.10
	Case 005	0.7770	20.77	63.90
	Case 008	0.8334	33.53	20.39

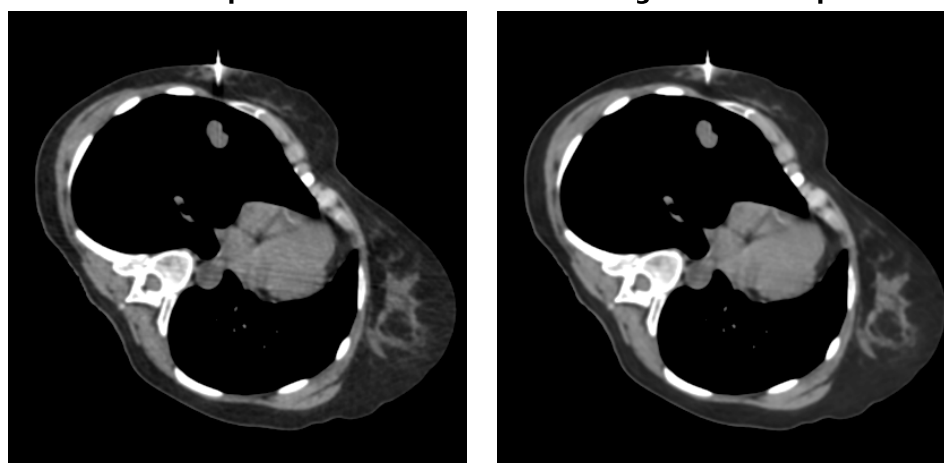
Cases were selected only by the extremes of SSIM(Input, Output). High SSIM denotes less structural change between input and output; this ranking does not define artifact-removal quality or clinical acceptability.

Table 5.33 Descriptive external cases with the highest and lowest input–output SSIM values.

Case 003 — Input-Output Pair
SSIM=0.9809; PSNR=41.19 dB; MAE=8.76 HU



Case 002 — Input-Output Pair
SSIM=0.9756; PSNR=42.35 dB; MAE=8.10 HU



Case 012 — Input-Output Pair
SSIM=0.9586; PSNR=37.91 dB; MAE=9.64 HU

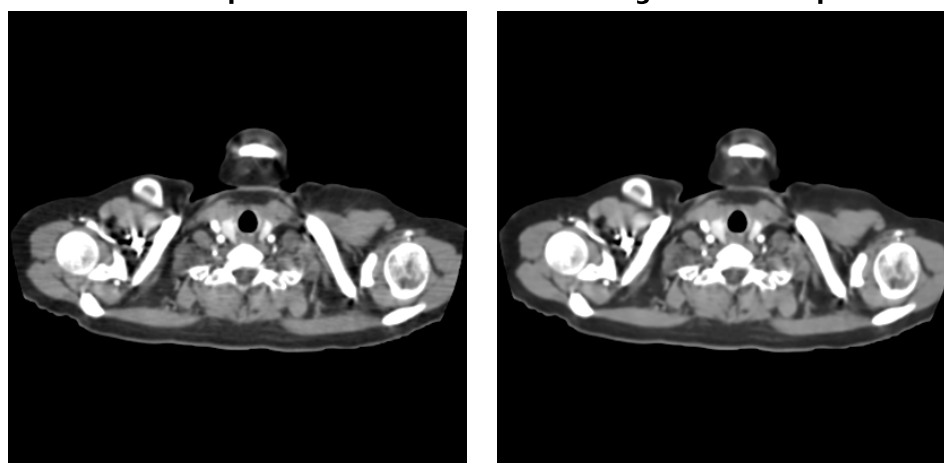
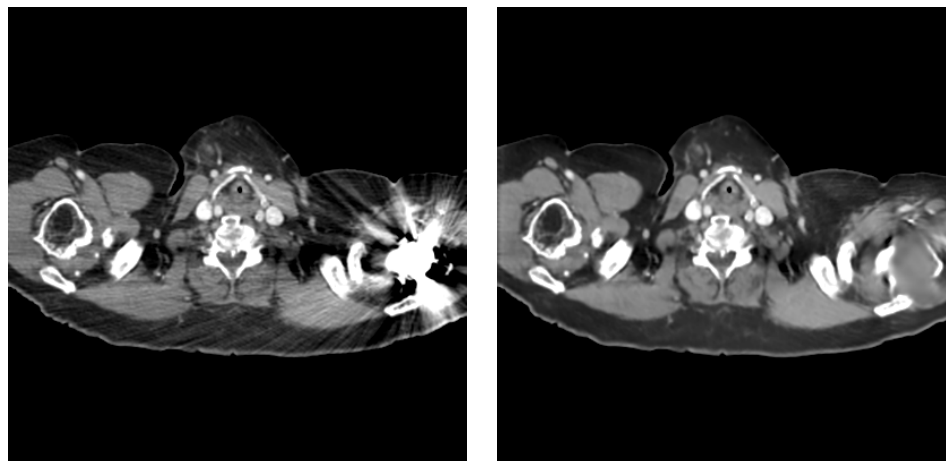
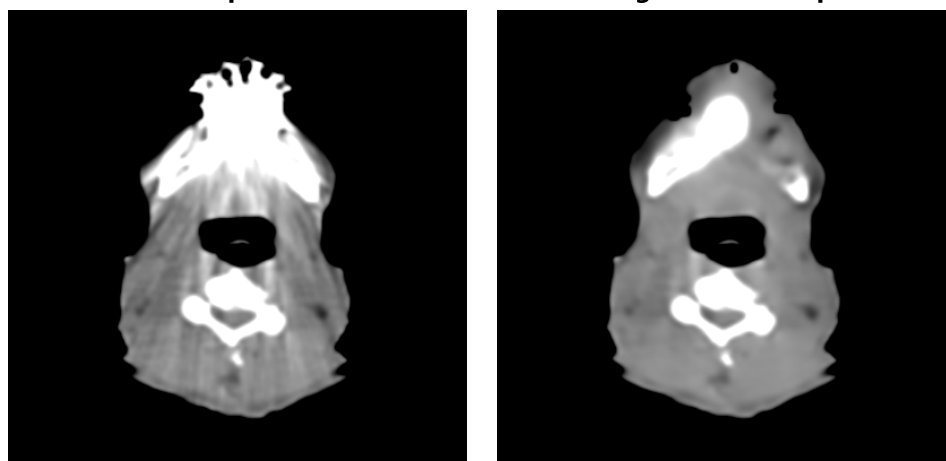


Fig.5.22 External input-output pairs with the three highest SSIM values, representing the least structural change. The soft-tissue window is identical for input and output within each case.

Case 015 — Input-Output Pair
SSIM=0.7278; PSNR=23.32 dB; MAE=43.10 HU



Case 005 — Input-Output Pair
SSIM=0.7770; PSNR=20.77 dB; MAE=63.90 HU



Case 008 — Input-Output Pair
SSIM=0.8334; PSNR=33.53 dB; MAE=20.39 HU
Input **Single Model output**

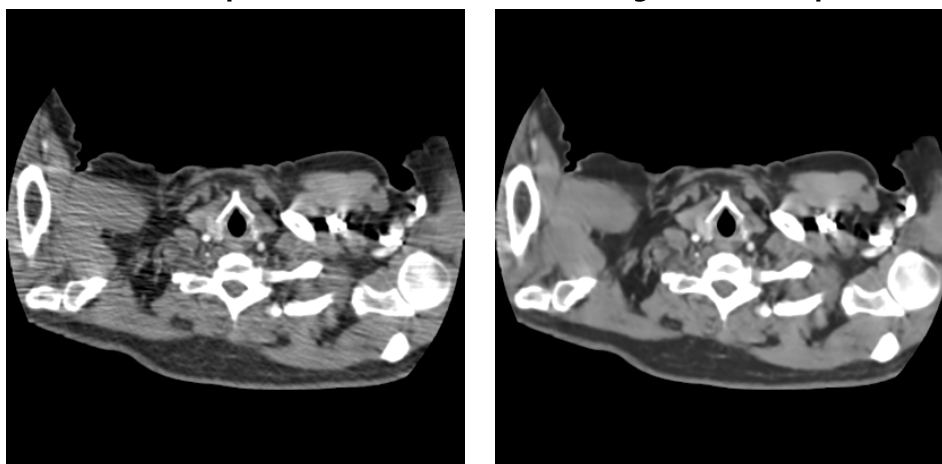


Fig.5.23 External input-output pairs with the three lowest SSIM values, representing the greatest structural change. The soft-tissue window is identical for input and output within each case.

5.5.3 Primary Outcome: Anatomical Visualization

The primary outcome is the paired preference, per case, regarding which image better preserves the anatomical region underlying the metal artifact (“Image A” / “Image B” / “No difference”). Answers are converted back to input/output using the left/right position recorded for each case, without the reader ever knowing this mapping while answering. Training cases and hidden repeats are not part of this analysis ($n = 15$ main cases per reader).

Because each doctor judges many cases, and each case is judged by many doctors, the 95% confidence intervals were obtained with a two-way cluster bootstrap that resampled readers and cases independently. This approach avoids treating the 75 responses as fully independent. The observed preference favored the Single Model in 42/75 presentations (56.0%), whereas the artifact input was preferred in 21/75 (28.0%). The remaining 12/75 presentations (16.0%) were judged equivalent.

Response	N	%	95% CI
Preferred output (Single Model)	42	56.0	[28.0; 82.7]
Preferred input (with artifact)	21	28.0	[10.7; 48.0]
No difference	12	16.0	[1.3; 37.3]

Pooled frequencies over 75 valid responses (5 readers $\times$ 15 main cases). 95% CIs are percentile intervals from a two-way cluster bootstrap over readers and cases (10,000 resamples; seed 2024).

Table 5.34 Paired preference regarding preservation of the anatomical region underlying the metal artifact.

Contrast	Difference (pp)	95% CI
Output preference – input preference	+28.0 pp	[−17.3; +70.7] pp

Difference between the two pooled proportions. 95% CI by the same two-way cluster bootstrap. A positive value favors the Single Model. No null-hypothesis test is reported because this small reader panel is interpreted descriptively.

Table 5.35 Cluster-bootstrap contrast for the primary outcome.

5.5.4 Secondary Outcomes and Safety Perception

The secondary outcomes use the same per-case format (“Image A” / “Image B” / “No difference”). They are reported descriptively as pooled frequencies with two-way cluster-bootstrap 95% confidence intervals. No correction for multiple comparisons was applied.

Response	N	%	95% CI
Preferred output (lower artifact intensity)	61	81.3	[65.3; 94.7]
Preferred input	8	10.7	[1.3; 22.7]
No difference	6	8.0	[0.0; 22.7]

$n = 75$ responses. 95% CIs by two-way cluster bootstrap over readers and cases (10,000 resamples; seed 2024).

Table 5.36 Paired preference regarding the perceived intensity of metal artifacts.

Response	N	%	95% CI
Preferred output (less distortion)	47	62.7	[38.7; 84.0]
Preferred input	17	22.7	[9.3; 38.7]
No difference	11	14.7	[1.3; 33.3]

$n = 75$ responses. 95% CIs by two-way cluster bootstrap over readers and cases (10,000 resamples; seed 2024). This item functions as an exploratory signal of possible harm: preference for the input suggests perceived distortions or artifacts introduced by the model outside the artifact zone.

Table 5.37 Paired preference regarding artifacts or anatomical distortions introduced in regions distant from the artifact.

Response	N	%	95% CI
Preferred output for clinical interpretation	42	56.0	[28.0; 82.7]
Preferred input	23	30.7	[12.0; 53.3]
No difference	10	13.3	[1.3; 32.0]

$n = 75$ responses. 95% CIs by two-way cluster bootstrap over readers and cases (10,000 resamples; seed 2024).

Table 5.38 Paired preference regarding the image chosen for clinical interpretation.

Additionally, at the end of the session, each reader answered once a 5-point ordinal question about the complementary usefulness of a version processed with this pattern relative to the original CT exam (Table 5.39).

Category	N	%
Definitely not	0	0
Probably not	1	20
Uncertain	1	20
Probably yes	2	40
Definitely yes	1	20

One response per reader (not per case). $n = 5$ completed readers.

Table 5.39 Perceived complementary usefulness of a processed version relative to the original CT exam.

Intra-reader consistency was assessed through two hidden repeats per reader: the same case is re-presented later in the session, without signaling, in the same left/right position as the first presentation. Because the responses are nominal and only 10 original–repeat pairs are available, the result is reported as exact agreement for each item (Table 5.40).

Outcome	Exact agreement	Total Repetitions
Artifact intensity	5/10 (50%)	10
Anatomical visualization	4/10 (40%)	10
Other-region distortion	4/10 (40%)	10
Clinical preference	5/10 (50%)	10

Two cases were repeated for each of the five readers, yielding 10 paired original–repeat responses per outcome.

Table 5.40 Intra-reader exact agreement in the hidden repeats.

5.5.5 Limited Interpretation to the Observational Design

In this completed panel, the preference frequencies describe how the five readers judged these 15 cases. They do not establish a causal effect on diagnostic or therapeutic outcomes and cannot be generalized beyond this reader–case sample. The absence of a clean reference in these external cases means that the assessment of anatomical accuracy depends entirely on readers' judgment, rather than on an objective error metric. The wide cluster-bootstrap intervals, the small number of readers, and the limited repeat agreement further constrain precision and preclude meaningful subgroup analyses by specialty or experience.

Chapter 6

Discussion

6.1 Summary of the Main Findings

The internal evaluation results support the primary hypothesis of structural improvement in the Artifact Zone relative to the corrupted input: across the six runs (two architectures, three seeds), **SSIM** in this zone increased by 0.322 to 0.328 (Table 5.8). For the run selected for external evaluation (Single Model, seed 2024), this improvement was statistically significant at the patient level ($\Delta\text{SSIM} = 0.3279$, 95% BCa CI [0.2963; 0.3548], $p_{\text{Holm}} = 2.73 \times 10^{-12}$, Table 5.17). An equivalent pattern held for the Dual Model ($\Delta\text{SSIM} = 0.3266$, $p_{\text{Holm}} = 2.73 \times 10^{-12}$). This localized structural improvement was accompanied by a consistent and equally significant global improvement: across all six runs, reconstruction increased global **PSNR** from 27.39 dB to 34.95–35.20 dB, global **SSIM** from 0.8361 to 0.9421–0.9442, and reduced global **MAE** from 36.6 HU to 17.3–17.8 HU (Table 5.11, Table 5.12, Table 5.13). At the patient level, the secondary global-improvement tests were significant for all three metrics in every run ($p < 0.001$, Table 5.14, Table 5.15, Table 5.16), with 95% confidence intervals excluding the null value.

Descriptive slice-level inspection (Figure 5.3.3) showed that unfavorable cases were rare: only 0.8% of slices had negative ΔPSNR and 1.6% had negative ΔSSIM , out of 1,478 test slices. This consistency is reinforced by the **ROI** analysis (subsection 5.3.1). It showed that correction was concentrated in the Artifact Zone ($\Delta\text{PSNR} \approx 11.8$ dB, $\Delta\text{SSIM} \approx 0.33$, **MAE** reduction ≈ 98 HU), with substantially smaller changes in the Clean Zone ($|\Delta\text{PSNR}| \approx 2$ dB, $|\Delta\text{SSIM}| \approx 0.06$, $|\Delta\text{MAE}| \approx 4$ HU). This contrast indicates that the model did not apply indiscriminate image smoothing, but rather concentrated changes where artifacts were present.

6.2 Comparison with Prior Work

Direct numerical comparison with previously published metal artifact reduction studies is inherently limited. Reported gains depend on the specific dataset, artifact severity distribution, evaluation region, and metric definition used by each study, and few works share a common benchmark. This reflects a lack of experimental standardization already noted in Chapter 3. With this caveat in mind, the results obtained here are broadly consistent, in direction and order of magnitude, with the improvements reported by image-domain deep learning approaches for CT-MAR. Gjestebj et al. [16] and Zhang and Yu [17] both reported that CNN-based correction outperformed classical sinogram-domain methods such as NMAR, without providing a global PSNR/ SSIM baseline directly comparable to this work. Lin et al.'s DuDoNet [20] reported significant improvements over single-domain approaches using a dual-domain design more complex than the single-domain U-Net variants compared here. Huang et al. [22], using a GAN-based approach evaluated on the DeepLesion dataset, reported an average relative increment of 11.3% in PSNR and 12.1% in SSIM. The corresponding relative increments obtained in this work are of comparable order of magnitude ($\approx 28\%$ in PSNR and $\approx 13\%$ in SSIM, computed from the global means in Table 5.12 and Table 5.11). However, the two studies differ in network architecture, loss function, and the exact definition of the evaluation region, so this comparison should be read as indicative rather than as a controlled benchmark.

The pre-specified primary hypothesis was evaluated specifically in the Artifact-ROI, directly targeting the region where the diagnostic risk of artifacts is concentrated, rather than relying exclusively on global image-quality metrics as most of the reviewed studies do. Finally, in line with the concern raised by Zech et al. [27] about performance degradation under domain shift, the selected model was additionally characterized on an independent external set of real exams from TCIA, a step not undertaken by any of the image-domain MAR studies reviewed in Chapter 3.

6.3 Architectural Comparison Interpretation

The paired comparison between architectures (section 5.4) addressed the second specific objective, whether auxiliary artifact-mask supervision improves reconstruction relative to the version without mask supervision, with a nuanced result. Globally, the Single Model showed a small but statistically significant advantage for all three metrics ($\Delta\text{PSNR} +0.054$ dB, ΔSSIM

+0.0012, ΔMAE +0.186 HU, with $p_{\text{Holm}} = 2.75 \times 10^{-2}$, 2.35×10^{-9} , and 8.62×10^{-4} respectively, Table 5.28). This pattern favored the Single Model in 28, 39, and 32 of the 41 patients, respectively. In the Artifact Zone, this advantage was limited to SSIM ($p_{\text{Holm}} = 0.0345$), with no significant difference in PSNR or MAE (Table 5.20).

Conversely, in the Clean Zone, the Dual Model produced smaller absolute changes from the input for all three metrics, with a statistically significant preservation advantage after Holm correction ($p_{\text{Holm}} = 3.37 \times 10^{-4}$ for SSIM, 0.0275 for PSNR, and 0.00865 for MAE, Table 5.24). This pattern can be interpreted as a trade-off. The additional segmentation supervision appears to spatially constrain correction, making the Dual Model marginally more conservative outside the Artifact Zone. This comes at the cost of a slightly smaller correction magnitude, both globally and within the Artifact Zone itself. However, in this work the decoder was kept almost entirely shared, with the segmentation output added only at the final layer. A deeper separation between reconstruction and segmentation decoders deserves better exploration in future works.

In neither direction do the observed differences appear clinically distinguishable on their own: they are tenths of a dB in PSNR, thousandths in SSIM, and fractions of a HU in MAE. Its statistical significance comes mainly from one architecture winning consistently across patients (sign tests with $n = 41$), not from a large difference in size. Thus, auxiliary mask supervision did not produce decisively superior reconstruction. Instead, it redistributed the correction-preservation trade-off in a measurable way.

6.4 Anatomical Preservation and Reconstruction Safety

Taken together, these additional descriptive analyses point to a consistent picture, though not one that is clinically conclusive on its own. The ROI partition already showed that changes in the Clean Zone were a small fraction of the correction observed in the Artifact Zone (Figure 5.11, Figure 5.12). Patient-level Bland–Altman agreement for mean body HU (Figure 5.14) showed a small but non-zero positive bias of +4.19 HU, with 95% limits of agreement from +1.45 to +6.93 HU. This is a slight systematic shift in mean intensity that should be monitored in future work, although it is small relative to the HU window range used.

As an additional characterization of the lower-performance tail, the 74 of 1,478 test slices (5%) with the lowest ΔPSNR (threshold 2.59 dB) retained positive mean improvement for all three metrics (mean $\Delta\text{PSNR} = 1.23$ dB, mean $\Delta\text{SSIM} = 0.0071$, mean Artifact-Zone $\Delta\text{MAE} = 62.0$ HU, Figure A.12). This held despite substantial dispersion and individual cases of degradation relative to the input. This suggests that unfavorable cases tend to be graded rather

than catastrophic. They nonetheless exist and warrant qualitative inspection, provided for the extreme Artifact-Zone Δ SSIM cases in [subsection 5.3.4](#).

A complementary exploratory analysis ([Figure A.9](#)) related artifact burden (artifact-area/body-area fraction, averaged per patient) to global improvement. It found a positive association: $\rho = 0.479$ for Δ PSNR (95% CI [0.156;0.717], $p = 0.00154$) and $\rho = 0.569$ for Δ SSIM (95% CI [0.225;0.794], $p = 0.000104$). In other words, the model tended to correct proportionally more in patients with a greater artifact burden, a reassuring finding from the perspective of targeted correction. However, this analysis was not one of the tests planned in advance, and it was not corrected for multiple comparisons. It should therefore be read as suggesting a pattern worth testing further, not as independent statistical proof.

Together, ROI stratification, Bland–Altman calibration, characterization of the lower-performance tail, and the association with artifact burden provide a convergent but still descriptive picture. Correction is concentrated where needed, clean tissue is largely preserved with a small calibration shift, and the lower-performing slices do not show systematic anatomical distortion. Definitive assessment of clinical safety cannot rest on these internal, reference-based analyses alone. The completed external reader evaluation ([section 5.5](#)) provides a further, complementary line of evidence, discussed later in this chapter, though it too remains descriptive and based on a small panel.

6.5 Generalization and External Evaluation

The fourth specific objective was to investigate the generalization capacity of the selected version on real data external to training, obtained from [TCIA](#). This objective can only be answered partially with the technical evidence currently available. The 15 main cases in the external set have no clean reference, so only the magnitude of change between input and output could be characterized ([Table 5.32](#)): mean SSIM of 0.8953, mean PSNR of 33.14 dB, and mean MAE of 21.12 HU, with meaningful variability between cases.

These values are not directly comparable with the internal improvement deltas, which require a clean reference to quantify correction accuracy. They only indicate that the model produced changes of plausible magnitude and neither collapsed to the identity nor produced extreme distortions in the sampled external cases. Picking out, purely for illustration, the three cases with the highest and the three with the lowest input-output SSIM shows this variability without being a formal statistical comparison.

Thus, the available external technical evidence supports only that model behavior on real data outside the training domain is qualitatively plausible and consistent in order of magnitude

with that observed internally. It does not establish artifact-removal accuracy or validated generalization without a clean reference. Whether human readers judge the external correction to be accurate and safe is addressed, within the descriptive limits of a small reader panel, by the completed blinded physician evaluation discussed in the next section.

6.6 Reader Evaluation and Clinical Implications

The blinded multi-reader evaluation ([section 5.5](#)) was designed with anatomical visualization as its primary outcome and artifact interference, paired preference, potential-harm flagging, complementary usefulness, and intra-reader consistency as secondary outcomes. This design suits a small panel of readers built specifically to compare input and output side by side.

Data collection was completed with five readers, two below the pre-specified target of seven. This was still at the minimum of five set in advance for the results to be considered useful, yielding 75 valid case-level responses ([Table 5.29](#)). For the primary outcome, anatomical visualization underlying the artifact, readers favored the model output in 56.0% of presentations, versus 28.0% favoring the artifact input and 16.0% judged equivalent. This is a contrast of +28.0 percentage points, whose two-way cluster-bootstrap 95% CI was wide and included zero ($[-17.3; +70.7]$ pp, [Table 5.34](#), [Table 5.35](#)). Because this interval does not exclude the no-difference value, the primary outcome is best read as a descriptive signal favoring the output on average across this small reader-case sample. It is not a statistically confirmed improvement in perceived anatomical preservation.

The secondary outcomes followed a broadly consistent, though not uniform, direction. Readers judged the output to show less visible metal artifact in 81.3% of presentations, against 10.7% favoring the input ([Table 5.36](#)), the largest and most consistent of the paired preferences recorded. For the harm-flagging item, which specifically probes for distortions introduced away from the artifact, the output was still preferred in 62.7% of presentations against 22.7% for the input ([Table 5.37](#)). This is the least consistent of the four items, and it is the one most relevant to a safety concern. The sizable minority of presentations favoring the input here should temper, rather than be absorbed into, the otherwise favorable pattern. Clinical-interpretation preference closely tracked the primary outcome (56.0% output, 30.7% input, [Table 5.38](#)). At the end of the session, three of the five readers considered a processed version “probably” or “definitely” a useful addition to a standard CT exam, one was uncertain, and one considered it probably not useful ([Table 5.39](#)). This was a small and split sample rather than a clear consensus.

Intra-reader exact agreement on the ten hidden-repeat pairs ranged from 40% to 50% across the four per-case questions (Table 5.40). For a three-category nominal item, this is only modestly above the one-third expected by chance. It indicates that individual readers did not always reproduce their own judgment when the same case reappeared, unannounced, later in the session. This limits the precision attributable to any single reader's response. It also reinforces that the pooled percentages above describe a panel-level tendency rather than an individually reliable signal.

Taken together, the completed evaluation offers descriptive evidence from a small and purposely built reader panel judging a purposely curated case set. The model output was preferred over the artifact input more often than not for anatomical preservation, artifact-intensity reduction, and clinical-interpretation preference. This preference was strongest and most consistent for reduced artifact intensity, and least consistent for the harm-flagging item. It does not, however, constitute a confirmatory clinical trial. The primary-outcome contrast interval included zero, the panel fell two readers short of the pre-specified target, intra-reader repeat agreement was only moderate, and no correction for multiple comparisons was applied across the four per-case outcomes. Consistent with the design set out in chapter 4, these results are interpreted here strictly as an observational, paired reader comparison. They support cautious, exploratory optimism about perceived anatomical preservation and complementary usefulness. However, they do not establish, on their own, that either architecture is causally better in clinical terms, nor do they rule out that the model introduces perceptible change in tissue away from the artifact for a relevant minority of cases.

6.7 Limitations

No patient paired metadata. The CT-MAR Dataset provides real slices with simulated metal artifacts, but no paired patient metadata to enable patient level partitioning. The reliability of the patient-level partitioning was strengthened by structural-similarity matching of CT-MAR slices to DeepLesion and UCLH Stroke EIT, audited using MS-SSIM and human visual review. This mitigated the risk of data leakage between training and evaluation. However, the matching is similarity-based retrieval rather than guaranteed pixel-level registration from the same tracked slice. Small residual anatomical differences between matched pairs cannot be fully excluded despite the audit.

Anatomical imbalance. The anatomical distribution was approximately 88% *body* and 12% *head* at the pair level, and 95.9% *body* and only 4.1% *head* (11 of 269 patients) at the

patient level. Conclusions specific to the head region therefore rely on a small patient subset and should be interpreted cautiously.

Patient-level sample size. The internal test set's statistical comparisons treated the patient as the unit of analysis ($n = 41$), which was enough for the main paired comparisons performed, but limited for more detailed subgroup analyses (by anatomical region, artifact severity, or implant type) beyond the exploratory association with artifact burden already identified as non-primary.

External evaluation. The external set is small and purposefully curated (15 main cases), has no clean reference, and is restricted to one model and seed selected after the internal test comparison of the validation-frozen candidates. The results describe change magnitude rather than correction accuracy.

Loss-function and fine-detail preservation. The reconstruction objective combined a pixel-wise L_1 term with local SSIM. Although this objective improved agreement with the clean internal reference, it did not explicitly optimize the preservation of noise texture, high-frequency detail, lesion conspicuity, or any task-specific diagnostic endpoint. One reader noted in an optional final comment that some processed images appeared blurred in a manner inconsistent with the density of the tissue under assessment and could be misleading if the processing-related origin of the alteration was not recognized. This single anonymous free-text observation is included as an illustrative qualitative concern, not as a quantified outcome or evidence that the loss function caused the effect.

Risk of misleading synthetic change. Neither the internal nor the external evaluation included lesion-level labels, microcalcification annotations, pathology reference, or a lesion-detection task. The reference-based internal analysis therefore quantified image fidelity but could not establish preservation of subtle pathological findings, while the external cases lacked a clean reference altogether. Consequently, this study cannot rule out the removal of subtle findings or the creation of anatomically plausible but misleading features by the network. Future safety validation should use lesion-annotated or pathology-linked data and a pre-specified diagnostic detection task.

Small external reader panel. The multi-reader evaluation compared input and output side by side and was completed with five readers and 75 case-level responses. The primary-outcome confidence interval was consequently wide and included the no-difference value. Intra-reader repeat agreement was only moderate (40–50%), and no multiplicity correction was applied across the four secondary outcomes, as discussed in the preceding section.

6.8 Implications and Future Work

The most immediate practical implications follow directly from the identified limitations. First, now that the label-blinded physician evaluation has been completed, any further claim of perceived clinical benefit or reconstruction safety should rest on replication with a larger, ideally multi-center multi-reader, multi-case panel. Such a panel would allow the primary-outcome confidence interval to exclude the no-difference value with adequate precision. The panel-level tendencies reported here are a basis for that replication, not a substitute for it. A task-specific diagnostic study should distinguish subjective image-quality outcomes, which may use ordinal rating scales, from lesion-detection or localization outcomes, for which [ROC](#) or [FROC](#) analysis requires an independent reference standard. Second, prospective validation in larger and more diverse external patient groups would test generalization and safety more robustly than the 15 external cases currently available. Such validation should give greater representation to the head region and to the harm-flagging scenario, where reader preference was least consistent.

From an architectural perspective, the identified trade-off between the Single Model (a slight advantage in correction magnitude) and the Dual Model (a slight preservation advantage in the Clean Zone) suggests that future work could explore ways to combine both effects. One option is adaptive weighting of reconstruction loss by the predicted artifact mask, rather than independent auxiliary supervision, to seek greater correction and greater preservation simultaneously. Testing progressively deeper separation of the reconstruction and segmentation decoders could also clarify its impact on correction and preservation.

Projection–image dual-domain MAR is a distinct future direction. The present pipeline begins with reconstructed image tensors and contains no raw projection data, sinograms, scanner geometry, or explicit physics-based data-consistency layer. A future dual-domain study would therefore require an acquisition-level dataset, a validated projection model, and a separate experimental protocol. It should not be presented as a simple modification of the current image-domain network. Such a design could combine projection-domain consistency with image-domain refinement, as in DuDoNet [\[20\]](#).

Transformer- and diffusion-based architectures could likewise be evaluated as alternative reconstruction backbones under the same patient-level split, reference-based metrics, and safety-oriented reader endpoints used here. Their ability to model longer-range context or richer image priors is a hypothesis to test, rather than evidence that they would necessarily preserve texture better than the present [U-Net](#). In particular, any generative model must be

assessed for preservation of subtle anatomy and for misleading synthetic detail, not only for improved perceptual appearance or global image-similarity metrics.

Finally, expansion of the observer protocol (more readers, calibration or prior-training rounds, and a larger external-case panel) should accompany any future extension of the clinical evaluation, while keeping the same care taken throughout this dissertation to separate exploratory findings from confirmed ones.

Chapter 7

Conclusions

7.1 Study Conclusions

This dissertation set out to develop, evaluate, and rigorously compare a [CNN](#)-based pipeline for metal artifact reduction in [CT](#), prioritizing anatomical preservation, robustness to domain shift, and reproducibility. The pre-specified primary hypothesis was that reconstruction produces a real, patient-level improvement in structural similarity within the Artifact Zone, while leaving the Clean Zone comparatively unchanged. This hypothesis was confirmed on the internal test partition for both the Single Model and the Dual Model, with correction concentrated where artifacts were present rather than distributed indiscriminately across the image. The pre-specified secondary hypotheses of global [PSNR](#), [SSIM](#), and [MAE](#) improvement over the corrupted input were likewise confirmed for both architectures. The direct architectural comparison did not identify a decisively superior design. The Single Model showed a small but consistent advantage in correction magnitude, globally and in the Artifact Zone, while the Dual Model showed a small but consistent Clean-Zone preservation advantage. This is a correction-preservation trade-off, not evidence that auxiliary mask supervision helps or hurts reconstruction outright.

The fourth objective, generalization to real external data from [TCIA](#), can only be answered in part. In the absence of a clean reference for these external cases, the internal technical evidence is limited to characterizing the magnitude of the input-to-output change, which was of a plausible order and did not indicate collapse to the input or gross distortion. The completed blinded, paired evaluation by five physicians on 15 external cases ([section 5.5](#)) adds a further, complementary line of evidence. Readers favored the model output over the artifact input for anatomical preservation, artifact-intensity reduction, and clinical-interpretation preference in a majority of presentations. This preference was most consistent for reduced artifact intensity and least consistent for the harm-flagging item that probes for distortion away from the

artifact. However, this panel fell two readers short of the pre-specified target. The 95% confidence interval for the primary output-versus-input contrast included the no-difference value, and intra-reader repeat agreement was only moderate. This reader evidence is therefore best described as a descriptive, directionally favorable signal from a small, purposely built panel, not as statistically confirmed proof of clinical benefit or of anatomical safety in tissue away from the artifact.

Considering the study as a whole, the technical evidence for internal reconstruction performance, reproducibility, and artifact-targeted structural correction is strong and consistent across seeds and architectures. The evidence for external generalization and perceived clinical benefit is present and directionally favorable, but preliminary. It rests on a small external case set and a small reader panel, and it does not, by itself, establish that either architecture is clinically safe or superior for use on unseen patients. Confirming that claim would require replication with a larger, ideally multi-center reader panel and a larger, more anatomically diverse external case set, as discussed in [chapter 6](#).

7.2 Dissertation Contributions

- A patient-level traceability procedure that recovered exam and patient identity absent from the original [CT-MAR](#) metadata, enabling patient-level partitioning designed to minimize cross-subset information leakage (187/41/41 patients), audited with [MS-SSIM](#) and human visual review.
- A controlled comparison between a single-supervision and a dual-supervision (auxiliary artifact-mask) U-Net, isolating the effect of the additional supervision under identical training conditions across three deterministic seeds per architecture.
- A patient-level, region-of-interest-aware statistical evaluation protocol that separates Artifact-Zone correction from Clean-Zone preservation, using pre-specified hypotheses, Holm-adjusted tests, and bootstrap confidence intervals, together with a fully reproducible pipeline of versioned configurations, environment snapshots, and checkpoint hashes.
- A blinded, paired external evaluation protocol combining quantitative input–output characterization on independent [TCIA](#) cases with a completed multi-reader clinical assessment, reported together with an explicit account of its scale, precision, and resulting limitations.

Bibliography

- [1] November 8, 1895: Roentgen's Discovery of X-Rays. <https://www.aps.org/apsnews/2001/11/1895-roentgens-discovery-xrays>. [Cited on page 1.]
- [2] Emico Okuno and Elisabeth Yoshimura. *Física das Radiações*. Oficina de Textos, 2010. [Cited on page 1.]
- [3] JOEL D. HOWELL. EARLY CLINICAL USE OF THE X-RAY. *Transactions of the American Clinical and Climatological Association*, 127:341–349, 2016. [Cited on page 1.]
- [4] Allen Taylor. Illuminating Health: The Evolution and Significance of X-ray Imaging. *Imaging in Medicine*, 15(6):133–134, December 2023. [Cited on page 1.]
- [5] Reza Vali. Unveiling the Marvels of Computed Tomography (CT) Imaging. *Imaging in Medicine*, 16(1):150–151, February 2024. [Cited on page 1.]
- [6] Mohd Noor. An Overview of Computed Tomography: What it is and How it Works? Applications of Computed Tomography. *Imaging in Medicine*, 15(5):93–94, October 2023. [Cited on page 1.]
- [7] Raymond A. Schulz, Jay A. Stein, and Norbert J. Pelc. How CT happened: The early development of medical computed tomography. *Journal of Medical Imaging*, 8(5):052110, September 2021. [Cited on pages 1 and 2.]
- [8] F. Edward Boas and Dominik Fleischmann. CT artifacts: Causes and reduction techniques. *Imaging in Medicine*, 4(2):229–240, April 2012. [Cited on page 1.]
- [9] Cecile E.J. Kleber, Ramez Karius, Lucas E. Naessens, Coen O. Van Toledo, Jochen A. C. Van Osch, Martijn F. Boomsma, Jan W.T. Heemskerk, and Aart J. Van Der Molen. Advancements in supervised deep learning for metal artifact reduction in computed tomography: A systematic review. *European Journal of Radiology*, 181:111732, December 2024. [Cited on pages 2, 6, 7, and 13.]

- [10] Sathyathas Puvanasunthararajah, Davide Fontanarosa, Marie-Luise Wille, and Saskia M. Camps. The application of metal artifact reduction methods on computed tomography scans for radiotherapy applications: A literature review. *Journal of Applied Clinical Medical Physics*, 22(6):198–223, May 2021. [Cited on page 2.]
- [11] Asutosh Sahu, Shobhit Mathur, Hiroyuki Takaoka, Joji Ota, Felix G. Meinel, Benjamin Böttcher, Benoît Magnin, Ankush Jajodia, and Corey T. Jensen. Transforming CT imaging with deep learning: Noise reduction, artifact management, and clinical applications – A comprehensive review. *European Journal of Radiology Artificial Intelligence*, 4:100042, December 2025. [Cited on pages 2, 6, and 7.]
- [12] Mohanad Alkhodari, Eman Alefishat, Herbert F. Jelinek, Ahmed Kaabneh, and Panos Liat-sis. Enhancing CCTA image quality: A review of deep learning approaches for advanced artifact correction and denoising. *Artificial Intelligence Review*, 58(10):331, August 2025. [Cited on page 2.]
- [13] Amel F. Alzain, Nagwan Elhussein, Ibtisam Abdallah Fadulelmulla, Amna Mohamed Ahmed, M. E. Elbashir, and Badria Awad Elamin. Common computed tomography artifact: Source and avoidance. *The Egyptian Journal of Radiology and Nuclear Medicine*, 52(1):151, 2021. [Cited on page 2.]
- [14] Julia F. Barrett and Nicholas Keat. Artifacts in CT: Recognition and Avoidance. *Radio-Graphics*, 24(6):1679–1691, November 2004. [Cited on page 6.]
- [15] Mark Selles, Jochen A.C. Van Osch, Mario Maas, Martijn F. Boomsma, and Ruud H.H. Wellenberg. Advances in metal artifact reduction in CT images: A review of traditional and novel metal artifact reduction techniques. *European Journal of Radiology*, 170:111276, January 2024. [Cited on page 6.]
- [16] Lars Gjestebj, Qingsong Yang, Yan Xi, Bernhard E. H. Claus, Yannan Jin, Bruno De Man, Ge Wang, and Hongming Shan. Deep learning methods for CT image-domain metal artifact reduction. In Bert Müller and Ge Wang, editors, *Developments in X-Ray Tomography XI*, page 31, San Diego, United States, September 2017. SPIE. [Cited on pages 6, 7, and 71.]
- [17] Yanbo Zhang and Hengyong Yu. Convolutional Neural Network Based Metal Artifact Reduction in X-Ray Computed Tomography. *IEEE Transactions on Medical Imaging*, 37(6):1370–1381, June 2018. [Cited on pages 6, 7, and 71.]

- [18] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation, May 2015. [Cited on pages 7 and 15.]
- [19] Esther Meyer, Frank Bergner, Rainer Raupach, Thomas Flohr, and Marc Kachelriess. Normalized metal artifact reduction (NMAR) in computed tomography. In *2009 IEEE Nuclear Science Symposium Conference Record (NSS/MIC)*, pages 3251–3255, Orlando, FL, October 2009. IEEE. [Cited on page 7.]
- [20] Wei-An Lin, Haofu Liao, Cheng Peng, Xiaohang Sun, Jingdan Zhang, Jiebo Luo, Rama Chellappa, and Shaohua Kevin Zhou. DuDoNet: Dual Domain Network for CT Metal Artifact Reduction. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10504–10513, Long Beach, CA, USA, June 2019. IEEE. [Cited on pages 7, 13, 71, and 77.]
- [21] Lequan Yu, Zhicheng Zhang, Xiaomeng Li, Hongyi Ren, Wei Zhao, and Lei Xing. Metal Artifact Reduction in 2D CT Images with Self-supervised Cross-domain Learning. *Physics in Medicine & Biology*, 66(17):175003, September 2021. [Cited on page 7.]
- [22] Zhiwei Huang, Guo Zhang, Jinzhao Lin, Yu Pang, Huiqian Wang, Tong Bai, and Lisha Zhong. Multi-modal feature-fusion for CT metal artifact reduction using edge-enhanced generative adversarial networks. *Computer Methods and Programs in Biomedicine*, 217:106700, April 2022. [Cited on pages 7 and 71.]
- [23] Xin Yi, Ekta Walia, and Paul Babyn. Generative Adversarial Network in Medical Imaging: A Review. *Medical Image Analysis*, 58:101552, December 2019. [Cited on page 7.]
- [24] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, April 2004. [Cited on pages 8, 11, and 13.]
- [25] Z. Wang, E.P. Simoncelli, and A.C. Bovik. Multiscale structural similarity for image quality assessment. In *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, pages 1398–1402, Pacific Grove, CA, USA, 2003. IEEE. [Cited on page 8.]
- [26] Alain Hore and Djemel Ziou. Image Quality Metrics: PSNR vs. SSIM. In *2010 20th International Conference on Pattern Recognition*, pages 2366–2369, Istanbul, Turkey, August 2010. IEEE. [Cited on page 8.]
- [27] John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model

- to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11):e1002683, November 2018. [Cited on pages 8 and 71.]
- [28] Martin J. Willemink, Wojciech A. Koszek, Cailin Hardell, Jie Wu, Dominik Fleischmann, Hugh Harvey, Les R. Folio, Ronald M. Summers, Daniel L. Rubin, and Matthew P. Lungren. Preparing Medical Imaging Data for Machine Learning. *Radiology*, 295(1):4–15, April 2020. [Cited on page 8.]
- [29] Christopher J. Kelly, Alan Karthikesalingam, Mustafa Suleyman, Greg Corrado, and Dominic King. Key challenges for delivering clinical impact with artificial intelligence. *BMC Medicine*, 17(1):195, December 2019. [Cited on page 8.]
- [30] Eri Haneda, Nils Peters, Jiayong Zhang, Grigorios Karageorgos, Wenjun Xia, Harald Paganetti, Ge Wang, Yi Guo, Jianhua Ma, Hyoung Suk Park, Kiwan Jeon, Fuxin Fan, Mareike Thies, and Bruno De Man. AAPM CT metal artifact reduction grand challenge. *Medical Physics*, 52(10):e70050, October 2025. [Cited on pages 9, 13, and 28.]
- [31] xcist. AAPM CT Metal Artifact Reduction (CT-MAR) Grand Challenge Benchmark Tool. https://github.com/xcist/example/tree/main/AAPM_datachallenge/. [Cited on pages 9 and 28.]
- [32] Nils Peters, Eri Haneda, Jiayong Zhang, Grigorios Karageorgos, Wenjun Xia, Joost Verburg, Ge Wang, Harald Paganetti, and Bruno De Man. A hybrid training database and evaluation benchmark for assessing metal artifact reduction methods for X-ray CT imaging. *Medical Physics*, 52(10):e70020, October 2025. [Cited on page 9.]
- [33] Mingye Wu, Paul FitzGerald, Jiayong Zhang, W. Paul Segars, Hengyong Yu, Yongshun Xu, and Bruno De Man. XCIST – An Open Access X-ray/CT Simulation Toolkit. *Physics in medicine and biology*, 67(19):10.1088/1361–6560/ac9174, September 2022. [Cited on page 9.]
- [34] Nir Goren, James Avery, Thomas Dowrick, Eleanor Mackle, Anna Witkowska-Wrobel, David Werring, and David Holder. Multi-frequency electrical impedance tomography and neuroimaging data in stroke patients. *Scientific Data*, 5:180112, July 2018. [Cited on pages 9 and 10.]
- [35] Nir Goren, Thomas Dowrick, James Avery, and David Holder. UCLH Stroke EIT Dataset - Radiology Data, August 2017. [Cited on page 9.]

- [36] Ke Yan, Xiaosong Wang, Le Lu, and Ronald M. Summers. DeepLesion: Automated mining of large-scale lesion annotations and universal lesion detection with deep learning. *Journal of Medical Imaging*, 5(3):036501, July 2018. [Cited on pages 9 and 10.]
- [37] Mustafa Umit Oner, Yi-Chih Cheng, Hwee Kuan Lee, and Wing-Kin Sung. Training machine learning models on patient level data segregation is crucial in practical clinical applications, April 2020. [Cited on page 10.]
- [38] Jiang Hsieh. *Computed Tomography: Principles, Design, Artifacts, and Recent Advances*. Wiley Interscience ; SPIE Press, Hoboken, N.J. : Bellingham, Wash, 2nd ed edition, 2009. [Cited on page 11.]
- [39] Saeid Asgari Taghanaki, Yefeng Zheng, S. Kevin Zhou, Bogdan Georgescu, Puneet Sharma, Daguang Xu, Dorin Comaniciu, and Ghassan Hamarneh. Combo Loss: Handling Input and Output Imbalance in Multi-Organ Segmentation, September 2021. [Cited on pages 14 and 20.]
- [40] Uğur Kesimal, Habip Eser Akkaya, Önder Polat, and Murat Sağlam. Use of Deep Learning Models in the Diagnosis of Proptosis Through Orbital Magnetic Resonance Imaging. *Medical Science Monitor*, 32, April 2026. [Cited on page 14.]
- [41] Yang Li, Bobo Yan, Jianxin Hou, Bingyang Bai, Xiaoyu Huang, Canfei Xu, and Limei Fang. UNet based on dynamic convolution decomposition and triplet attention. *Scientific Reports*, 14(1):271, January 2024. [Cited on page 15.]
- [42] Rich Caruana. Multitask Learning. [Cited on page 15.]
- [43] Sebastian Ruder. An Overview of Multi-Task Learning in Deep Neural Networks, June 2017. [Cited on page 15.]
- [44] Hang Zhao, Orazio Gallo, Iuri Frosio, and Jan Kautz. Loss Functions for Image Restoration With Neural Networks. *IEEE Transactions on Computational Imaging*, 3(1):47–57, March 2017. [Cited on page 19.]
- [45] Lin Zhang, Lei Zhang, Xuanqin Mou, and D. Zhang. FSIM: A Feature Similarity Index for Image Quality Assessment. *IEEE Transactions on Image Processing*, 20(8):2378–2386, August 2011. [Cited on page 20.]
- [46] Kenneth Clark, Bruce Vendt, Kirk Smith, John Freymann, Justin Kirby, Paul Koppel, Stephen Moore, Stanley Phillips, David Maffitt, Michael Pringle, Lawrence Tarbox, and

- Fred Prior. The Cancer Imaging Archive (TCIA): Maintaining and operating a public information repository. *Journal of Digital Imaging*, 26(6):1045–1057, December 2013. [Cited on page 24.]
- [47] Brad Albertina, Mark Watson, Chandra Holback, Rose Jarosz, Shanah Kirk, Yueh Lee, Kimberly Rieger-Christ, and John Lemmerman. The Cancer Genome Atlas Lung Adenocarcinoma Collection (TCGA-LUAD) (Version 4) [Data set], 2016. [Cited on page 25.]
- [48] Samuel G. Armato, Geoffrey McLennan, Luc Bidaut, Michael F. McNitt-Gray, Charles R. Meyer, Anthony P. Reeves, Binsheng Zhao, Denise R. Aberle, Claudia I. Henschke, Eric A. Hoffman, Ella A. Kazerooni, Heber MacMahon, Edwin J. R. Van Beek, David Yankelevitz, Alberto M. Biancardi, Peyton H. Bland, Matthew S. Brown, Roger M. Engelmann, Gary E. Laderach, Daniel Max, Richard C. Pais, David P. Y. Qing, Rachael Y. Roberts, Amanda R. Smith, Adam Starkey, Poonam Batra, Philip Caligiuri, Ali Farooqi, Gregory W. Gladish, C. Matilda Jude, Reginald F. Munden, Iva Petkovska, Leslie E. Quint, Lawrence H. Schwartz, Baskaran Sundaram, Lori E. Dodd, Charles Fenimore, David Gur, Nicholas Petrick, John Freymann, Justin Kirby, Brian Hughes, Alessi Vande Casteele, Sangeeta Gupte, Maha Sallam, Michael D. Heath, Michael H. Kuhn, Ekta Dharaiya, Richard Burns, David S. Fryd, Marcos Salganicoff, Vikram Anand, Uri Shreter, Stephen Vastagh, Barbara Y. Croft, and Laurence P. Clarke. Data from LIDC-IDRI [Data set], 2015. [Cited on page 25.]
- [49] The Research for Precision Oncology Program and the Applied Proteogenomics Organizational Learning and Outcomes (APOLLO) Research Network. VA Research Precision Oncology Program – APOLLO (VAREPOP-APOLLO) (Version 3) [Dataset], 2024. [Cited on page 25.]
- [50] National Cancer Institute Clinical Proteomic Tumor Analysis Consortium (CPTAC). The Clinical Proteomic Tumor Analysis Consortium Sarcomas Collection (CPTAC-SAR) (Version 10) [Data set], 2019. [Cited on page 25.]
- [51] National Cancer Institute Clinical Proteomic Tumor Analysis Consortium (CPTAC). The Clinical Proteomic Tumor Analysis Consortium Pancreatic Ductal Adenocarcinoma Collection (CPTAC-PDA) (Version 15) [Dataset], 2018. [Cited on page 25.]

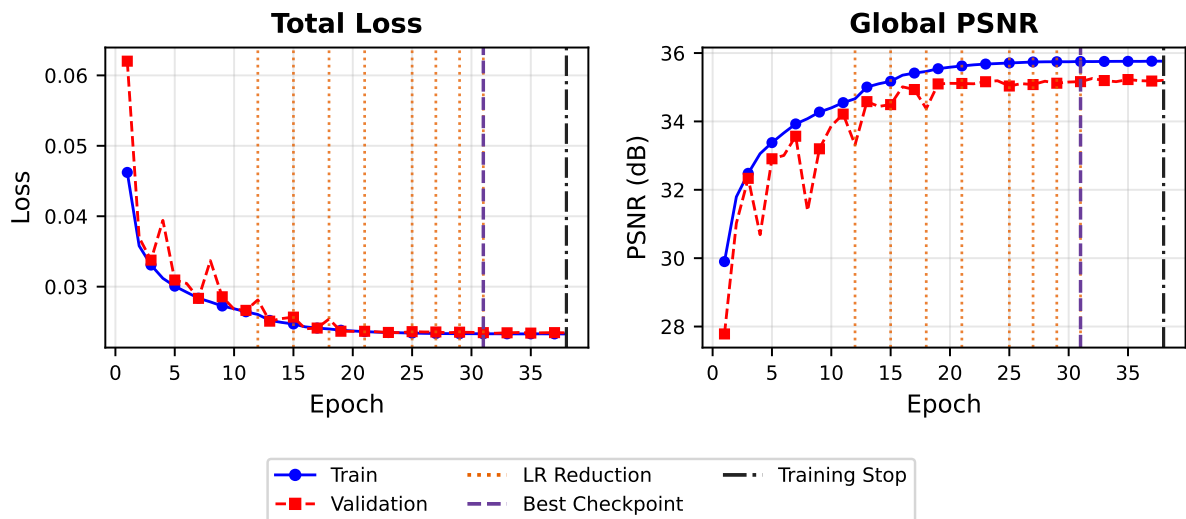
Appendix A

Supplementary Internal Evaluation

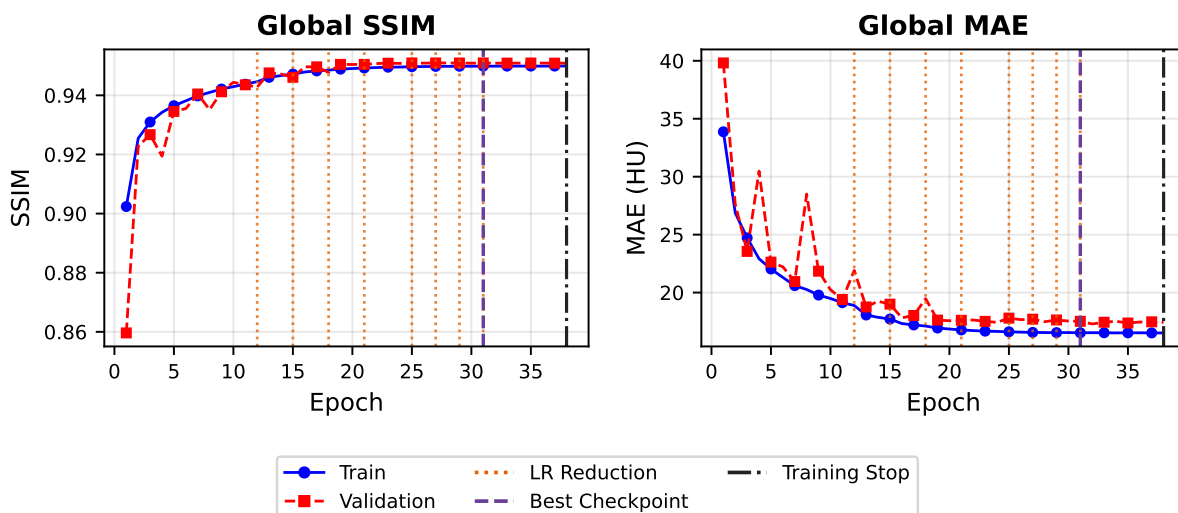
A.1 Training Convergence Across Runs

This section provides the complete convergence documentation for the six deterministic training runs. The figures and tables below document per-run checkpoint selection and did not, by themselves, determine the final external-evaluation candidate.

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

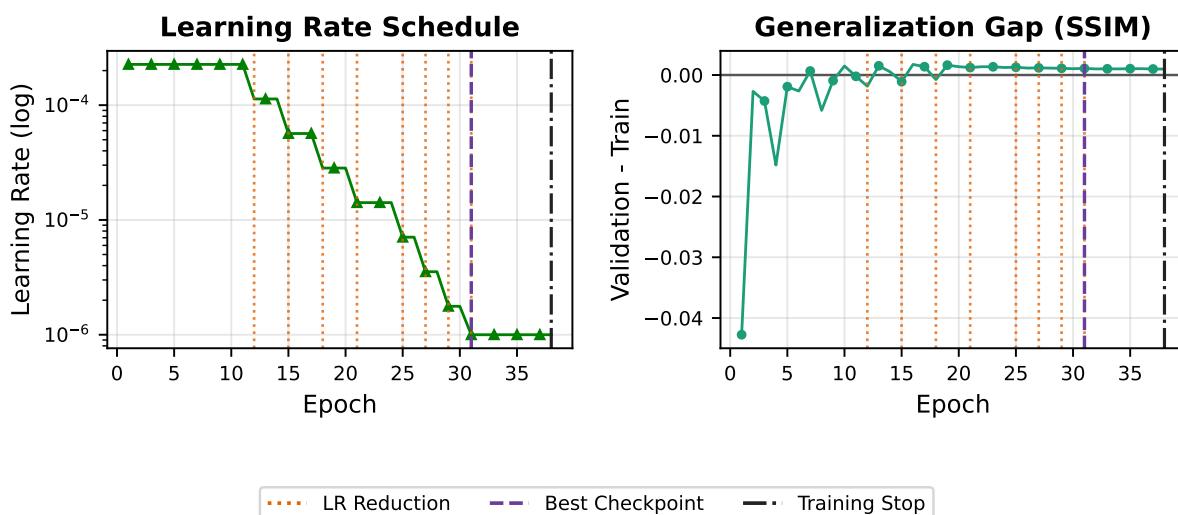
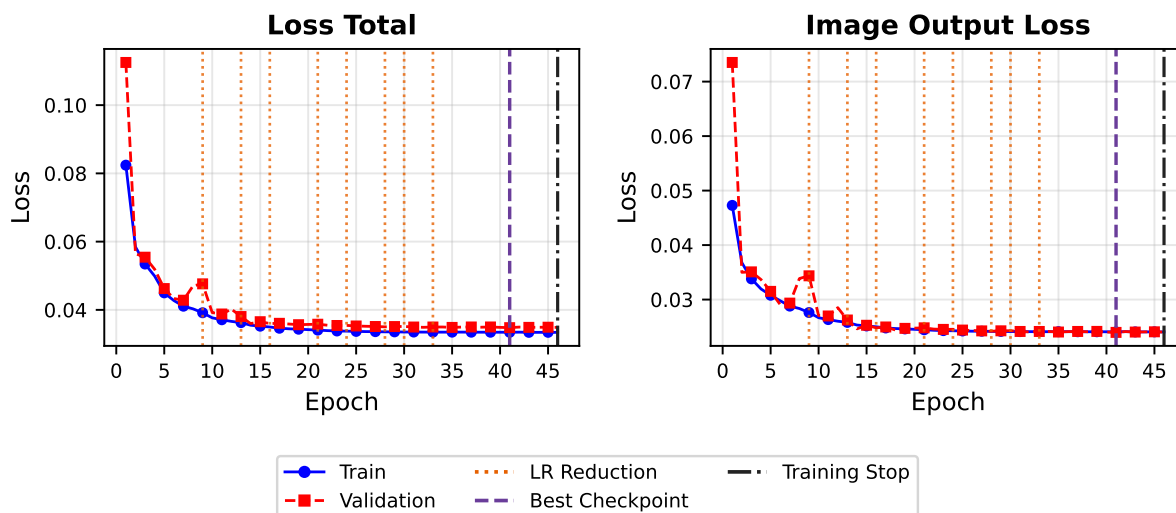
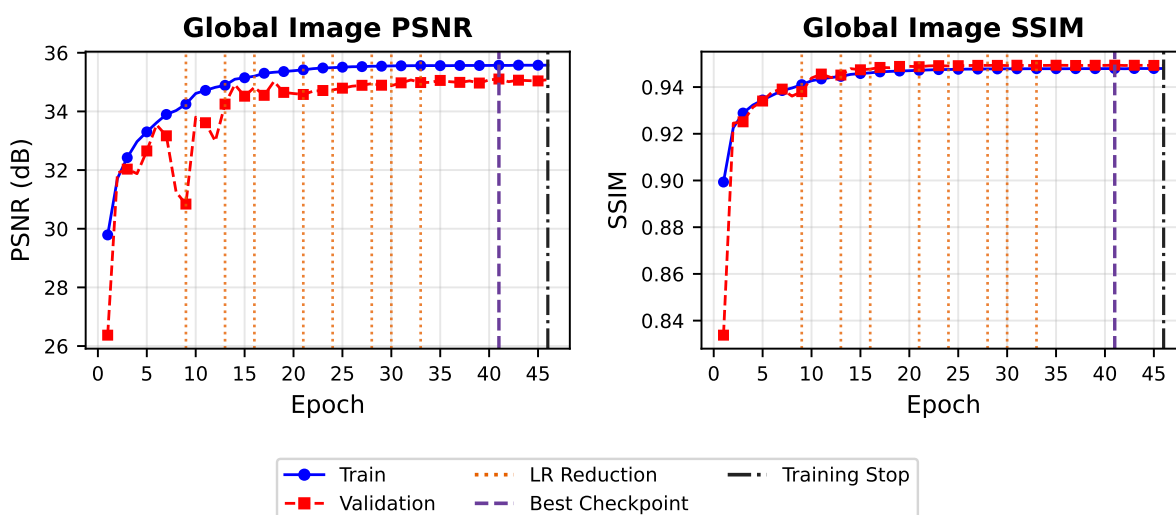


Fig.A.1 Training convergence for the Single Model, seed 42. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

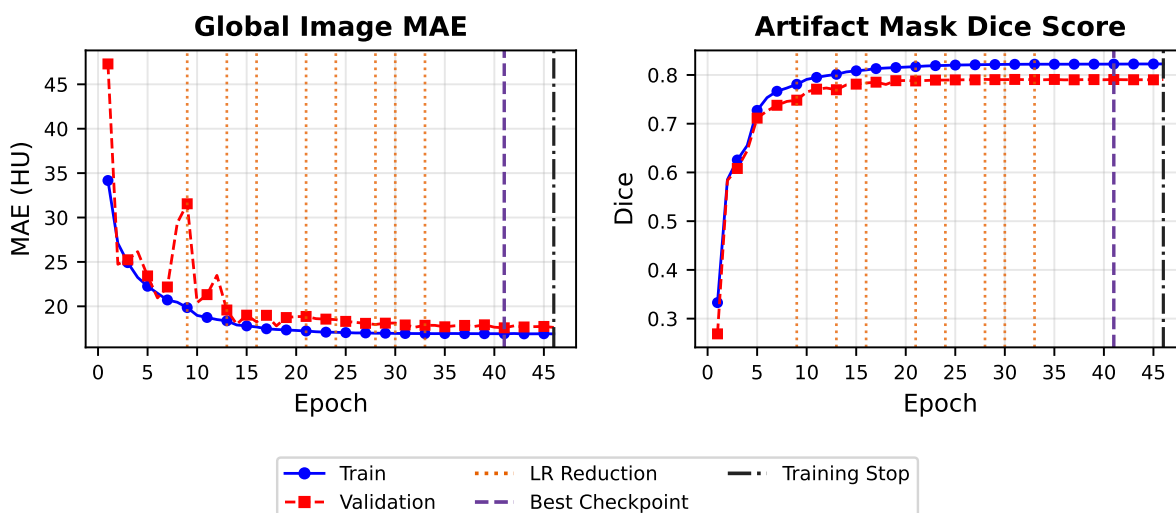
Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence

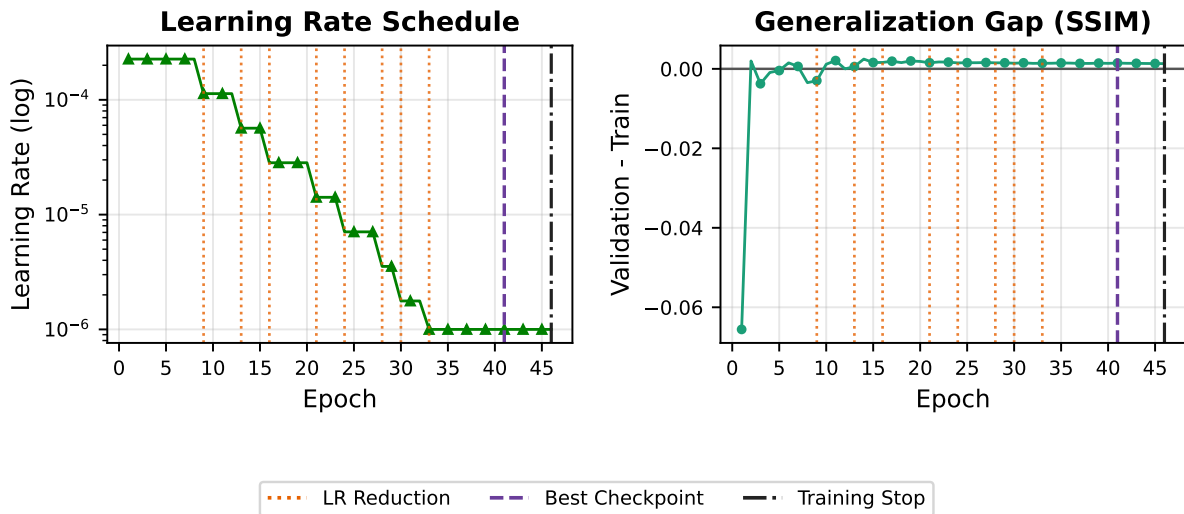
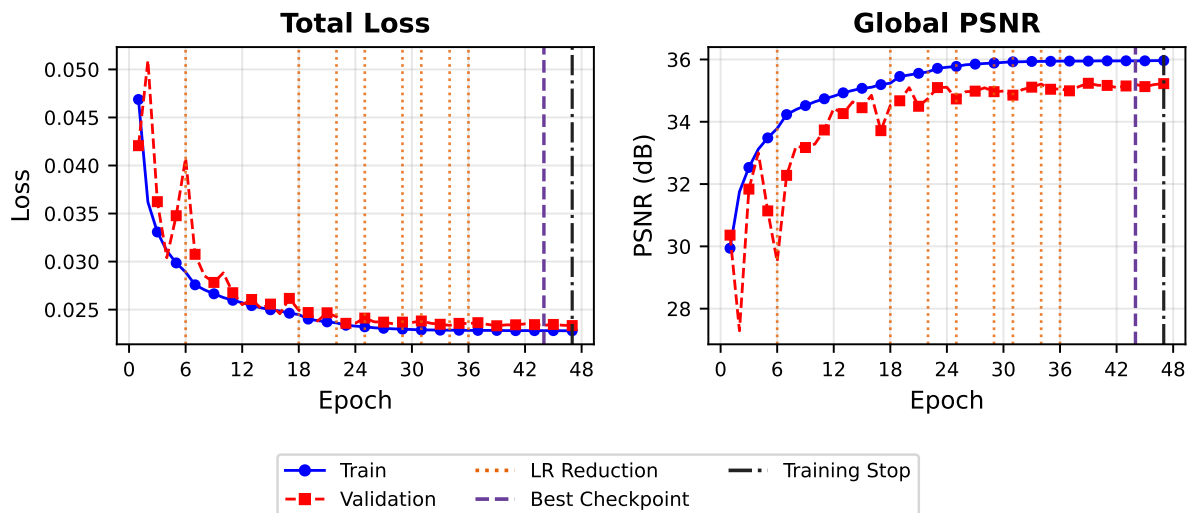
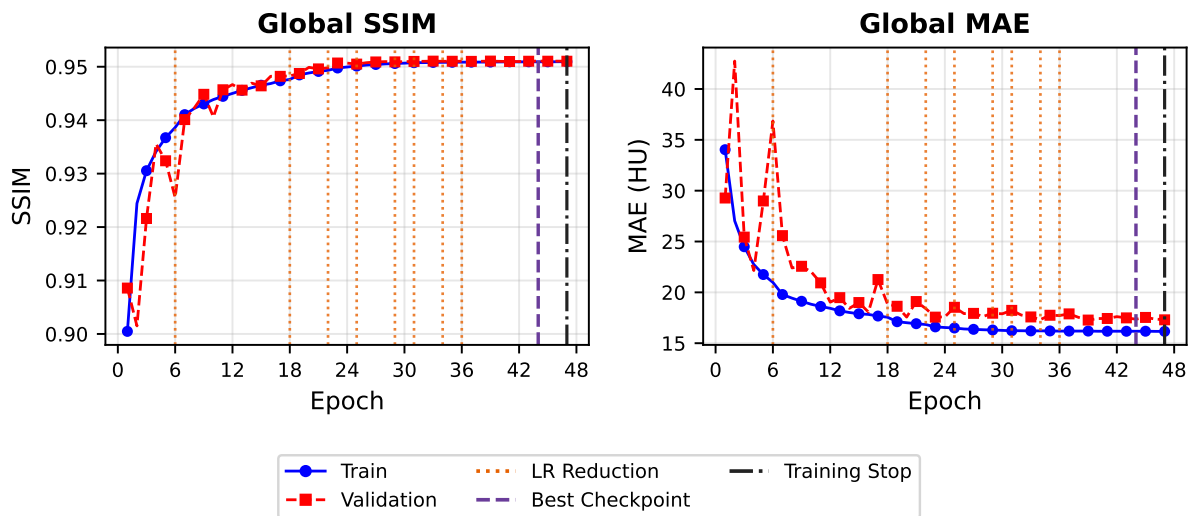


Fig.A.2 Training convergence for the Dual Model, seed 42. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

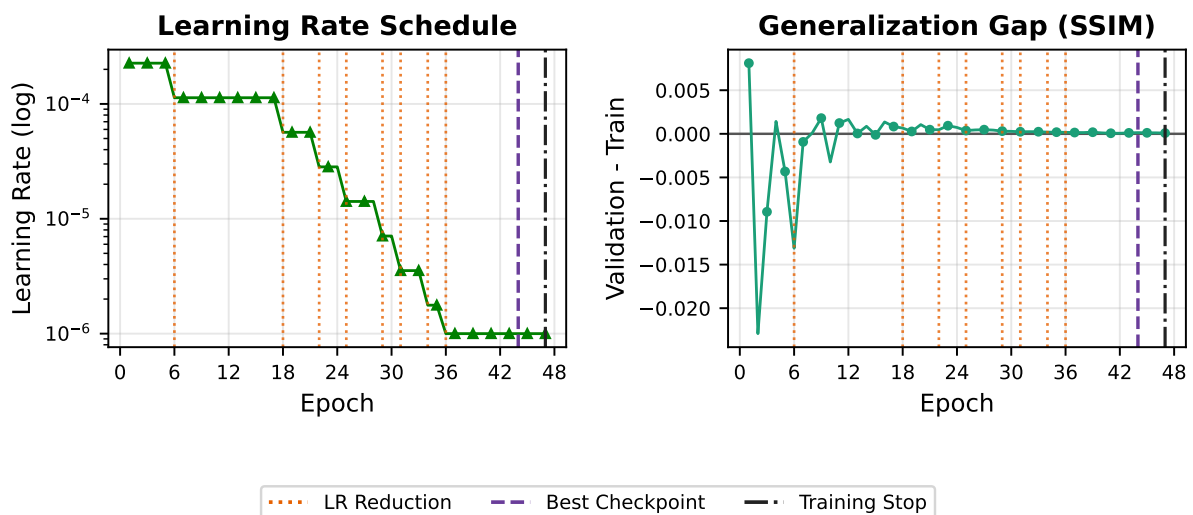
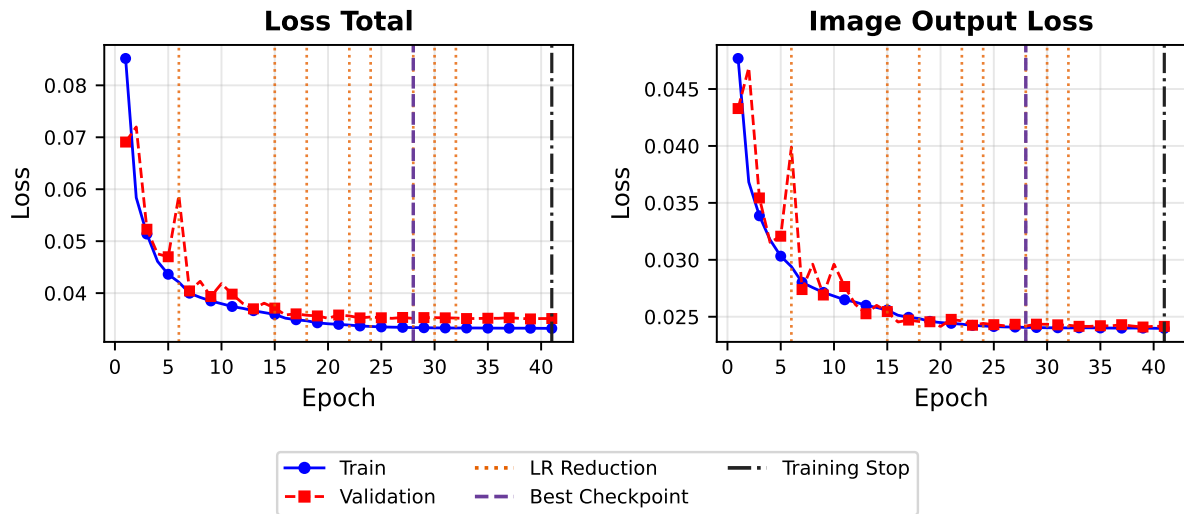
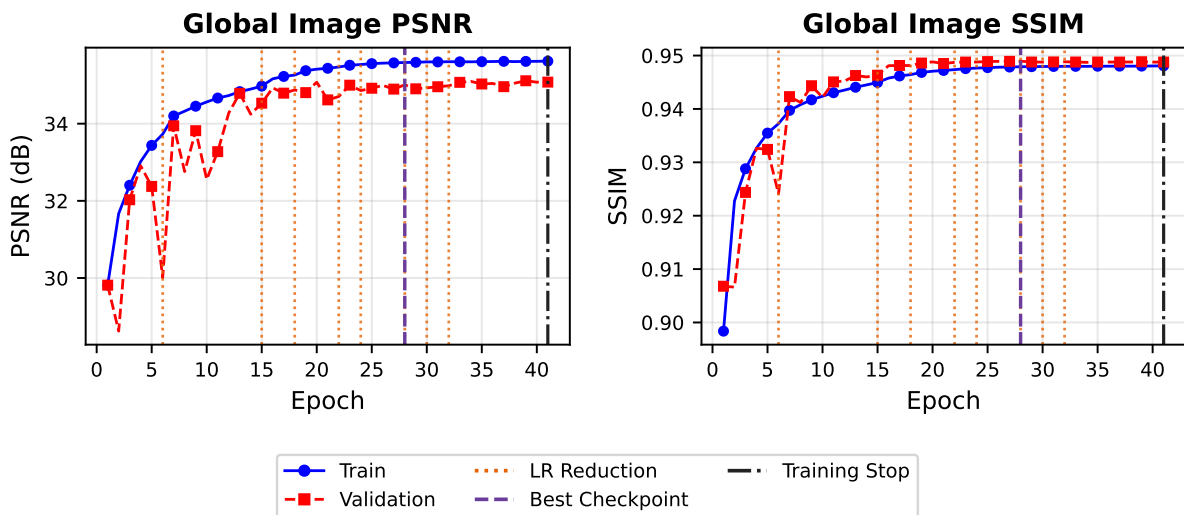


Fig.A.3 Training convergence for the Single Model, seed 1337. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

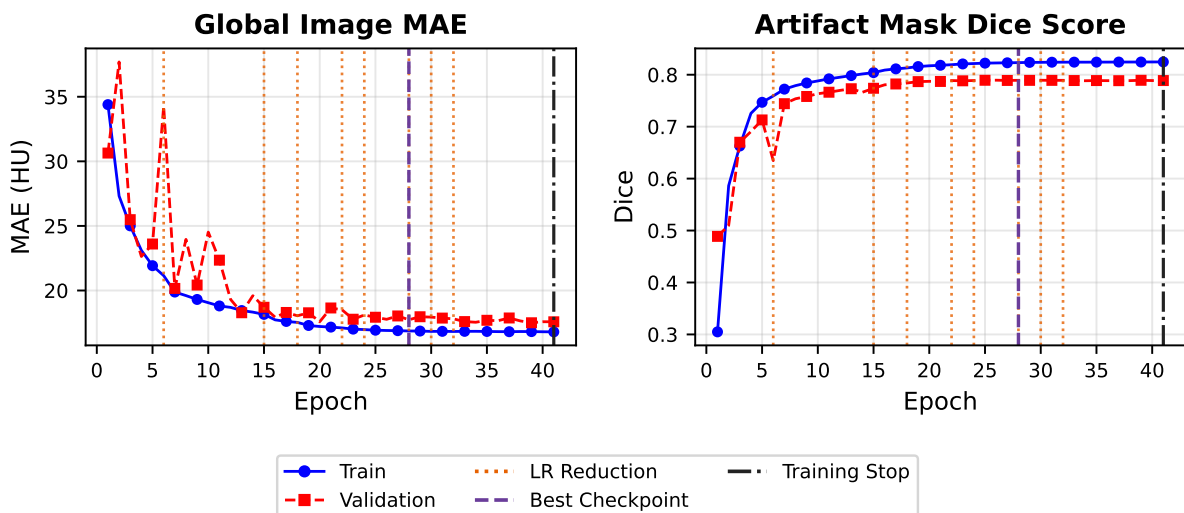
Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence

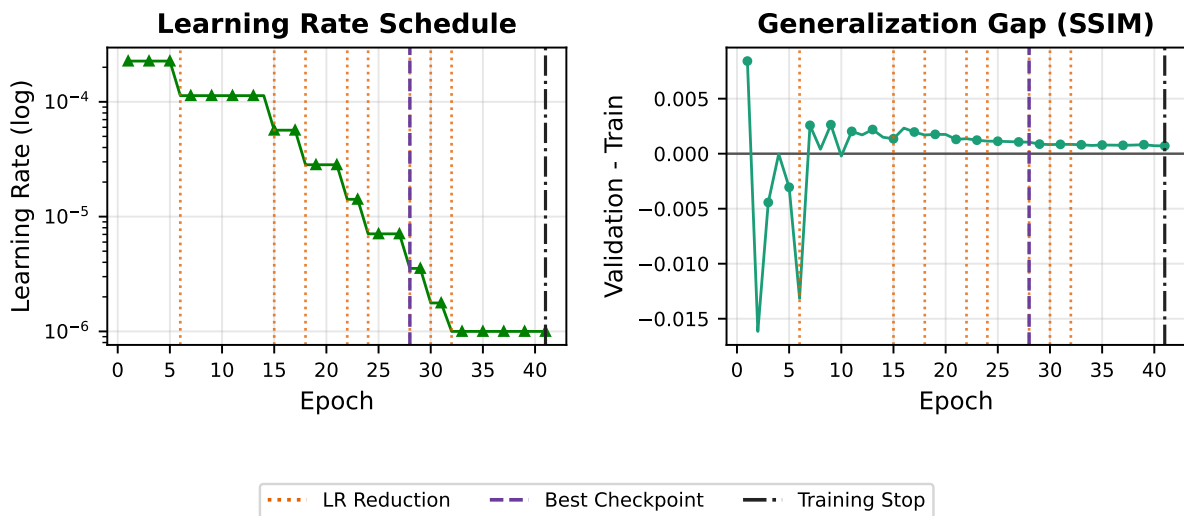
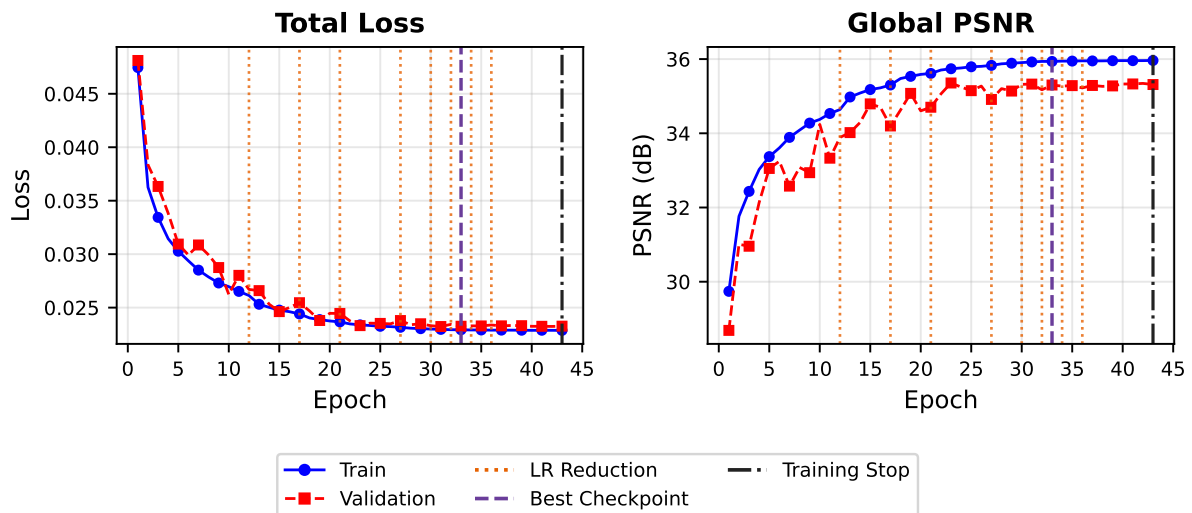
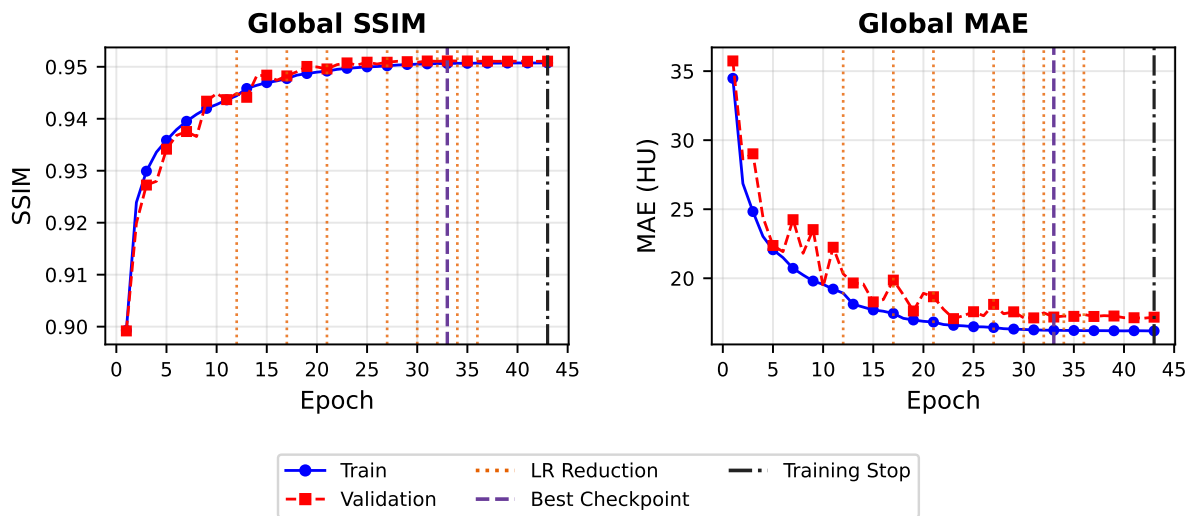


Fig.A.4 Training convergence for the Dual Model, seed 1337. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

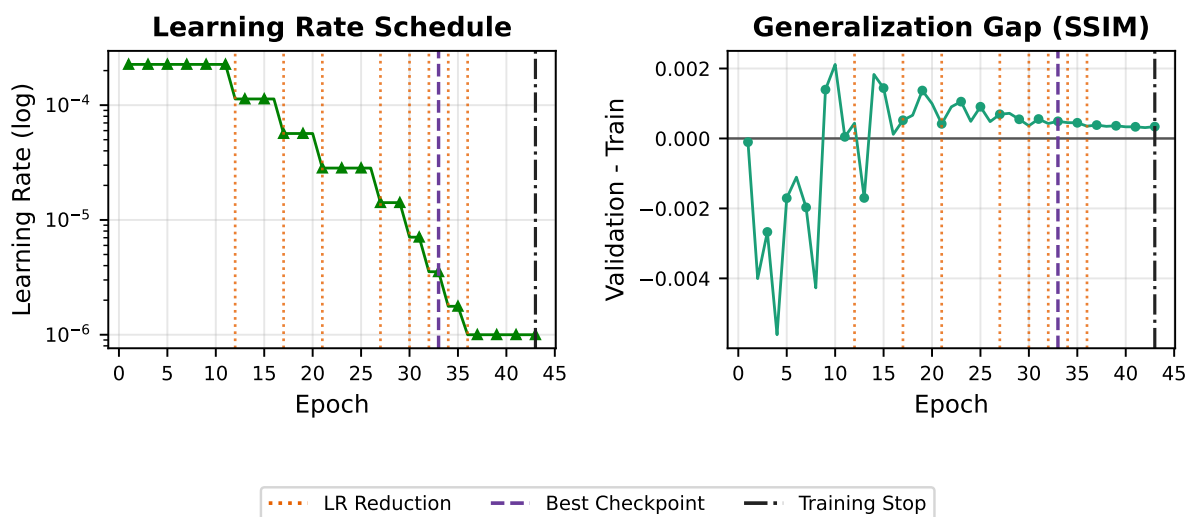
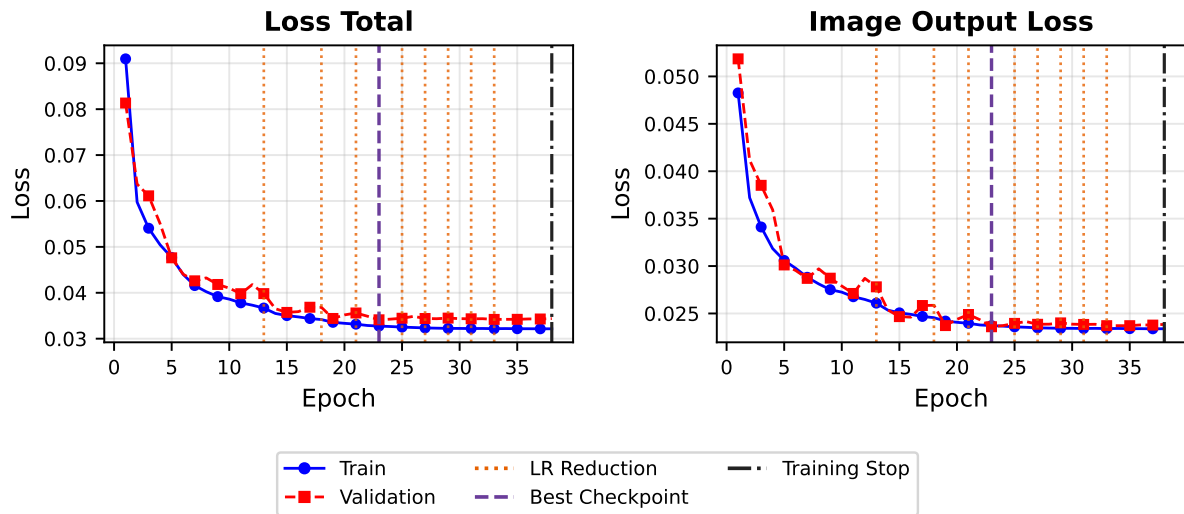
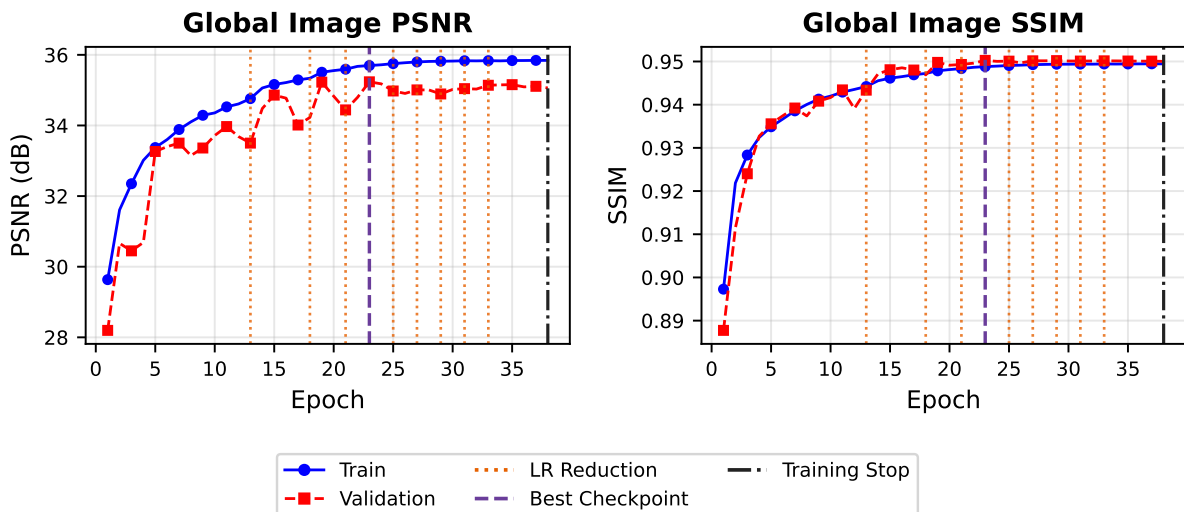


Fig.A.5 Training convergence for the Single Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

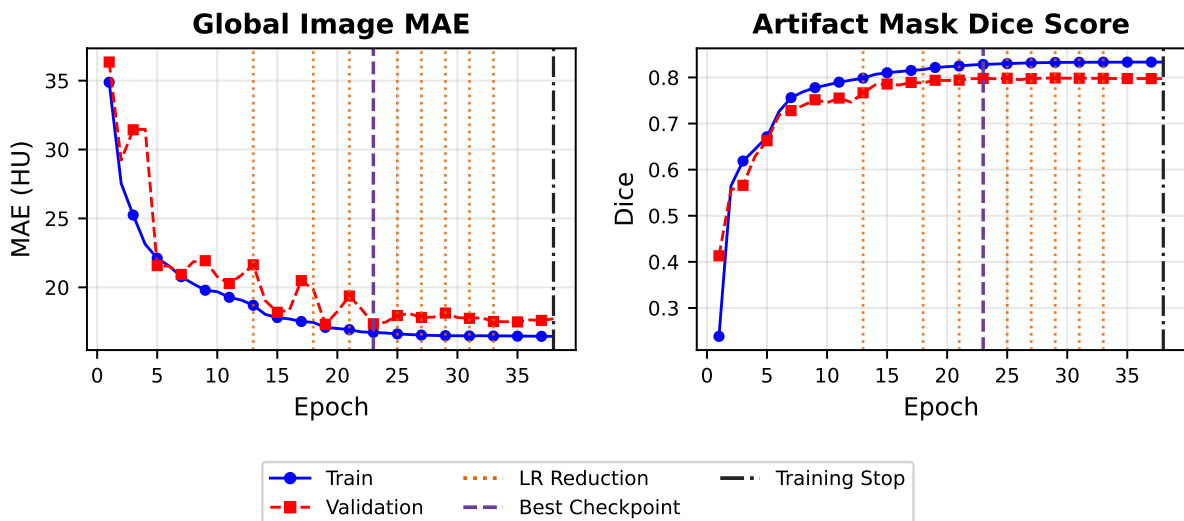
Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence

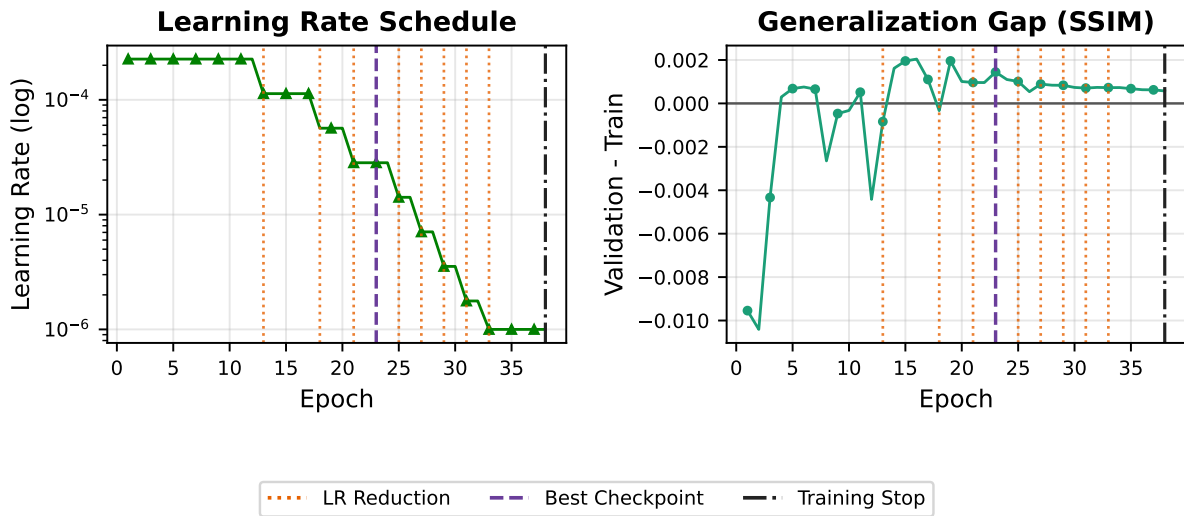


Fig.A.6 Training convergence for the Dual Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

Best SSIM (HU)			
Seed	42	1337	2024
Single Model	0.95096 (31)*	0.95106 (44)	0.95113 (33)
Dual Model	0.94938 (41)	0.94895 (28)	0.95025 (23)

*Values in parentheses indicate the corresponding epoch.

Table A.1 Highest validation SSIM by model and seed, with the corresponding epoch. For the Dual Model, SSIM was calculated on the reconstructed-image output.

PSNR (dB) - Same Best SSIM Epoch			
Seed	42	1337	2024
Single Model	35.16365 (31)	35.20020 (44)	35.30099 (33)
Dual Model	35.10422 (41)	35.00705 (28)	35.23622 (23)

Table A.2 Validation PSNR at the epoch selected by the highest validation SSIM, by model and seed. For the Dual Model, both metrics were calculated on the reconstructed-image output.

MAE (HU) - Same Best SSIM Epoch			
Seed	42	1337	2024
Single Model	17.52002 (31)	17.36296 (44)	17.19180 (33)
Dual Model	17.58006 (41)	17.73584 (28)	17.35387 (23)

Table A.3 Validation MAE at the epoch selected by the highest validation SSIM, by model and seed. For the Dual Model, both metrics were calculated on the reconstructed-image output.

This appendix presents complementary quantitative checks for the internal evaluation. The following figure is included solely as a visual diagnostic of pixel-level radiometric agreement

and does not constitute inferential analysis, because sampled pixels are not independent observations.

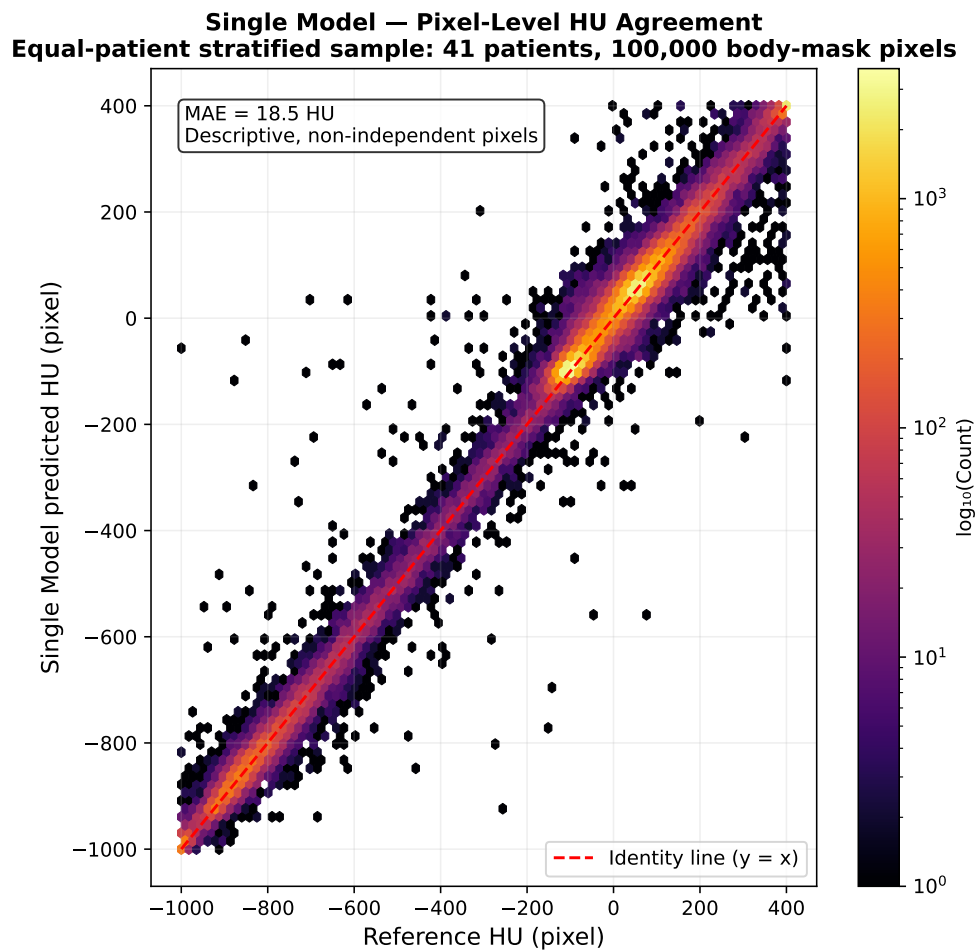


Fig.A.7 Pixel-level HU agreement between the Single Model reconstruction and the clean reference. The hexbin uses a fixed equal-patient stratified sample of 100,000 valid body-mask pixels from 41 test-set patients. It is a descriptive calibration diagnostic. No hypothesis test or confidence interval is derived from the sampled pixels.

The following plots complement the global-delta histograms presented in the main text. Each point represents one test slice, so these plots were not used for statistical inference.

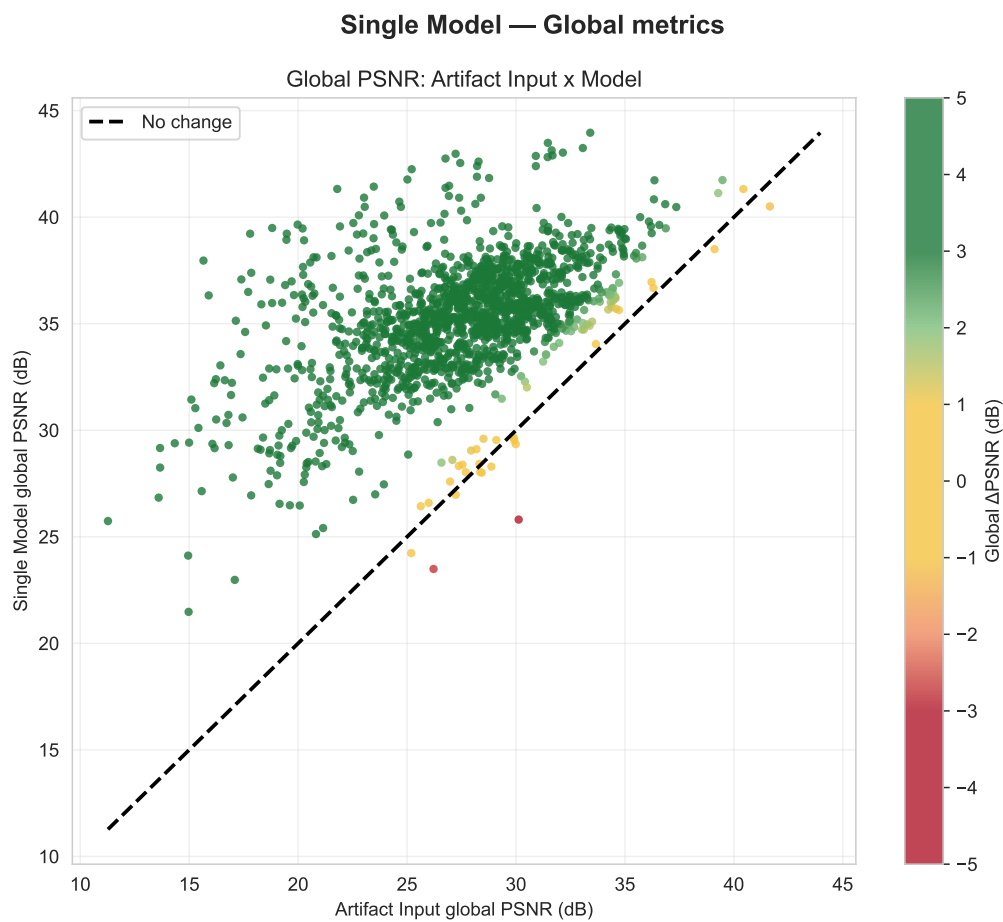
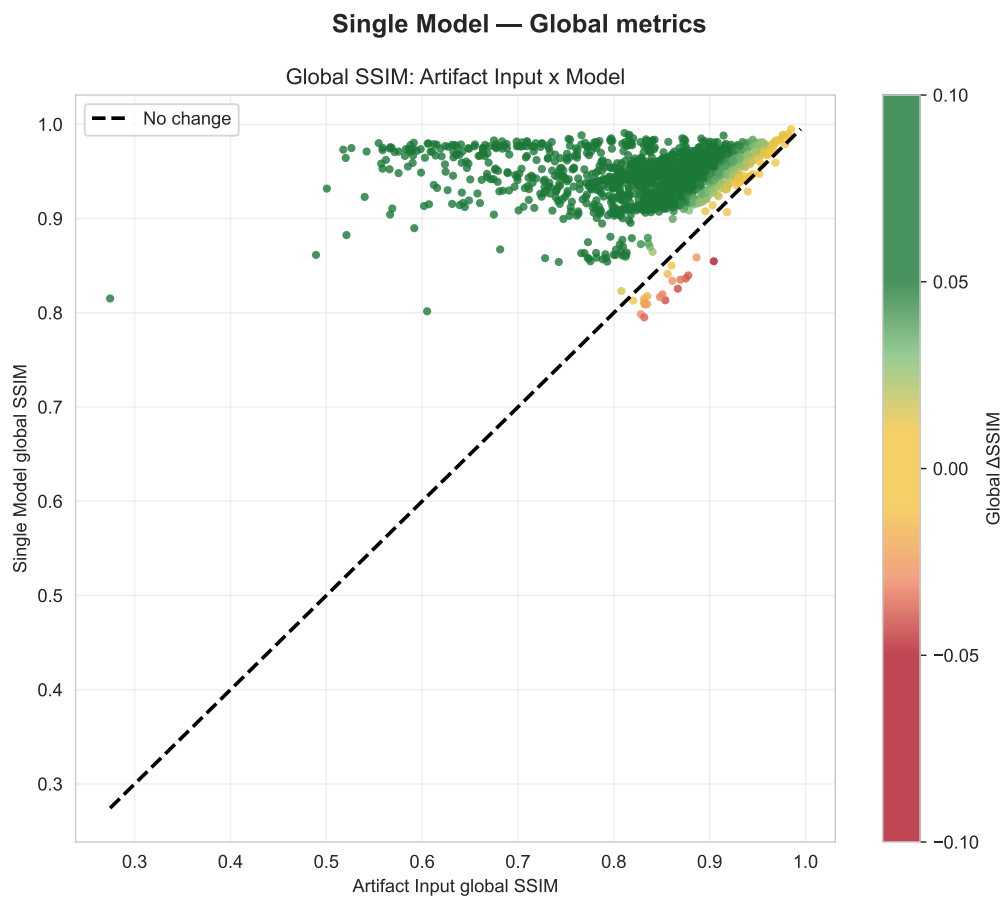
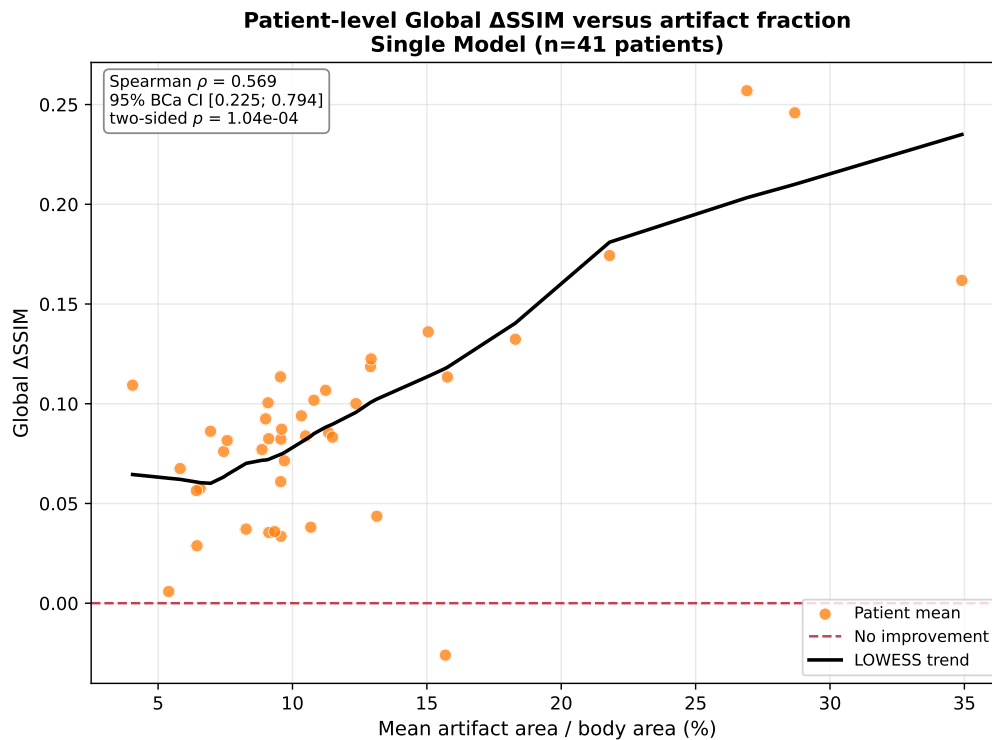


Fig.A.8 Per-image global SSIM (top) and PSNR (bottom) for the corrupted input and the Single Model reconstruction, seed 2024. The identity line denotes no change, marker color encodes the corresponding global delta. These slice-level relationships are descriptive, whereas statistical inference used patient-level aggregation.

As an exploratory analysis of heterogeneity, [Figure A.9](#) relates artifact burden to global improvement. Each point represents the mean across all slices of one patient ($n = 41$), giving each patient equal weight. Artifact burden was defined as the mean artifact-area/body-area fraction per slice. Deltas correspond to the reconstruction metric minus the input metric, both calculated against the clean reference. Two-sided Spearman correlations were calculated with paired BCa bootstrap 95% confidence intervals (10 000 resamples; seed 1337), and the trend curve is a LOWESS estimate from *Statsmodels*. This analysis was not part of the primary hypothesis-testing family and did not receive Holm adjustment. A positive association was observed between artifact burden and improvement: $\rho = 0.479$ for ΔPSNR (95% CI [0.156; 0.717], $p = 0.00154$) and $\rho = 0.569$ for ΔSSIM (95% CI [0.225; 0.794], $p = 0.000104$).



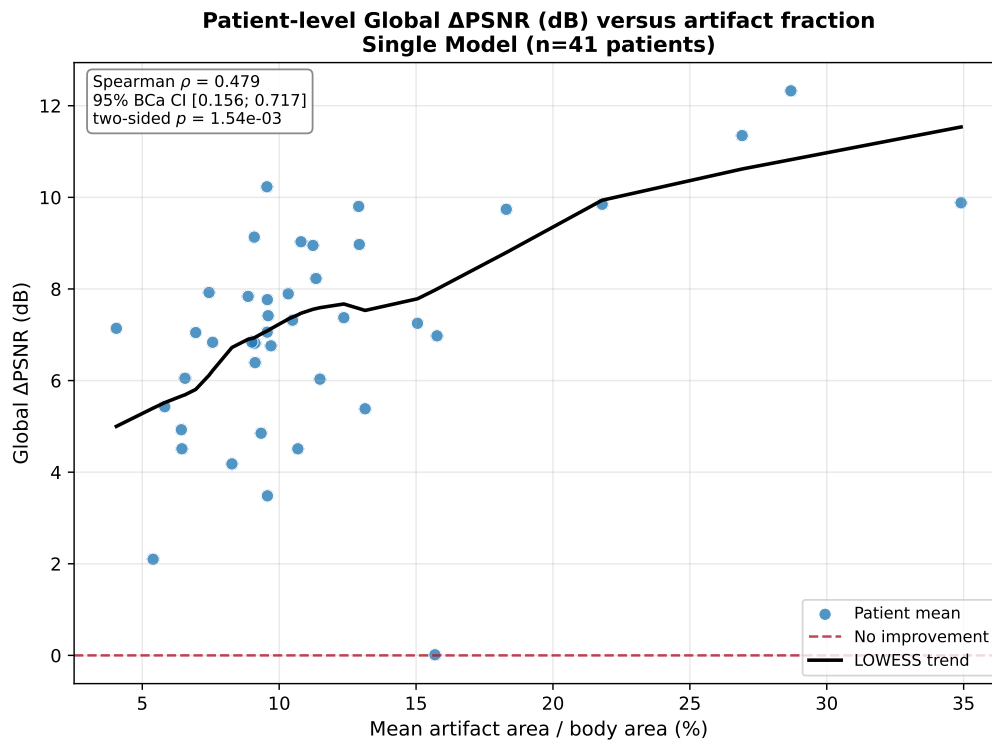


Fig.A.9 Exploratory patient-level association between mean artifact fraction and global improvement for the Single Model, seed 2024: Δ SSIM (top) and Δ PSNR (bottom). Each point is one patient's mean across its test-set slices ($n = 41$). Lines show Statsmodels LOWESS trends. Annotations report two-sided Spearman correlations and paired BCa bootstrap 95% confidence intervals.

A.2 Worst Performance Cases

As a complementary characterization of lower-performing slices, two slice-level groups were identified for the Single Model, seed 2024, in the test set: first by the bottom fifth percentile of global Δ SSIM and then, complementarily, by the bottom fifth percentile of global Δ PSNR. This slice-level classification is purely descriptive: it does not constitute a hypothesis test or reproduce the patient-level candidate-selection analysis, whose primary endpoint was Artifact-ROI Δ SSIM, and it serves only to quantitatively situate the slice groups whose individual visual examples are discussed in [subsection 5.3.4](#).

Using the global Δ SSIM criterion ($N = 74/1478$ slices; threshold = 0.0146), mean Δ PSNR improvement remained positive (1.88 dB), but mean Δ SSIM improvement was slightly negative (-0.0019). The mean Artifact-Zone MAE reduction was 65.2 HU ([Figure A.10](#)). Thus, the most unfavorable subset according to this descriptive global criterion included a small mean SSIM degradation, even though most remaining slices continued to improve substantially.

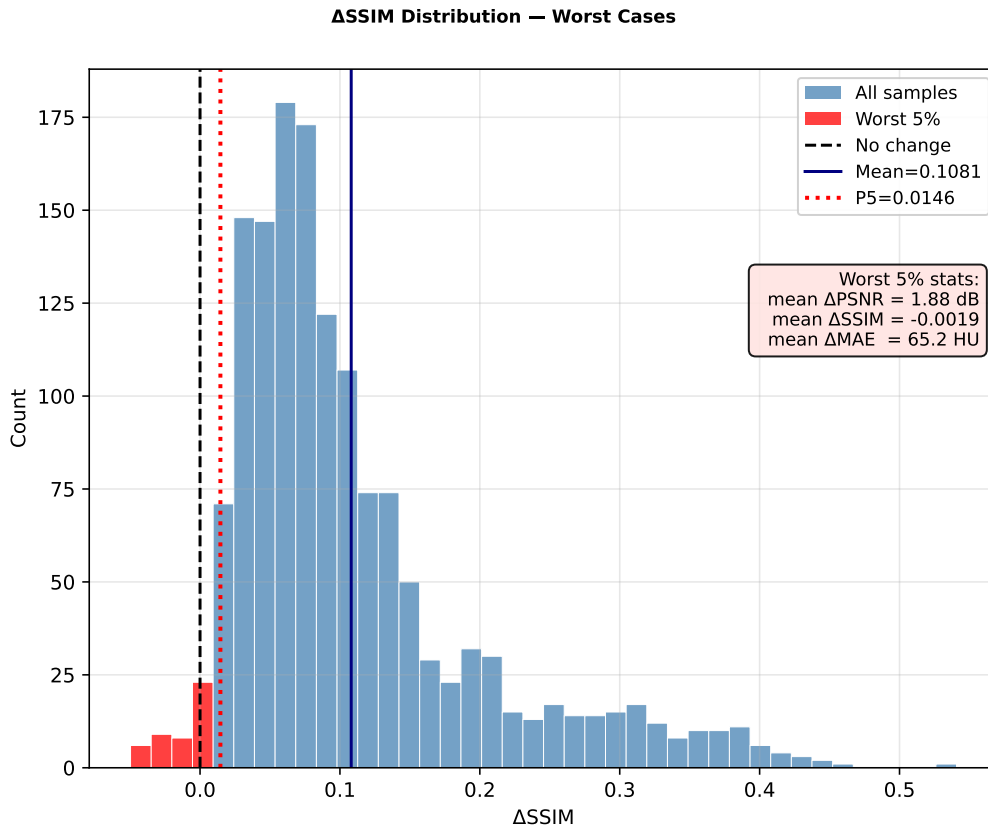


Fig.A.10 Bottom 5th-percentile slices by global Δ SSIM for the Single Model, seed 2024, test split ($N = 74/1478$, threshold = 0.0146, highlighted in red). Descriptive. No inferential test is derived from this subset.

Using the complementary Δ PSNR criterion ($N = 74/1478$ slices; threshold = 2.59 dB), this group was not identical to that defined by Δ SSIM. Mean improvement remained positive for all three metrics (mean Δ PSNR = 1.23 dB, mean Δ SSIM = 0.0071, mean Artifact-Zone Δ MAE = 62.0 HU), although with substantial dispersion and some individual cases of degradation relative to the input, visible in the left tail of [Figure A.11](#).

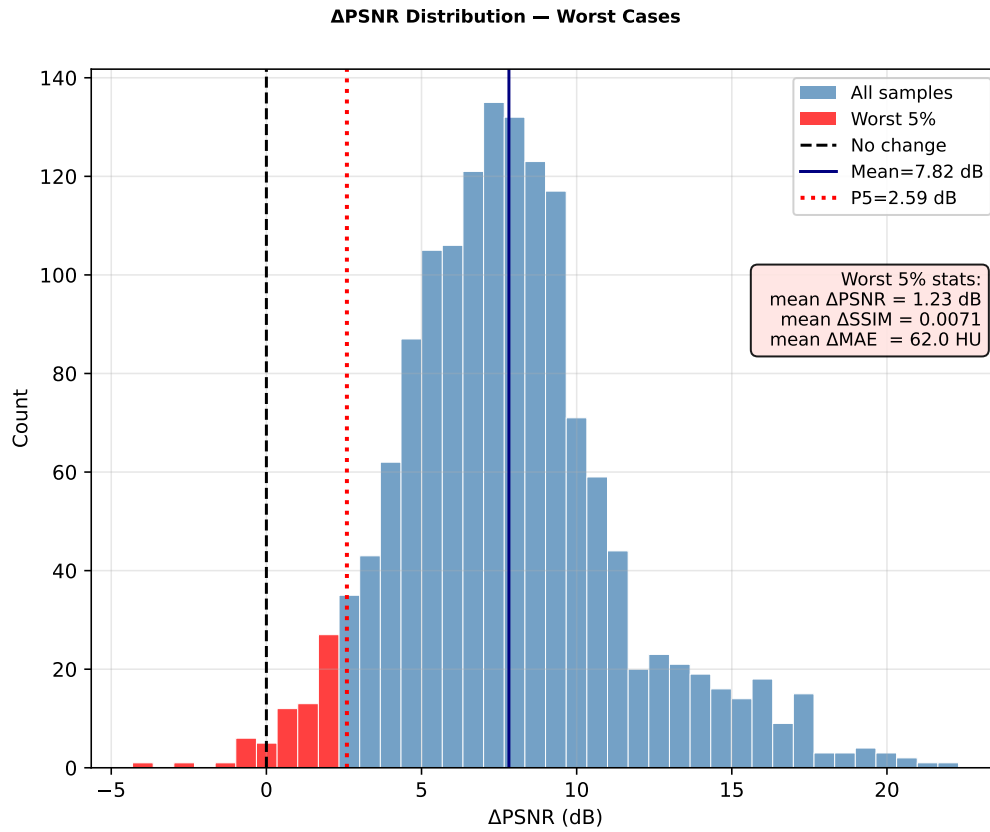


Fig.A.11 Bottom 5th-percentile slices by global Δ PSNR for the Single Model, seed 2024, test split ($N = 74/1478$, threshold = 2.59 dB, highlighted in red). Descriptive. No inferential test is derived from this subset.

Finally, [Figure A.12](#) jointly relates slice-level Δ PSNR and Δ SSIM, highlighting the lower-performance tail defined by Δ PSNR in the preceding paragraph and allowing visualization of the bivariate dispersion of unfavorable cases.

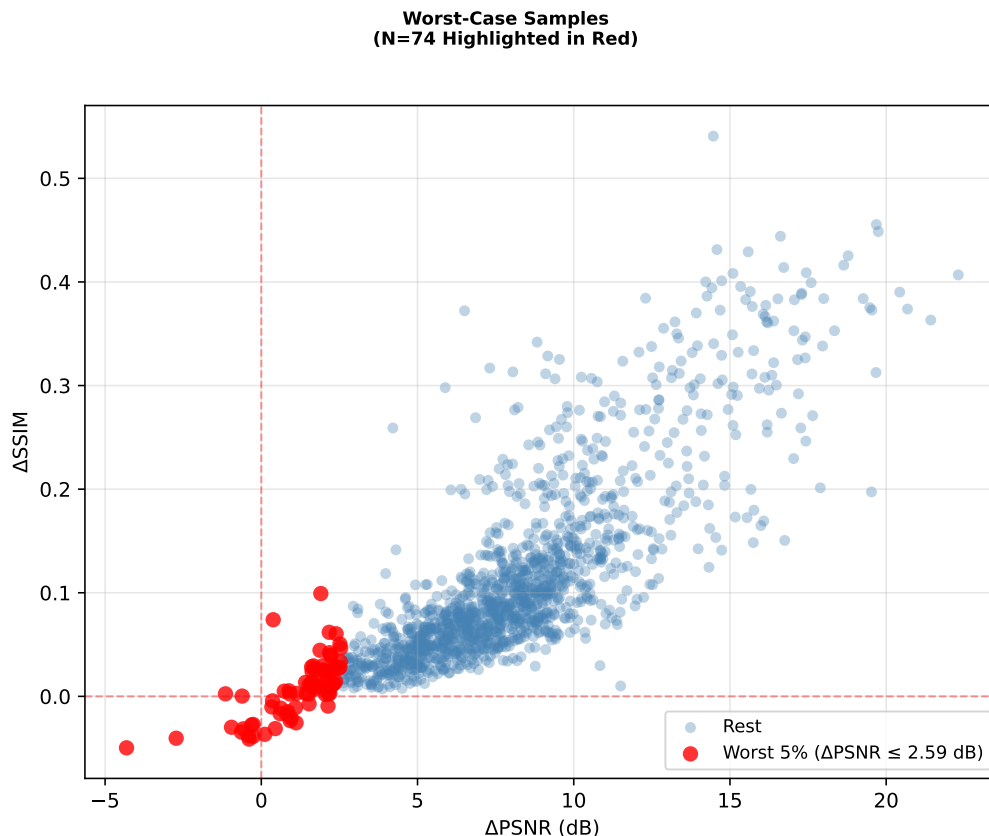


Fig.A.12 Bottom 5th-percentile slices by global ΔPSNR for the Single Model, seed 2024, test split ($N = 74/1478$, threshold = 2.59 dB, highlighted in red). ΔPSNR vs. ΔSSIM scatter. Descriptive. No inferential test is derived from this subset.

A.3 Computational Performance

Runtime performance was quantified for the checkpoint of the model selected for external evaluation, on the internal test partition ($n = 1478$ slices). The benchmark was run in the environment described in Materials and Methods, using an NVIDIA RTX A4500 GPU, batch size one, and five warm-up passes. For each slice, latency was measured from the model call to explicit synchronization of the output with the CPU. Consequently, this measurement includes inference execution and output synchronization, but excludes image loading and external preprocessing or result-presentation stages.

Mean latency was 24.13 ms per slice (standard deviation: 0.69 ms, median: 24.30 ms), corresponding to a throughput of 41.4 slices per second. The 95th and 99th percentiles were 24.66 ms and 27.53 ms per slice, respectively. The proximity of the mean, median, and 95th percentile indicates latency stability for most runs. The complete distribution and slice-level variation are shown in [Figure A.13](#). These values characterize the experimental configuration

used and do not, by themselves, represent the response time of a complete clinical workflow or of another computational environment.

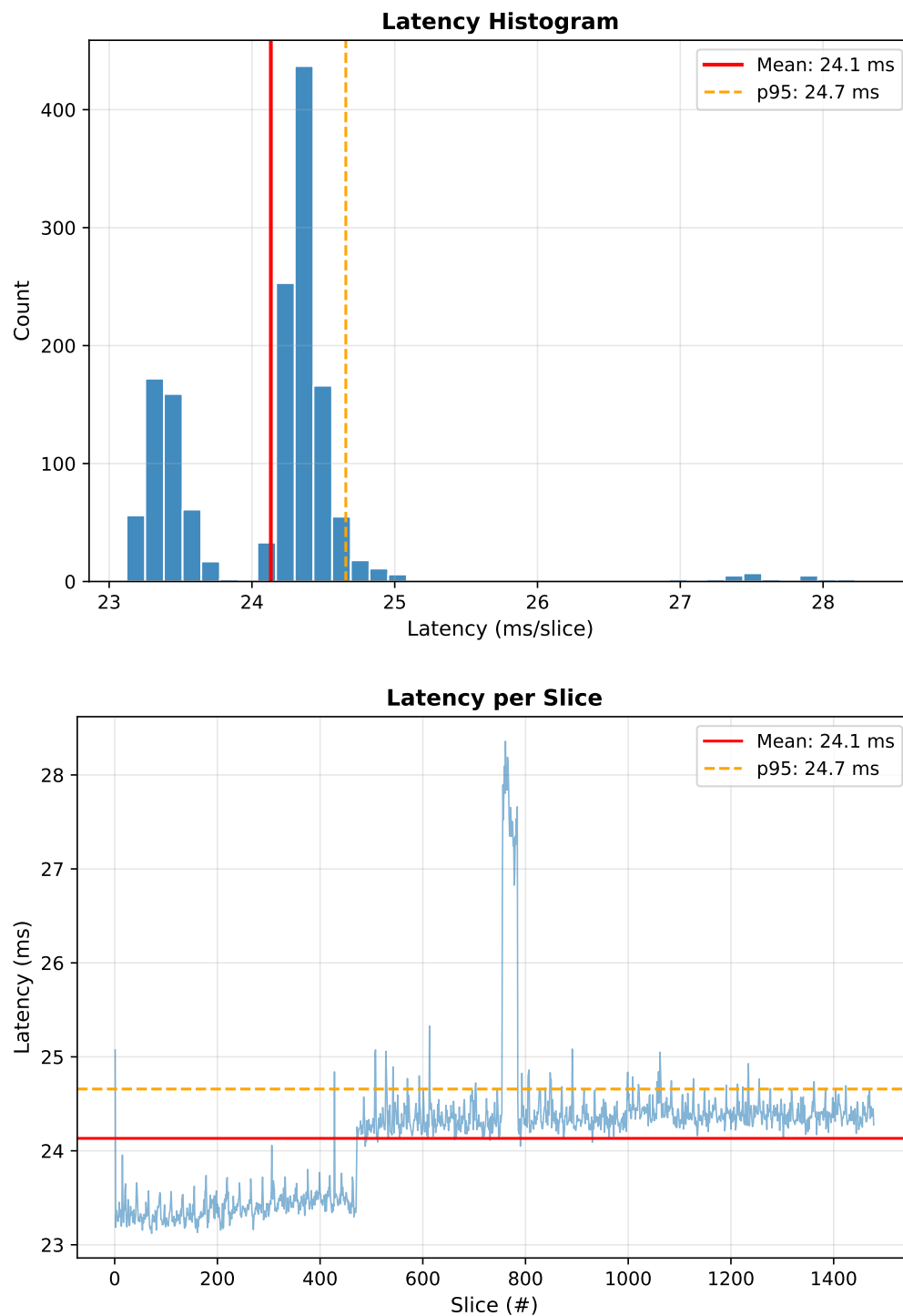


Fig.A.13 Inference benchmark: latency histogram and latency per slice.